Asymptotic Minimax Optimality of Set Estimation

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Abstract

Parameters of interest are often expressed as transformations of partially identified raw parameters. In such settings, the identified set of the targeted objects can be non-convex and there are no theoretical results on how well one can learn it from data. This paper fills part of this gap by studying the asymptotic efficiency of set estimation for a general class of identified sets. I first derive the asymptotic minimax efficiency bound for set estimators under risk functions based on the Hausdorff metric. A key ingredient in the analysis is a limit experiment framework tailored to the local structure of the identified set. I then show that a set estimator based on a criterion function is efficient under mild conditions. The analysis requires deriving the limiting distribution of the set estimator in the Hausdorff metric, which I obtain without globally characterizing the identified set via its support functions. I examine the finite-sample performance of a bootstrap-based confidence set motivated by the obtained limiting distribution. My framework is illustrated with an application to optimal taxation with behavioral agents.

Keywords: partial identification, set estimation, asymptotic optimality.
JEL Classification Codes: C10, C13, C14, C18.

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1. Introduction

Since Manski (1989), the literature on partial identification has shown that informative conclusions can often be drawn without assumptions (e.g., exclusion restrictions, functional form, equilibrium selection, support conditions) that are needed to pin down a parameter uniquely but are hard to justify in practice; instead, the data combined with reliable assumptions lead to an *identified set*, which provides more credible information[1]. Despite its conceptual soundness, a difficulty observed in its empirical applications is that the estimated set and the confidence set are often too large to be informative. Unlike in the point-identified case, the size of these objects in a partially identified setting is determined by two components: the true identified set in the population and statistical noise. The size of the identified set, or more precisely the size of the "sharp" identified set[2], is itself informative to researchers; it indicates how effective the data and the assumptions are in narrowing down the parameters of interest and provides guidance for the choice of assumptions. The size of the statistical noise, on the other hand, reflects the effectiveness of statistical procedures rather than the econometric model or the choice of assumptions, implying that there may be room for improvement.

Regarding the statistical noise for set objects, little is known about how well one can handle it. This is in sharp contrast to the point-identified case. When a parameter is point identified, classical theory provides a lower bound on such noise; examples include the Cramér–Rao bound, the Hájek–Le Cam convolution and the local asymptotic minimax theorems (e.g., Hájek (1970), Hájek (1972), and Le Cam (1972)), and their counterparts for moment condition models (e.g., Hansen (1982) and Chamberlain (1987)). These lower bounds serve as a criterion for an efficient estimator. Comparable benchmarks are scarce for partially identified models. An exception is Kaido and Santos (2014), who derive a semiparametric efficiency bound for a convex identified set. Outside this class, there is no available lower bound on the statistical noise, which makes it difficult to attribute the size of estimated objects purely to that of the sharp identified set. Note that the lack of optimality results matters in practice because parameters of interest or policy-relevant objects are often expressed via transformation of partially identified raw parameters. In such situations, the set of target objects can be non-convex; moreover, since the Hausdorff metric, which is a distance between sets commonly used in the literature, is not invariant under general transformations, a naive two-step approach to compute the target set can be highly suboptimal.

A goal of this paper is to fill part of this gap by investigating the asymptotic efficiency of set estimation. This can be formalized as the following statistical decision problem. Let $\Theta_I \subset \mathbb{R}^{d_\theta}$ be the identified set of interest, which belongs to a certain family of sets

[1]This approach appears in many areas of empirical research: the bounding approach (Lee (2009)) to deal with attrition; moment inequalities derived from equilibrium or economic agents' optimizing behavior in models of entry games, network formation, and export decisions (e.g., Ciliberto and Tamer (2009), Ho (2009), Pakes et al. (2015), and Dickstein and Morales (2018)); and sign restrictions in structural vector autoregressions (e.g., Uhlig (2005), Moon and Schorfheide (2012), Baumeister and Hamilton (2015), Granziera et al. (2018), and Giacomini and Kitagawa (2021)).

[2]The sharp identified set means the smallest region implied by the maintained assumptions.

$\mathcal{K}^*_\Theta$. For any set-valued estimator $T_n : \mathcal{X} \mapsto \mathcal{K}^*_\Theta$ defined on a sample space with probability measure P, its distance to Θ_I is naturally measured by the Hausdorff distance $d_H(T_n(X), \Theta_I)$. Accordingly, I can define the risk based on a loss function $l : \mathbb{R}_+ \mapsto \mathbb{R}_+$ and d_H. The central question then concerns the asymptotic risk

$$\liminf_{n\to\infty} \mathbb{E}_P \, l \circ d_H(T_n(X), \Theta_I) \geq B,$$

and my goal is to find the lower bound B. In this framework, I set $\mathcal{K}^*_\Theta$ to be the class containing the family of singletons as well as convex and non-convex sets; this flexibility allows the above framework to cover a wide range of situations, such as the point-identified case $\Theta_I = \{\theta_0\}$ and those not considered in the efficiency arguments in the literature.

The paper makes two main contributions. I first derive the generic lower bound for set estimators. My approach consists of three steps. First, I extend the limit experiment framework of Le Cam (1972) to the situation where multiple parameters are "observationally equivalent" in a parametric model. The resulting limit experiment takes the form of a random process or field indexed by the parameters on the boundary $\partial\Theta_I$. Second, using this limit experiment representation, I derive the asymptotic minimax lower bound for set estimators. Third, I extend the analysis in the parametric regime to the semiparametric one via the approach of Chamberlain (1987).

Building on the bound, I next derive the Hausdorff limiting distribution of the extremum-type estimator $\widehat{\Theta}_I$ based on a criterion function. This type of estimator has been studied since Manski and Tamer (2002) and Chernozhukov et al. (2007). To the best of my knowledge, there is no study that establishes the limiting distribution except Kaido (2016), who uses the duality between the criterion-function-based approach and the support-function-based approach; this framework necessarily requires that the target set be convex (if Θ_I is non-convex, then the target is its convex hull), whereas my framework covers non-convex sets as well. Under mild conditions on the smoothness of the moment restrictions, which are satisfied in a wide range of moment inequality models, the limiting distribution of $\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I)$ is shown to be expressed as a functional of a Gaussian process or field indexed by the parameters on the boundary $\partial\Theta_I$. This finding suggests the usefulness of bootstrap-based confidence sets, which are typically considered for convex Θ_I (Beresteanu and Molinari, 2008; Kaido, 2016).

My framework can be useful in situations where parameters of interest are functions of partially identified raw parameters. As an example, I illustrate my framework with an application to the optimal top tax rate with behavioral agents. The formula for the optimal top tax rate in Farhi and Gabaix (2020) reflects behavioral biases of taxpayers, which sharpens theoretical understanding of optimal taxation. However, when empirically computing optimal top tax rates based on the formula, one faces the difficulties that the incomes of these taxpayers are bracketed and that these biases are hard to uniquely pin down. I show that my result on efficient set estimation provides credible information on optimal taxation.

The key mechanism in deriving the generic lower bound is to track the local information at each $\theta \in \partial\Theta_I$, which is potentially direction-dependent. This dependence on direction arises when the (moment) restrictions defining the identified set are only

(Hadamard) directionally differentiable. The standard differentiability (in the sense of Fréchet) can be treated as a special case of directional differentiability, and I cover this general scenario. As in the classical point-identified case, the efficiency analysis focuses on the local behavior of the set estimator, i.e., how a local perturbation of the parameter affects the risk. In my context, the local perturbation of the set Θ_I needs to be considered. Efficiency is often considered via the minimax criterion in the literature on statistical decision theory, and I follow this convention. Throughout the analysis, I work with the local nature of Θ_I and do not use its global features.

The limiting distribution of a set estimator is typically derived using support functions of the identified set. In this approach, the identified set is globally embedded into the space of continuous functions on the unit sphere and the quality of estimation reduces to how well one can estimate the support function that characterizes Θ_I (e.g., Beresteanu and Molinari (2008), Bontemps et al. (2012), and Kaido and Santos (2014)). To maintain the generality of the class of sets, I work directly with the stochastic process or field that represents the Hausdorff distance $d_H(\widehat{\Theta}_I, \Theta_I)$, thereby bypassing the global characterization of the identified set Θ_I. This stochastic process is not asymptotically equicontinuous, implying that the usual arguments for weak convergence of stochastic processes cannot be applied. Similar to Chernozhukov et al. (2007) and Kaido (2016), I overcome this issue by using a type of "*weak epiconvergence*" technique to track the behavior of the stochastic process.

1.1. **Literature.** This paper is related to several strands of the literature. The first is the large literature on estimation and inference in partially identified models, which is now mature; see Molinari (2020) for details. The most related paper in this strand is Chernozhukov et al. (2007), who develop a general criterion function approach to set estimation and confidence regions for identified sets, establishing consistency and rates in the Hausdorff metric; the estimator I study belongs to their class. The intuition behind this study stems from the literature on (unconditional) moment inequalities (e.g., Romano and Shaikh (2010), Andrews and Guggenberger (2009), Andrews and Soares (2010), Bugni (2010), and Canay (2010)).

This paper also contributes to the literature on various notions of optimality in partially identified models. Most of this work concerns hypothesis testing, where optimality is expressed in terms of power. Andrews and Soares (2010) compare the asymptotic power of competing tests of moment inequalities; Armstrong (2014) studies power against local alternatives for conditional moment inequalities; Bugni et al. (2012) analyze how local misspecification distorts the size of these procedures. The paper closest to mine in this strand is Kaido and Santos (2014). The authors consider identified sets defined by moment inequalities that are convex in the parameter, with the slopes of any linear constraints known, so that the identified set is convex and characterized by its support function; this characterization via the support function is used in Beresteanu and Molinari (2008), Beresteanu, Molchanov, et al. (2011), and Bontemps et al. (2012) in the econometrics literature. They apply the techniques of semiparametric efficient estimation (e.g., Bickel et al. (1998)) to support function estimation and derive the efficiency bound. My approach is different in that I do not embed the identified set, via

its support function, into the space of continuous functions indexed by the unit vectors; this global characterization of the identified set cannot distinguish two sets whose convex hulls coincide. Instead, I focus on local deviations at each point on the boundary and connect them to their implications for the Hausdorff distance. This local nature widens the scope of the efficiency bound to include finite sets of points, disconnected sets, and sets with outward kinks as special cases.

The basis of this paper's technical development is the limit experiment and statistical decision frameworks (e.g., Le Cam (1972), Hájek (1970), and van der Vaart (1991)). Both have been widely used in econometrics to evaluate estimators, tests and decision rules (e.g., Hirano and Porter (2009), Müller (2011), and Andrews and Mikusheva (2022)); see Hirano and Porter (2020) for a detailed account of the framework and its recent developments. In this literature, the object of interest is a point in a finite- or infinite-dimensional space, and efficiency results hinge on the derivative of some criterion function at that point. This approach does not extend directly to the set-identified context. I tailor the underlying limit experiment framework to partially identified models so that set estimation can be analyzed in a manner similar to the classical point-identified case.

1.2. **Organization.** The rest of the paper is organized as follows. Section 2 presents a motivating application of my framework to optimal taxation with behavioral agents. Section 3 presents the general setting and assumptions, including the class of identified sets covered in the analysis. Section 4 develops the tailored limit experiment framework and derives the lower bound for set estimators. I also extend the analysis to the case where the identified set is defined via general moment restrictions. Section 5 studies the asymptotic properties of the set estimator based on a criterion function and derives its limiting distribution in the Hausdorff metric. Section 6 presents Monte Carlo simulations. Section 7 concludes. All proofs are collected in the Appendix.

2. Motivating example

I here present the optimal top tax rate with behavioral agents as an example in which this paper can be useful for economic questions studied in the empirical literature. The logic behind this example is general enough to apply to other contexts, such as computing markups or deriving optimal unemployment benefits. The unifying theme in these examples is as follows. Many parameters of interest in applied work, such as optimal tax rates and markups, are nonlinear functions of a few primitive quantities. Two points arise. First, point identification of the primitives often relies on assumptions that are not motivated by economic theory, and dropping these assumptions leaves the primitives partially identified. Second, a nonlinear map can turn a convex set of primitives into a non-convex set of the parameter of interest. Existing optimality results for set estimation do not cover this case. Nor can one estimate the set of primitives and then map it: the Hausdorff metric is not preserved under nonlinear transformations, so optimality of the first step does not carry over to the second.

Policy parameter of interest. I start from formally defining parameter of interest. Consider taxpayers whose income is above a threshold z^* and who face a constant marginal tax rate τ on income above z^*. A small increase in τ raises revenue from income already reported in the bracket and lowers it through the behavioral response of reported income. Saez (2001) shows that the revenue-maximizing rate is

$$\tau^* = \frac{1}{1+\pi\cdot e}, \quad \pi := \frac{z_m(z^*)}{z_m(z^*)-z^*}, \quad z_m(z^*) := \mathbb{E}_P[Z \mid Z \geq z^*], \tag{1}$$

where Z denotes reported income, π is the Pareto parameter of the upper tail, and e is the elasticity of reported income with respect to the net-of-tax rate $1-\tau$[3]. The formula (1) is a sufficient statistic result in the sense of Chetty (2009).

The formula in Saez (2001) presumes that taxpayers respond to their true marginal rate. Motivated by the large literature in behavioral economics[4], Farhi and Gabaix (2020) extend Saez (2001) and derive the optimal top rate when taxpayers misperceive the tax schedule. In their framework, a taxpayer with income z perceives the marginal rate

$$T'_{,s}(z) = \omega\cdot T'(z) + (1-\omega)\left[\int_0^\infty T'(uz)\,\psi(u)\,du + b(z)T(0)\right],$$

with $\int_{\mathbb{R}_+}\psi(u)\,du = 1$ and $\lim_{z\to\infty} b(z) = 0$, that is, a weighted average of the true marginal rate, with attention weight $\omega\in[0,1]$, and of a weighted average of the marginal rates at other income levels uz, plus a term in $T(0)$ that vanishes for large z. This represents the situation where a taxpayer could respond to a rate that is far from the true marginal rate. In the revenue-maximizing case without income effects, Proposition 12 in Farhi and Gabaix (2020) gives

$$\bar\tau = \frac{1}{1+\bar\zeta_{c,r}\cdot\pi\cdot\kappa(A)}, \quad \kappa(A) := \omega + (1-\omega)A, \quad A := \int_0^\infty u^{\pi-1}\psi(u)\,du, \tag{2}$$

where $\bar\zeta_{c,r}$ is the compensated elasticity of a taxpayer who perceives the schedule correctly. The factor κ is the fraction of a top-rate reform to which taxpayers respond, and A measures the shape of misperception. If taxpayers iron, the misperceived rate is the average rate $T(z)/z = \int_0^1 T'(uz)\,du$, so that ψ is uniform on $[0,1]$ and $A = 1/\pi < 1$. A misperceived rate that puts weight on incomes above the taxpayer's own, such as a salient statutory top rate, gives $A > 1$. Note that either $\omega = 1$ or $A = 1$ yields $\kappa = 1$; hence (2) reduces to (1) with $e = \bar\zeta_{c,r}$.

Given the setting above, an interesting question would be how the deviation of the perceived rate from the true one affects the optimal tax rate. Since A captures the shape of the misperception, in empirical work this question boils down to finding the

[3] Saez (2001) denotes the Pareto parameter by a. I use π to be consistent with the notation of Farhi and Gabaix (2020).

[4] De Bartolome (1995) finds in the laboratory that many subjects use the average rate as if it were the marginal rate. Feldman et al. (2016) find that households reduce reported wage income when the Child Tax Credit expires, a lump-sum change that leaves the marginal rate unchanged. Rees-Jones and Taubinsky (2020) find in a survey experiment that taxpayers commonly "iron", that is, act on their average rate, while few "spotlight" a single bracket. Attention to taxes also varies across individuals (Taubinsky and Rees-Jones, 2018).

pairs $(\bar{\tau}, A)$ that are consistent with the available data. The object of interest is then formally expressed as the pair that is supported by the data, i.e.,

$$\theta = (\bar{\tau}, A)^t \in \Theta \subset \mathbb{R}^{d_\theta}, \quad d_\theta = 2,$$

which records which form of misperception is required for a given rate to be optimal[5]. If all ingredients in equation (2) can be uniquely pinned down from the data, then the pair $(\bar{\tau}, A)$ of interest is uniquely determined, and the identified set is a singleton in $\mathbb{R}^{d_\theta}$. However, the assumptions required for point identification of these ingredients are stringent in practice as discussed below. In this case, instead of assuming a unique pair of $(\bar{\tau}, A)$, focusing on the "profile" of the set of pairs $(\bar{\tau}, A)$ that is supported by the data and by weaker assumptions provides robust policy implications.

Partial identification. As illustration, I here focus on two inputs of (2) that are not point identified: the tail of the income distribution and the misperception shape.

The first is the tail. The mean income $z_m(z^*)$ is an average over an unbounded region, which is determined by the observations that data sources withhold; published tax tabulations have an open top bracket, and public-use survey micro-data are top-coded. Piketty and Saez (2003) address this by Pareto interpolation, which assumes the functional form whose parameter is at issue. There is no consensus regarding this point. Auten and Splinter (2024) find that top income shares are lower, and have risen less since 1980, than those in Piketty, Saez, and Zucman (2018), and the difference turns on the allocation of income missing from tax returns.

The second is the misperception shape. While Rees-Jones and Taubinsky (2020) provide empirical evidence on A from samples representing the general population, no paper exists that directly measures A for taxpayers in the top bracket. There are reasons to expect top earners to differ from the general population in terms of misperception shape: the gap between the average and the marginal rate is largest at the top, which strengthens the effect of ironing, whereas the top statutory rate is a salient and widely discussed number, which strengthens over-perception.

Identified set. Based on the argument above, the set of pairs $\theta = (\bar{\tau}, A)^t$ supported by the data, i.e., the identified set for θ, is derived as follows. Suppose that reported incomes are observed with topcoding at a known cap $z_c > z^*$, so that the mean income $z_m(z^*)$ is bounded but not identified, and let $[\pi_L, \pi_U]$ denote the resulting range of the tail parameter π. Under appropriate conditions on the data and the behavioral parameters, the identified set for θ can be expressed as

$$\Theta_I = \left\{ (\bar{\tau}, A)^t \in \Theta : \; A_L \leq A \leq A_U, \; \frac{1}{1 + \bar{\zeta}_{c,r} \pi_U \kappa(A)} \leq \bar{\tau} \leq \frac{1}{1 + \bar{\zeta}_{c,r} \pi_L \kappa(A)} \right\}, \tag{3}$$

where $[A_L, A_U]$ is a maintained range for the misperception shape. The formal set-up of the moment restrictions and the derivation of Θ_I are given in Section 3.3.

The set Θ_I has specific geometric features. First, it is not convex. Its upper boundary, $A \mapsto 1/(1 + \bar{\zeta}_{c,r} \cdot \pi_L \cdot \kappa(A))$, is the reciprocal of a positive affine function of A and hence

[5]Here I fix ω at a known value ω_0, for which salience studies provide estimates (e.g., Chetty et al., 2009; Taubinsky and Rees-Jones, 2018; Lardeux, 2023), although these come from different contexts.

strictly convex for $\omega_0 < 1$, so the region below it is not convex. Second, the set is the intersection of four sets defined by smooth inequalities, so that its corners are outward kinks, and the slopes of the two inequalities on $\bar{\tau}$ depend on the data through π_L and π_U. These points suggest an alternative approach from the existing one which utilize the convexity of identified set. The support function approach (e.g., Beresteanu and Molinari (2008), Kaido and Santos (2014)) characterizes the convex hull of Θ_I, which contains pairs $(\bar{\tau}, A)$ that no admissible value of π generates; hence the dependence of the optimal rate on the misperception shape, which is the object of this exercise, cannot be inferred from this approach.

3. Modeling framework

3.1. **Problem settings.** Let $(\mathcal{X}, \mathcal{B}_\mathcal{X}, P)$ denote a probability space, where P belongs to a family of probability measures $\mathcal{P}$ on $(\mathcal{X}, \mathcal{B}_\mathcal{X})$. Let $\Theta \subset \mathbb{R}^{d_\theta}$ with $d_\theta < \infty$ be a suitably large compact set in the d_θ-dimensional Euclidean space; without loss of generality, take $\Theta = [-M_\theta, M_\theta]^{d_\theta}$ for some $M_\theta < \infty$. Consider a moment function $m : \Theta \times \mathcal{X} \mapsto \mathbb{R}^{d_m}$ with $d_\theta \leq d_m$, which is assumed to belong to a family $\mathcal{M}$ of measurable functions. Given a moment function m and a probability measure P, one obtains a moment restriction $\Phi : \mathbb{R}^{d_\theta} \times \mathcal{P} \mapsto \mathbb{R}^{d_q}$ defined as $\Phi(\theta, P) := q\left(\int_\mathcal{X} m(\theta, \cdot)\, dP(\cdot)\right)$, where $q : \mathbb{R}^{d_m} \mapsto \mathbb{R}^{d_q}$ with $d_\theta \leq d_q \leq d_m$ is some known function.

My interest is the "identified set" of Φ, i.e., the set of all parameters that take certain values of Φ,

$$\begin{aligned} \Theta_I &= \Theta_I(\Phi, P) \\ &:= \left\{ \theta \in \Theta^o \subset \mathbb{R}^{d_\theta} : \Phi(\theta, P) = q\left(\int_\mathcal{X} m(\theta, x)\, dP(x) \right) \leq 0 \in \mathbb{R}^{d_q} \right\}. \end{aligned} \tag{4}$$

In section 4 I derive the asymptotic efficiency bound for Θ_I defined in (4). Based on the result, in section 5 I consider the set-valued estimator T_n based on criterion function $Q : \Theta \mapsto \mathbb{R}$, where T_n is defined as the contour set or level set of the sample analogue Q_n. This formulation also incorporates moment inequality models; $q(s) = (s_1 \vee 0, \cdots, s_{d_m} \vee 0)$ for any $s = (s_1, \cdots, s_{d_m})$ as in Chernozhukov et al. (2007) for example. This potentially multi-valued set Θ_I is assumed to belong to a certain class $\mathcal{K}^*_\Theta$ of sets specified in detail later. A key observation here is that properties of Θ_I depend on the moment restriction Φ and a probability measure P.

One of my objectives is to derive and characterize the efficiency bound of set estimators for Θ_I, $T_n : \mathcal{X}^n \mapsto \mathcal{K}^*_\Theta$, among a certain class of estimators. In the point identified case, one needs proper restrictions on the model $\mathcal{P}$ and the class of estimators. In particular, the model $\mathcal{P}$ is assumed to be "smooth" often measured in the L^2-norm of its density (the so-called "differentiable in quadratic mean"), while the class of estimators is assumed to behave the same way in neighborhoods of parameters ("locally regular"); see Bickel et al. (1998) for textbook treatment.

The key difficulties in my context are (1) how these conditions on $\mathcal{P}$ and the class of estimators in the point identified case are transferred to the set-identified case and (2)

what conditions on $\mathcal{K}_\Theta^*$ are appropriate. For (2), I set up $\mathcal{K}_\Theta^*$ to allow non-convex sets there; this widens the scope of Kaido and Santos (2014), where the target class $\mathcal{K}_\Theta^*$ is restricted to convex sets. I study these conditions in detail in the set-identified case.

3.2. **Class of compact sets $\mathcal{K}_\Theta^*$.** I specify the target class of sets $\mathcal{K}_\Theta^*$ to which the identified set Θ_I is assumed to belong. The point here is that the class $\mathcal{K}_\Theta^*$ should be nice enough such that analysis to derive efficiency bound and limiting distribution of estimators in point-identified case can be naturally extended to set case, while at the same time the class should be flexible enough for application.

Let $\mathcal{K}$ be the family of all non-empty compact sets in $\mathbb{R}^{d_\theta}$. The space $(\mathcal{K}, d_H)$, where $d_H : \mathcal{K} \times \mathcal{K} \mapsto \mathbb{R}_+$ is the Hausdorff metric, is then a complete separable metric space[6]. The target class $\mathcal{K}_\Theta^*$ is assumed to be $\mathcal{K}_\Theta^* \subset \mathcal{K}_\Theta := \mathcal{K} \cap \Theta$ with $\Theta = [-M_\theta, M_\theta]^{d_\theta}$, whose characteristic is specified as follows.

First, let $\mathcal{K}_1 \subset \mathcal{K}_\Theta$ be such that

$$\mathcal{K}_1 := \{K \in \mathcal{K}_\Theta \text{ in } \mathbb{R}^{d_\theta} : |K| \leq N_1 < \infty\}.$$

The class $\mathcal{K}_1$ captures compact sets with up to N_1-number of finite points. Next, define a class $\mathcal{K}_2 \subset \mathcal{K}_\Theta$ as

$$\mathcal{K}_2 := \{K \in \mathcal{K}_\Theta \text{ in } \mathbb{R} : K = \bigcup_{i=1}^{N_2} K_i,\ K_i \in \mathcal{K}_\Theta, K_i \cap K_j = \emptyset, K_i^o \neq \emptyset \text{ for each } i \neq j\},$$

for some $N_2 < \infty$. The class $\mathcal{K}_2$ includes compact sets which consist of up to N_2-number of compact intervals.

Now I define the last compact class $\mathcal{K}_3 \subset \mathcal{K}_\Theta$. The class is designed so that its members have a piecewise smooth boundary and may have a dimension lower than d_θ. A geometric notion that captures these properties is manifold. To formalize this intuition, I follow the definitions in Lee (2012). Let $\mathbb{H}^k := \{x = (x^1, \ldots, x^k) \in \mathbb{R}^k : x^k \geq 0\}$ denote the closed upper half-space and $\overline{\mathbb{R}}_+^k := \{x \in \mathbb{R}^k : x^1 \geq 0, \ldots, x^k \geq 0\}$ the closed positive orthant, which is the model for corners.

Definition 3.1 (Smooth manifold with corners)**.** Let $k \geq 1$ be an integer.

(a) A *topological k-manifold with boundary* is a second-countable Hausdorff space M in which every point has a neighborhood homeomorphic either to an open subset of $\mathbb{R}^k$ or to a relatively open subset of $\mathbb{H}^k$.

(b) Let M be a topological k-manifold with boundary. An *interior chart* for M is a pair (U, φ), where $U \subset M$ is open and φ is a homeomorphism from U onto an open subset of $\mathbb{R}^k$. A *chart with corners* for M is a pair (U, φ), where $U \subset M$ is open and φ is a homeomorphism from U onto a relatively open subset of $\overline{\mathbb{R}}_+^k$.

(c) Two such charts (U, φ) and (V, ψ) are *smoothly compatible* if the transition map $\varphi \circ \psi^{-1} : \psi(U \cap V) \mapsto \varphi(U \cap V)$ is smooth, i.e., it admits a C^∞ extension to an open neighborhood of each point of $\psi(U \cap V)$.

[6]More generally, the space of non-empty compact sets is complete with the Hausdorff metric if the underlying space is locally compact or complete (Schneider (2013), Molchanov (2017)). Separability also holds if the underlying space is a separable Banach space (Giné et al. (1983)).

(d) A *smooth structure with corners* on M is a maximal collection of smoothly compatible interior charts and charts with corners whose domains cover M. A topological k-manifold with boundary together with a smooth structure with corners is called a *smooth k-manifold with corners*.

Parts (a) and (b)–(d) are the definitions in Lee (2012, p. 25) and Lee (2012, p. 415), respectively. Note that every smooth manifold with or without boundary is also a smooth manifold with corners and that smooth maps, tangent spaces and differentials on smooth manifolds with corners are defined through these charts in exactly the same way as on smooth manifolds with boundary (Lee, 2012, p. 417). Accordingly, following Lee (2012, p. 85), a map $\iota : M \mapsto \mathbb{R}^{d_\theta}$ from a smooth manifold with corners M is a *smooth embedding* if it is a smooth immersion (i.e., its differential is injective at every $x \in M$) and a homeomorphism onto its image $\iota(M)$ with the subspace topology.

Now I define a target set based on Definition 3.1 in the following manner.

Definition 3.2 (Embedded submanifold with corners)**.** Let $k \in \{1, \dots, d_\theta\}$. A subset $K \subset \mathbb{R}^{d_\theta}$ is a *k-dimensional embedded submanifold with corners* of $\mathbb{R}^{d_\theta}$ if K, endowed with the subspace topology, is a topological k-manifold with boundary equipped with a smooth structure with corners such that the inclusion map $\iota_K : K \hookrightarrow \mathbb{R}^{d_\theta}$ is a smooth embedding.

Definition 3.2 is the analogue of an embedded submanifold with boundary, with charts with corners in place of boundary charts (Lee, 2012, p. 120). Note that the requirement that ι_K be a smooth embedding is important[7]. I now define

$$\mathcal{K}_3 := \big\{K \in \mathcal{K}_\Theta : K \text{ is a compact } k\text{-dimensional embedded submanifold with corners of } \mathbb{R}^{d_\theta} \text{ for some } k \in \{1, \dots, d_\theta\}\big\}.$$

The class $\mathcal{K}_3$ includes full-dimensional sets ($k = d_\theta$), such as a closed ball or a polytope with non-empty interior, as well as lower-dimensional sets ($k < d_\theta$), such as a sphere or a curve segment in $\mathbb{R}^3$[8].

For the target class $\mathcal{K}_\Theta^*$, I focus on the union of each compact class described above.

Assumption 3.3 (Class of compact sets $\mathcal{K}_\Theta^*$)**.**

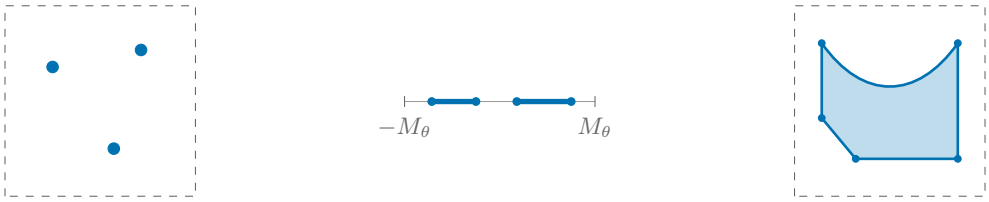
$$\Theta_I \in \mathcal{K}_\Theta^* = \begin{cases} \mathcal{K}_1 \cup \mathcal{K}_2 & \text{if } d_\theta = 1, \\ \mathcal{K}_1 \cup \mathcal{K}_3 & \text{if } d_\theta \geq 2. \end{cases}$$

The class $\mathcal{K}_1$ roughly captures the point identified case and the case with multiple distinct singletons. The class $\mathcal{K}_2$ corresponds to the (multiple of) interval identified set.

[7] Without this requirement, Definition 3.1 only requires that *some* smooth structure with corners exists on K, which places no restriction on how K sits in $\mathbb{R}^{d_\theta}$. For example, the Koch snowflake curve in $\mathbb{R}^2$ is homeomorphic to the unit circle and hence admits a smooth structure, although it has no tangent line at any point.

[8] Note that throughout the paper, ∂K and K^o denote the *topological* boundary and interior of K as a subset of $\mathbb{R}^{d_\theta}$, which are in general different from the manifold boundary and interior of K (Lee, 2012, p. 26).

The class $\mathcal{K}_3$ reflects the case where the identified set has piecewise smooth boundary, which is often observed in moment inequality models.



(a) $\mathcal{K}_1$ (b) $\mathcal{K}_2$ (c) $\mathcal{K}_3$

FIGURE 1. Illustrative members of the classes $\mathcal{K}_1$, $\mathcal{K}_2$ and $\mathcal{K}_3$.

Remark 3.4. Note that the classes $\mathcal{K}_1$ and $\mathcal{K}_2$ can be seen as special cases of $\mathcal{K}_3$ if $k = 0$ is allowed in Definitions 3.1 and 3.2. A compact 0-dimensional manifold is a finite set, so every $K \in \mathcal{K}_1$ is a compact 0-dimensional embedded submanifold of $\mathbb{R}^{d_\theta}$; and for $d_\theta = 1$, every $K \in \mathcal{K}_2$ is a compact 1-dimensional embedded submanifold with corners of $\mathbb{R}$, since smooth 1-manifolds with corners are the same as smooth 1-manifolds with boundary (Lee, 2012, p. 416). I keep the three classes separate for ease of exposition; the bounds N_1 and N_2 make $\mathcal{K}_1$ and $\mathcal{K}_2$ smaller than the corresponding parts of $\mathcal{K}_3$.

3.3. **Example revisited: optimal top tax rate.** I here cast the example of Section 2 in the framework above. Recall from (2) that, for a known attention weight ω_0, the optimal top rate is $\bar{\tau} = 1/(1 + \bar{\zeta}_{c,r} \cdot \pi \cdot \kappa(A))$ with $\kappa(A) = \omega_0 + (1 - \omega_0)A$, and that the parameter is $\theta = (\bar{\tau}, A)^t$, $d_\theta = 2$.

Data. Let $Z \geq 0$ denote reported income, which is not observed when it exceeds a known cap $z_c > z^*$. The observed variable is $X = (\tilde{Z}, D)$ with $\tilde{Z} := Z \wedge z_c$ and $D := \mathbb{1}\{Z \geq z_c\}$, and $\{X_i\}_{i\in[n]}$ is a random sample from P on $(\mathcal{X}, \mathcal{B}_\mathcal{X})$. Define the identified quantities

$$
\begin{aligned}
S &:= P(\tilde{Z} \geq z^*), \\
M_L &:= \mathbb{E}_P\left[\tilde{Z} \cdot \mathbb{1}\{\tilde{Z} \geq z^*\}\right], \\
M_U &:= \mathbb{E}_P\left[\left(\tilde{Z} \cdot (1 - D) + \bar{z} \cdot D\right) \cdot \mathbb{1}\{\tilde{Z} \geq z^*\}\right],
\end{aligned}
$$

where $\bar{z}$ is given in Assumption 3.5(ii) below. M_L replaces each topcoded income by the cap, and M_U by the largest admissible mean income above the cap.

Assumption 3.5 (Top tax rate)**.** (i). $A \in [A_L, A_U]$ for a known range with $0 < A_L < A_U$. (ii). $\mathbb{E}[Z \mid Z \geq z_c] \leq \bar{z}$ for a known $\bar{z} > z_c$. (iii). The attention weight $\omega_0 \in (0, 1)$ and the elasticity $\bar{\zeta}_{c,r} > 0$ are known. (iv). $P(\tilde{Z} > z^*) > 0$ and $P(D = 1) > 0$.

Assumption 3.5 (i) is the only restriction on misperception. For example, if every taxpayer in the bracket either irons or responds at least as strongly as a correctly

perceiving taxpayer, then A_L can be taken at the ironing value $1/\pi$ evaluated at the largest admissible π, while $A_U > 1$ bounds the degree of over-perception. Assumption 3.5 (ii) is used only on the topcoded cell, which implies that the mean income above z^* is finite so that $\pi > 1$. It bounds the mean income above the cap, not the support of Z, and hence unbounded Pareto tails are allowed. Note that the expectation is taken with respect to the distribution of Z, which is well defined since $P(D = 1) > 0$ by Assumption 3.5 (iv). Assumption 3.5 (iii) treats the elasticity as known for simplicity[9]. Assumption 3.5 (iv) rules out a degenerate bracket and ensures that π is not point identified.

Parameter space. Let $\Theta := [\tau_L^*, \tau_U^*] \times [A_L^*, A_U^*]$ with $0 < \tau_L^* < \tau_U^* < 1$ and $0 < A_L^* < A_L < A_U < A_U^*$, where τ_L^* and τ_U^* are chosen so that Θ_I below lies in the interior of Θ. Note that $\kappa(A) > 0$ on Θ. Then inverting (2) gives the tail parameter that supports θ,

$$\pi(\theta) := \frac{1}{\bar{\zeta}_{c,r} \cdot \kappa(A)} \left(\frac{1}{\bar{\tau}} - 1 \right), \tag{5}$$

which is C^∞ on Θ, with $\partial \pi(\theta)/\partial \bar{\tau} = -1/(\bar{\zeta}_{c,r} \cdot \kappa(A) \cdot \bar{\tau}^2) < 0$. Note that the map $\theta \mapsto (\pi(\theta), A)$ is a change of coordinates rather than a projection, since the coordinate A is retained.

Moment restrictions. Let q be the identity and $m = (m_1, \cdots, m_4) : \Theta \times \mathcal{X} \mapsto \mathbb{R}^4$ be given by

$$\begin{aligned}
m_1(\theta, X) &= \left[(1 - \pi(\theta)) \left(\tilde{Z} \cdot (1 - D) + \bar{z} \cdot D \right) + \pi(\theta) \cdot z^* \right] \cdot \mathbb{1}\{\tilde{Z} \geq z^*\}, \\
m_2(\theta, X) &= \left[(\pi(\theta) - 1)\, \tilde{Z} - \pi(\theta) \cdot z^* \right] \cdot \mathbb{1}\{\tilde{Z} \geq z^*\}, \\
m_3(\theta, X) &= A_L - A, \quad m_4(\theta, X) = A - A_U,
\end{aligned}$$

so that

$$\Phi_1(\theta, P) = (1 - \pi(\theta))\, M_U + \pi(\theta) z^* S, \quad \Phi_2(\theta, P) = (\pi(\theta) - 1)\, M_L - \pi(\theta) z^* S.$$

Both stochastic constraints are affine in the moments (S, M_L, M_U) with coefficients that depend on θ, and they are defined on all of Θ; the restrictions $\Phi_3(\theta)$ and $\Phi_4(\theta)$ do not depend on P.

Under the setting above, the identified set Θ_I is expressed as in (3). The formal statement is the following.

Proposition 3.6 (Identified set for the top tax rate)**.** *Suppose Assumption 3.5 holds. Let $\mathcal{Q}(P)$ be the set of distributions Q of Z on $[0, \infty)$ with $\mathbb{E}_Q[Z \,|\, Z \geq z_c] \leq \bar{z}$ under which $(Z \wedge z_c, \mathbb{1}\{Z \geq z_c\}) \sim P$. For $Q \in \mathcal{Q}(P)$, let $z_{m,Q} := \mathbb{E}_Q[Z \,|\, Z \geq z^*]$ and $\pi_Q := z_{m,Q}/(z_{m,Q} - z^*)$. Finrally define*

$$\Theta_I := \{\theta \in \Theta : A_L \leq A \leq A_U,\ \pi(\theta) = \pi_Q \text{ for some } Q \in \mathcal{Q}(P)\}.$$

Then the following statements hold.

[9] If $\bar{\zeta}_{c,r}$ is estimated, it enters Φ below through q. I abstract from this here and leave it for future research.

(a) $\{\pi_Q : Q \in \mathcal{Q}(P)\} = [\pi_L, \pi_U]$ *with*

$$\pi_L := \frac{M_U/S}{M_U/S - z^*}, \quad \pi_U := \frac{M_L/S}{M_L/S - z^*}, \quad 1 < \pi_L < \pi_U < \infty.$$

(b) $\Theta_I = \{\theta \in \Theta : \Phi_j(\theta, P) \leq 0,\ j = 1, \cdots, 4\}$, *and* Θ_I *equals the set in (3).*

Proof. See Appendix B.1. □

4. Efficiency bounds for identified set

My purpose here is to derive the lower bound of the asymptotic risk

$$\liminf_{n\to\infty} R_{P,l}(T_n; A) := \liminf_{n\to\infty} \int_{\mathcal{X}^n} l \circ d_H\left(T_n(x^n), A\right)\ dP(x^n), \tag{6}$$

where $l \circ d_H : \mathcal{K}_\Theta \times \mathcal{K}_\Theta \mapsto \mathbb{R}_+$ denotes a loss function which is some composition of the Hausdorff distance d_H, a set $A \in \mathcal{K}_\Theta$ is of interest, and T_n is a (possibly scaled) $\mathcal{K}_\Theta$-valued estimator. The case $A = \Theta_I$ with $d_H(T_n, \Theta_I) = o_P(1)$ (and hence $R_{P,l}(T_n; \Theta_I) = o(1)$ for well-behaved l such as continuous and bounded ones) is a minimal consistency result obtained in the literature. I aim to go further to characterize the efficiency of T_n by focusing on its behavior in the local neighborhood of Θ_I.

To analyze the asymptotic properties of the risk in such a situation, I start my analysis with the case where identified set is defined via a likelihood function. In section 4.1 I first establish the asymptotic representation for set-valued statistic T_n via the limit experiment framework that is tailored to my context. Based on this result, in section 4.2 I derive two types of results on the asymptotic lower bound. In the process of deriving the results, I discuss how the key notions used in point identified case are naturally extended to the set-identified context. In section 4.3 I extend these results to the general moment function on m.

From now on, the Borel σ-algebra $\mathcal{B}_{\mathcal{K}_\Theta}$ generated by the Hausdorff metric d_H is defined on $\mathcal{K}_\Theta$, where recall $\mathcal{K}_\Theta := \mathcal{K} \cap [-M_\theta, M_\theta]^{d_\theta}$ with $M_\theta > 0$ and $\mathcal{K}$ is the family of all non-empty compact sets in $\mathbb{R}^{d_\theta}$. Similarly $\mathcal{B}_\mathcal{K}$ denotes the Borel σ-field on $\mathcal{K}$. Since the space $(\mathcal{K}, d_H)$ is complete and separable, so is $(\mathcal{K}_\Theta, d_H)$. The measurability of "set-valued" random variables $T_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$ are hence accordingly defined with respect to the measurable space $(\mathcal{K}_\Theta, \mathcal{B}_{\mathcal{K}_\Theta})$[10].

4.1. Limit experiment for identified set.

Consider a parametric likelihood function $p_\theta = dP_\theta/d\mu$ on the sample space $(\mathcal{X}, \mathcal{B}_\mathcal{X})$ with a σ-finite measure μ. The model is indexed by a finite dimensional parameter $\theta \in \Theta^o \subset \mathbb{R}^{d_\theta}$. For simplicity, I consider the situation where potentially multiple parameters induce observational equivalence, i.e., the equivalence $p_{\theta_1}(X_i) = p_{\theta_2}(X_i)$ holds μ-a.e. even when $\theta_1 \neq \theta_2$. Following the discussion in Chen et al. (2018), given a true density $p_0 = dP_0/d\mu$, there exist potentially multiple parameters $\{\theta_l\}_{l\in L}$ that satisfy $\mathrm{KL}(p_0|p_{\theta_l}) := \int_\mathcal{X} \log(p_0/p_{\theta_l}) \cdot p_0\ d\mu = 0$ for any

[10]The topology on the family of non-empty compact sets induced by the Hausdorff metric is equivalent to *Myopic topology* if the underlying space is a separable metric space; see Appendix A.2.

$l \in L$, where L is an arbitrary set. The likelihood p_θ is then maximized by these parameters, a set of which corresponds to $\Theta_I = \{\theta_l\}_{l\in L} \subset \mathbb{R}^{d_\theta}$. I want to estimate, as well as possible, the set Θ_I of parameters that are consistent with the model $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ and the available data $\{X_i\}_{i\in[n]}$.

To set up the main idea, suppose that the likelihood p_θ satisfy the "*differentiable in quadratic mean* (DQM)" condition—recall that a set of probability densities (or "experiment") $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ is said to satisfy the DQM at $\theta_0 \in \Theta^o$ if there exists a measurable function $\dot{\ell}_{\theta_0}$ such that

$$\int_{\mathcal{X}} \left(\sqrt{p_{\theta_0+h}} - \sqrt{p_{\theta_0}} - \frac{1}{2}\dot{\ell}_{\theta_0} h \sqrt{p_{\theta_0}} \right)^2 d\mu = o(\|h\|^2), \tag{7}$$

as $h \to 0$ in $\mathbb{R}^{d_\theta}$. The measurable function $\dot{\ell}_\theta$ is the score at θ; it can be shown that $P_\theta \dot{\ell}_\theta = 0$ and that there exists the Fisher information $I_\theta = P_\theta \dot{\ell}_\theta^t \dot{\ell}_\theta$ (e.g. van der Vaart (1998)). Two features can be observed in the point identified case. First, the score at an identified point θ_0 is uniquely determined as $\dot{\ell}_{\theta_0}$ regardless of the direction of h, which plays key roles to derive (asymptotic) efficiency bound. Second, the non-singularity of the Fisher information $I_\theta > 0$ is typically assumed; this is a type of regularity condition which holds for standard models. As shown below, in general these features no longer hold in set identified cases.

When the object of interest is the set Θ_I that yields observational equivalence, it is natural to consider the Minkowski sum $\Theta_I \oplus h = \{\theta + h : \theta \in \Theta_I\}$ as the local deviation of Θ_I via h[11]. Accordingly, this is where the difference from the point identified scenario arises and some care is required. The information of local asymptotics, i.e., $\lim_{h\to 0} p_{\theta+h}/p_\theta$, is the key quantity, but this quantity in my setting depends on the direction of $h \in \mathbb{R}^{d_\theta}$. More precisely, suppose that the model $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ satisfies the DQM in (7) at any $\theta \in \Theta_I$. On one hand, the score yields the degenerate score $\dot{\ell}_\theta$, which necessarily leads to the degenerate Fisher information $I_\theta = 0$ for any $\theta \in \Theta_I^o$ and for any $h \in \mathbb{R}^{d_\theta}$[12]. On the other hand, given $h \in \mathbb{R}^{d_\theta}$, one would have a non-zero score $\dot{\ell}_\theta$ for $\theta \in \partial\Theta_I$ if $\theta + h \notin \Theta_I^o$ whereas $\dot{\ell}_\theta = 0$ otherwise. These are intuitive given that the score $\dot{\ell}_\theta$ in the widely used assumtpion of DQM in (7) is the Fréchet derivative of the map $\phi : \Theta \to L^2(\mu)$ with $\phi(\theta) = \sqrt{p_\theta}$. This observation implies that each direction expressed by h could provide different information, i.e. the score $\dot{\ell}_{\theta,h}$ and the Fisher information $I_{\theta,h} = P_\theta \dot{\ell}_{\theta,h} \dot{\ell}_{\theta,h}^t$ depend on the location of $\theta \in \Theta_I$ and direction h, which is a sharp contrast to the point identified case with Fréchet differentiability of the model $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ where any local deviation leads to a single I_θ.

[11]Formally, it is required that $\Theta_I \oplus h \subset \Theta^o \subset \mathbb{R}^{d_\theta}$ and hence the norm of h needs to be suitably adjusted for that purpose. To avoid notational complexity, I hereafter assume that the requirement is satisfied.

[12]This can be seen due to the equivalence $p_{\theta+h} = p_\theta$, μ-a.e. for $h \in \mathbb{R}^{d_\theta}$ whose norm is small enough. Specifically, the score $\dot{\ell}_\theta$ is defined as the Fréchet derivative of the map $\theta \mapsto \sqrt{p_\theta}$ in $L^2(\mu)$ but the observational equivalence yields $\dot{\ell}_\theta = 0 \in L^2(\mu)$

I now formalize and generalize the intuition built above to include a situation where observational equivalence is characterized not only equality but also inequality constraint. Given that the model has different scores depending on the location of parameters and direction, it is not appropriate to characterize scores via its Fréchet derivative because it requires uniqueness regardless of how its limit is taken (i.e. invariant of direction). I instead assume the following condition.

Assumption 4.1 (Directional scores)**.** A model $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ is said to have the directional score at $\theta \in \Theta^o$ for $h \in \mathbb{R}^{d_\theta}$, denoted by $\dot{\ell}_{\theta,h}$, if there exists a non-empty set $H_\theta \subset \mathbb{R}^{d_\theta}$ such that $\dot{\ell}_{\theta,h}h \neq 0$ for any $h \in H_\theta$ and

$$\int_{\mathcal{X}} \left(\frac{\sqrt{p_{\theta+t_nh_n}} - \sqrt{p_\theta}}{t_n} - \frac{1}{2}\dot{\ell}_{\theta,h}h\sqrt{p_\theta} \right)^2 d\mu = o(1), \tag{8}$$

for any $\{h_n\}_{n\in\mathbb{N}} \subset H_\theta$ and any $\{t_n\}_{n\in\mathbb{N}} \subset \mathbb{R}_+$ with $h_n \to h \in H_\theta$ and $t_n \to 0$ as $n \to \infty$.

The crucial difference from (7) is that the score $\dot{\ell}_{\theta,h}$ depends on h. Accordingly, for any $\theta \in \Theta_I$ the Fisher information $I_{\theta,h} := P_\theta \dot{\ell}^t_{\theta,h}\dot{\ell}_{\theta,h}$ $(= P_0\dot{\ell}^t_{\theta,h}\dot{\ell}_{\theta,h}$ if the identified set is defined via equality $p_\theta = p_0$ μ-a.e.) is formally defined to be direction dependent. In this sense, both $\dot{\ell}_{\theta,h}$ and $I_{\theta,h}$ can be viewed as maps on $\Theta^o \times \mathbb{R}^{d_\theta}$, capturing interaction of location of θ and direction h. Note that this definition implies $\dot{\ell}_{\theta,h}h$ being continuous and homogeneous of degree one in $h \in H_\theta$. In short, Assumption 4.1 states that the map $\theta \mapsto \sqrt{p_\theta}$ is directionally differentiable at $\theta \in \Theta^o$ tangentially to H_θ in the sense of Hadamard; see Shapiro (1990)[13]. To sharpen the characteristic of H_θ, I focus the geometric structure of Θ_I with the following quantities for $\theta \in \partial\Theta_I$;

$$\begin{aligned} T(\theta) &:= \{h \in \mathbb{R}^{d_\theta} : \theta + t_nh_n \in \Theta_I \text{ for some } t_n \searrow 0 \text{ and } h_n \to h \}, \\ N(\theta) &:= \{v \in \mathbb{R}^{d_\theta} : v^th \leq 0 \text{ for all } h \in T(\theta)\}. \end{aligned} \tag{9}$$

$T(\theta)$ is the tangent cone of Θ_I at θ and $N(\theta)$ is the normal cone (Rockafellar and Wets, 1998, Section 6). Using these notions of directional scores and geometric characteristics, I show the following result.

Lemma 4.2 (Limiting Gaussian field)**.** *Suppose that the model $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ satisfies Assumption 3.3 and Assumption 4.1 at all $\theta \in \partial\Theta_I$. Assume that (i) $\mathbb{R}^{d_\theta} \setminus T(\theta) \subset H_\theta$ for any $\theta \in \partial\Theta_I$ and (ii) for any $h \in \mathbb{R}^{d_\theta}$ there exists a partition $\partial\Theta_I = \cup_{k=1}^K S_k$ such that $\theta \mapsto \dot{\ell}_{\theta,h}$ is continuous with respect to the operator norm on each S_k. Then the following results hold:*

(a). *there exists a non-empty compact set $\partial\Theta_{I,h} \subset \partial\Theta_I$ such that*
  (i). *uniformly over $\theta \in \partial\Theta_{I,h}$ the local Fisher information $I_{\theta,h}$ is continuous and $I_{\theta,h} \neq 0$ (i.e. non-degenerate),*
  (ii). *there exist a set $H \subset \mathbb{R}^{d_\theta}$ satisfying $\partial\Theta_I = \cup_{h\in H}\partial\Theta_{I,h}$,*

[13]The Hadamard directional differentiability is often employed to analyze inference of parameters which can be characterized via directionally differentiable maps and hence a standard Delta method based on full differentiability cannot be applied (e.g. Fang and Santos (2019)).

(b). *for any* $h \in \mathbb{R}^{d_\theta}$*, it follows that*

$$\log \prod_{i=1}^{n} \left(\frac{p_{\theta+h/\sqrt{n}}}{p_\theta} \right) (X_i) \rightsquigarrow \mathcal{G}_{\alpha_h, \Sigma_h} \quad \text{as } n \to \infty, \tag{10}$$

in the space of continuous functions $(\mathcal{C}(\partial\Theta_{I,h}), \|\cdot\|_\infty)$*, where* $\mathcal{G}_{\alpha_h,\Sigma_h}$ *is the Gaussian field on* $\partial\Theta_{I,h}$ *with the mean* $\alpha_h : \partial\Theta_{I,h} \to \mathbb{R}$ *and the covariance kernel* $\Sigma_h : \partial\Theta_{I,h} \times \partial\Theta_{I,h} \mapsto \mathbb{R}_+$ *such that*

$$\alpha_h(\theta) = -\frac{1}{2} h^t I_{\theta,h} h, \quad \Sigma_h(\theta, \theta') = h^t (P_0 \dot{\ell}_{\theta,h} \dot{\ell}^t_{\theta',h}) h.$$

Proof. See Appendix B.2. □

The intuition behind Lemma 4.2 is as follows. For models satisfying Assumption 4.1, there always exist deviation (or direction) h and some part of identified sets where the local Fisher information $I_{\theta,h}$ are all non-degenerate; on this part the limit of the likelihood ratio weakly converges to a Gaussian field (conditional on h) whose mean and covariance are non-zero due to the non-degenerate local Fisher information $I_{\theta,h}$. Furthermore, the collection of these subsets can cover the original identified set entirely if the identified set belongs to the class specified in Assumption 3.3. In this sense, Lemma 4.2 is suggestive in that the information derived from the local deviation of the set Θ_I by h is governed by the local deviation of the boundary $\partial\Theta_{I,h}$ by h. I exploit this structure to get the efficiency bound of set estimators $T_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$. Note also that the weak convergence in (10) holds uniformly over $\partial\Theta_{I,h}$.

Remark 4.3. The viewpoint of Lemma 4.2 has potentially wider applications. More specifically, even when the identified set Θ_I does not belong to the class in Assumption 3.3, it could be possible to find a collection of compact sets $\{\partial\Theta_{I,h}\}_{h\in H}$ such that $\nu(\partial\Theta_I \setminus \cup_{h\in H}\partial\Theta_{I,h}) < \epsilon$ for an arbitrary $\epsilon > 0$, where ν is some finite measure. In this case, rather than dealing with the entire set $\partial\Theta_I$ under the efficiency argument as in this paper, it may be enough to focus on a particular subset $\partial\Theta_{I,h}$ depending on specific applications. Regarding this point, Lemma 4.2 opens the possibility of estimating functionals which depend on a part of identified sets as in Fang and Santos (2019).

The main result in this section builds on Lemma 4.2. Consider a model $\{p^n_\theta := \prod_{i=1}^n p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ which yields the set of observationally equivalent parameters Θ_I. I focus on estimators $T_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$ in the locally deviated model $\{p^n_{\partial\Theta_{I,h}\oplus h/\sqrt{n}} : h \in \mathbb{R}^{d_\theta}\}$ defined by

$$p^n_{\partial\Theta_{I,h}\oplus h/\sqrt{n}} = \int_{\partial\Theta_{I,h}} p^n_{\theta+h/\sqrt{n}} \, d\lambda_h(\theta), \tag{11}$$

where λ_h is a probability measure which is absolutely continuous with respect to either the counting measure ($\sharp$) or the Lebesgue measure (Leb) depending on the nature of the

set $\partial\Theta_{I,h}$[14]. Formally, letting Λ be the family of all Borel probability measures on $\partial\Theta_I$, I take $\lambda_h \in \Lambda_h \subset \Lambda$, where Λ_h is the collection of all Borel probability measures on $\partial\Theta_{I,h}$. The quality of T_n is measured by the (scaled) Hausdorff metric $\sqrt{n} \cdot d_H(T_n, \partial\Theta_{I,h} \oplus h/\sqrt{n})$. Note here that the object of interest can be viewed as an $\mathbb{R}_+$-valued random variable $\sqrt{n} \cdot (d_H \circ T_n) : \mathcal{X}^n \mapsto \mathbb{R}_+$. As shown in Lemma 4.2, the model in local neighborhood of the observationally equivalent parameters Θ_I yields the Gaussian field on $\partial\Theta_{I,h}$ as limit. I show that the law of $\sqrt{n} \cdot (d_H \circ T_n)$ can be asymptotically equivalent to that of $d_H \circ T$, where $T : \mathcal{X} \mapsto \mathcal{K}_\Theta$ is a set-valued statistic in the Gaussian mixture indexed by $\theta \in \partial\Theta_{I,h}$. This is formally stated in the following theorem.

Theorem 4.4 (Asymptotic representation for set estimators)**.** *Suppose that the following conditions hold:*

(i). *the same conditions of Lemma 4.2 hold,*

(ii). *a set-valued statistic $T_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$ defined in $\{p^n_{\partial\Theta_{I,h} \oplus h/\sqrt{n}} : h \in \mathbb{R}^{d_\theta}\}$ satisfies*

$$\mathcal{L}\left(\sqrt{n} \cdot d_H(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n})\right) \rightsquigarrow S_h$$

for a probability measure S_h on $(\mathbb{R}_+, \mathcal{B}_{\mathbb{R}_+})$, where $T_n^h := T_n|_{\mathcal{N}(\partial\Theta_{I,h})}$ is the restriction to the d_H-local neighborhood of $\partial\Theta_{I,h}$, i.e., for any $n \in \mathbb{N}$,

$$\begin{aligned}\mathcal{N}(\partial\Theta_{I,h}) &:= \mathcal{K} \cap \left[\cup_{r \in \mathbb{R}_+} (\partial\Theta_{I,h})^{d_H(r)/\sqrt{n}}\right] \\ &= \bigcup\nolimits_{r \in \mathbb{R}_+} \left\{K \in \mathcal{K} : d_H(K, \partial\Theta_{I,h}) \leq r/\sqrt{n}\right\}.\end{aligned}$$

Then there exist an independent random variable U taking values in $(\mathcal{U}, \mathcal{B}_\mathcal{U}, \mathcal{L}(U))$ and a randomized set-valued statistic $T^h : \mathcal{X} \times \mathcal{U} \mapsto \mathcal{K}_\Theta$ defined in $(\mathcal{X} \times \mathcal{U}, \mathcal{B}_\mathcal{X} \otimes \mathcal{B}_\mathcal{U}, G_h \otimes \mathcal{L}(U))$ such that

$$\mathcal{L}\left(\sqrt{n} \cdot d_H(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n})\right) \rightsquigarrow S_h = \mathcal{L}\left(d_H(T^h, \partial\Theta_{I,h} \oplus h)\right) \quad \text{as } n \to \infty, \tag{12}$$

where, with $I^\dagger_{\theta,h}$ representing the Moore-Penrose inverse of $I_{\theta,h}$, it holds that

$$G_h = \int_{\partial\Theta_{I,h}} N(h, I^\dagger_{\theta,h})\, d\lambda_h(\theta) \quad \text{for any } h \in \mathbb{R}^{d_\theta}. \tag{13}$$

Proof. See Appendix B.2. □

The implication of Theorem 4.4 is similar to the point identified case. For any set-valued estimator T_n^h defined in a locally deviated experiment $\{p^n_{\partial\Theta_{I,h}+h/\sqrt{n}} : h \in \mathbb{R}^{d_\theta}\}$, the law of the (scaled) Hausdorff metric $\sqrt{n} \cdot d_H(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n})$ is expressed by that of a set-valued statistic T^h defined in a Gaussian mixture experiment, where the mixing elements lie on the boundary of the identified set $\partial\Theta_{I,h}$. Thus, the asymptotic risk of any estimator T_n^h is bounded below by that of the estimator T^h in the Gaussian mixture experiment, which serves as the lower bound of the limit experiment. Based on this result, I establish the lower bound of the risk consisting of $d_H(T^h, \partial\Theta_{I,h} \oplus h)$ in section 4.2.

[14]The device of mixing measure that is similar to λ_h here has been used, for example, in the context of testing (e.g. the "weighted average" testing in Andrews and Ploberger (1994))

I end here by commenting on the philosophy of my analysis in section 4.1. My emphasis is the observational equivalence deduced from the model $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ and the data $\{X_i\}_{i\in[n]}$, both of which are available to the researcher. Accordingly, my result on the limit experiment and the resulting efficiency results are all with respect to the model and data available at hand. This point clearly appears in using the relation $\mathrm{KL}(p_0|p_\theta) = 0$ for any $\theta \in \Theta_I$ as a starting point of my analysis. An alternative perspective would be the following; assuming the uniqueness of the true p_0, which cannot necessarily be point identified via conditions on $\{p_\theta : \theta \in \Theta^o \subset \mathbb{R}^{d_\theta}\}$ and $\{X_i\}_{i\in[n]}$, I could still proceed with my analysis while emphasizing $p_0 \neq p_\theta$ for $\theta \in \Theta_I$. This case can be formally handled in section 4.3 by treating p_θ with $\theta \in \Theta_I$ as pseudo-likelihood.

4.2. **Asymptotic lower bound.** I here derive the lower bound of the asymptotic risk arising from $\sqrt{n} \cdot d_H(T_n, \partial\Theta_I \oplus h/\sqrt{n})$, where, as seen in section 4.1 it is reasonable to focus on the boundary of the identified set. The classical results for the identified point $\Theta_I = \{\theta_0\}$ have a long history (e.g. Hájek (1970), Hájek (1972)). Similar to this literature, the driving force here is the appropriate limit experiment obtained in Theorem 4.4. An intuition from the point identified case suggests that, for a part $\partial\Theta_{I,h} \subset \partial\Theta_I$ which corresponds to the given direction $h \in \mathbb{R}^{d_\theta}$, the limit experiment

$$\mathcal{L}\left(\sqrt{n} \cdot d_H\left(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n}\right)\right) \overset{a}{\sim} \mathcal{L}(d_H(T^h(X,U), \partial\Theta_{I,h} \oplus h)),$$
$$X \sim G_h = \int_{\partial\Theta_{I,h}} N(h, I^\dagger_{\theta,h})\, d\lambda_h(\theta)$$

provides the lower bound of the corresponding asymptotic risk. The departure from the point identified case is that the limit experiment in Theorem 4.4 contains information of $\partial\Theta_{I,h}$ as Gaussian field defined on it. This intuition plays a key role when deriving the asymptotic lower bound of the risk.

I proceed with my analysis from the local minimax perspective, i.e., the central question here is "how accurately can one estimate the set $\partial\Theta_I$ when the set is locally deviated to $\partial\Theta_I \oplus h/\sqrt{n}$?" and I focus on the worst case of this accuracy. The minimax risk of the entire region $\partial\Theta_I$ is necessarily bounded below from the risk of $\partial\Theta_{I,h}$ for any $h \in \mathbb{R}^{d_\theta}$, each of which can be represented via the appropriate Gaussian mixture limit experiment. This intuition is formalized in Theorem 4.5.

Theorem 4.5 (Local asymptotic minimax bound for Θ_I)**.** *Suppose that the same conditions in Theorem 4.4 hold. Then for any such set-valued estimators $T_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$ in $\{p^n_{\partial\Theta_{I,h}\oplus h/\sqrt{n}} : h \in \mathbb{R}^{d_\theta}\}$ and for any subconvex loss function $l : \mathbb{R} \mapsto \mathbb{R}$*[15]*, it holds that*

$$\sup_{D\in\mathcal{D}} \liminf_{n\to\infty} \sup_{h\in D} \mathbb{E}_{P^n_h}\left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I \oplus h/\sqrt{n})\right)\right]$$
$$\geq \sup_{h\in\mathbb{R}^{d_\theta}} \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l \circ \|x\|\, dN(0, I^\dagger_{\theta,h})(x)\, d\lambda_h(\theta),$$

[15]Generally, a function $l : \mathbb{R}^s \mapsto \mathbb{R}$ is called *subconvex* if its sublevel sets $\{x \in \mathbb{R}^s : l(x) \leq c\}$ are closed, convex and symmetric about the origin for any $c \in \mathbb{R}$.

where $dP_h = p_h d\mu$ *with* $p_h = p_{\partial\Theta_{I,h} \oplus h/\sqrt{n}}$ *is as in (11) and* $\mathcal{D} := \{D \subset \mathbb{R}^{d_\theta} : 0 < |D| < \infty\}$ *is a directed set with the partial order* $\subset$.

Proof. See Appendix B.3. □

Corollary 4.6 (Local asymptotic minimax bound for Θ_I)**.** *Under the same conditions as in Theorem 4.5, it holds that*

$$\sup_{D \in \mathcal{D}} \liminf_{n \to \infty} \sup_{h \in D} \sup_{\lambda \in \Lambda} \mathbb{E}_{P_h^n} \left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I \oplus h/\sqrt{n}) \right) \right]$$
$$\geq \sup_{h \in \mathbb{R}^{d_\theta}} \sup_{\theta \in \partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l \circ \|x\| \, dN(0, I^\dagger_{\theta,h})(x).$$

Proof. See Appendix B.3. □

Theorem 4.5 shows that the supremum of Gaussian mixture experiments in (13) provides a generic lower bound of the asymptotic risk of $\sqrt{n} \cdot d_H(T_n, \partial\Theta_I \oplus h/\sqrt{n})$ in a minimax sense. Roughly speaking, the lower bound is governed by the local Fisher information $I_{\theta,h}$; the smaller local information $I_{\theta,h}$ make it harder to estimate the corresponding region $\partial\Theta_{I,h}$ and the lower bound is obtained by looking at the most difficult direction h.

Remark 4.7. Theorem 4.5 nests the classical result for the point identified case. In particular, when $\Theta_I = \{\theta_0\} = \partial\Theta_I$ the locally deviated model in (11) reduces to $\{p^n_{\theta_0 + h/\sqrt{n}} : h \in \mathbb{R}^{d_\theta}\}$ with the counting measure $\lambda_h = \sharp$. Restricting the range of estimators T_n to $\mathcal{K}_\Theta = \mathcal{K}_1' := \{K \in \mathcal{K}_1 \text{ in } \mathbb{R}^{d_\theta} : |K| = 1\}$, with the loss function taking the form of $l \circ \|.\|$, corresponds to the local minimax result for the point estimation of θ_0 (e.g. Hájek (1972)).

Remark 4.8. Note that my lower bound is based on the local asymptotic minimax criteria. In point identified cases, there is another argument to derive the lower bound via convolution (e.g. Hájek (1970)). Compared with the minimax criteria, this approach necessarily restrict the scope of estimators to those satisfying "local regularity" conditions. As pointed out in Hirano and Porter (2012), however, this requirement is not directly applicable to the set identified context where the local parameter (direction) matters in asymptotic distributions.

4.3. **General moment conditions.** I here consider the semiparametric situation, where no likelihood function fully describes the model but moment restrictions are available, based on which the finite-dimensional parameters of interest are characterized. To derive the asymptotic efficiency bound in this case, I here extend the previous analysis of a parametric regime in section 4.2 to a semiparametric one. My approach is an extension of Chamberlain (1987), where semiparametric efficiency bound of GMM estimator is derived for a point identified parameter.

A principle idea is as follows. Given a moment restriction $\Phi = q \circ (\mathbb{E}_P m)$ and corresponding identified set Θ_I defined as in (4), I set up an "auxiliary parametric model" P_γ, where $(\theta^t, \xi^t)^t = \gamma \in \Gamma \subset \mathbb{R}^{d_\gamma}$ and ξ serving as nuisance parameter. I first derive the lower bound of set estimators for this class of parametric models $P_\gamma \in \mathcal{P}_\gamma$ by suitably applying the result in section 4.2. Care is needed in constructing this parametric model P_γ so that the lower bound obtained in section 4.2 can be applied. I then extend this result to a wider class of models $\mathcal{P} \supset \mathcal{P}_\gamma$ via expanding the parameter space $\Xi := \Gamma \setminus \Theta$ so that $P_\gamma \in \mathcal{P}_\gamma$ approximates each $P \in \mathcal{P}$ in a suitable topology while keeping the efficiency bound maintained. Each step requires careful treatment of identified set Θ_I unlike the point identified case.

I first state the requirements for the pair $(\Phi, \mathcal{P})$ of the class $\mathcal{P}$ to which the true model P_0 is assumed to belong and of a moment restriction $\Phi = q \circ (\mathbb{E}_P m)$ based on which Θ_I is defined. In this section I explicitly write $\Theta_I(\Phi, P)$ to recall the dependence of Θ_I on Φ and P. The regularity conditions on $(\Phi, \mathcal{P})$ are formally summarized in Assumption 4.9 as follows.

Assumption 4.9 (Regularity conditions on $(\Phi, \mathcal{P})$)**.** For any $P \in \mathcal{P}$, the pair (Φ, P) satisfies the following conditions;

(i). $\emptyset \neq \Theta_I(\Phi, P) \subset \Theta^o$,

(ii). for each $\theta \in \Theta_I(\Phi, P)$, there exist a map $\phi_{\theta,h} : \mathbb{R}^{d_\theta} \mapsto \mathbb{R}^{d_q}$ and a set $H_\theta \subset \mathbb{R}^{d_\theta}$ such that $\mathbb{R}^{d_\theta} \setminus T(\theta) \subset H_\theta$, $\phi_{\theta,h} h \neq 0$ for $h \in H_\theta$ and

$$\lim_{n\to\infty} \left\| \frac{\Phi(\theta + t_n h_n, P) - \Phi(\theta, P)}{t_n} - \phi_{\theta,h} h \right\| = 0,$$

where $\{h_n\}_{n\in\mathbb{N}} \subset H_\theta$ and $\{t_n\}_{n\in\mathbb{N}} \subset \mathbb{R}_+$ with $h_n \to h \in H_\theta$ and $t_n \to 0$ as $n \to \infty$,

(iii). for any $h \in \mathbb{R}^{d_\theta}$ there exists a partition $\partial\Theta_I(\Phi, P) = \cup_{k=1}^K S_k$ such that $\theta \mapsto \phi_{\theta,h}$ is continuous with respect to the operator norm on each S_k,

(iv). for any $\theta \in \partial\Theta_I(\Phi, P)$, it holds that $\mathbb{E}_P[(\partial/\partial P)\Phi(\theta, P)] = 0$ and $\mathbb{E}_P[(\partial/\partial P)\Phi(\theta, P)]^{\otimes 2} > 0$, where $(\partial/\partial P)\Phi(\theta, P) : \mathcal{P} \mapsto \mathbb{R}^{d_q}$ is a map with a non-empty connected set $H_{P,\theta} \subset \mathcal{P}$ satisfying $P \in H_{P,\theta}$ and

$$\lim_{n\to\infty} \left\| \frac{\Phi(\theta, P_{\eta_n}) - \Phi(\theta, P)}{\eta_n} - \left[\frac{\partial}{\partial P}\Phi(\theta, P)\right](P') \right\| = 0,$$

for any path $\{P_{\eta_n}\}_{n\in\mathbb{N}} \subset \mathcal{P}$ and $\{\eta_n\}_{n\in\mathbb{N}} \subset [0, 1]$ such that, as $n \to \infty$, $\eta_n \to 0$ and $P_{\eta_n} = (1 - \eta_n)P + \eta_n P'_n \to P$ with $P'_n \to P' \in H_{P,\theta}$.

Assumption 4.9 can be interpreted as a set version of regularity conditions. It is clear from Assumption 4.9 (ii) that the map $\phi_{\theta,h}$ is continuous and positively homogeneous of degree one in $h \in H_\theta$; see also Shapiro (1990). The intuition built in the parametric case in secion 4.1 continues to hold. For example, it is easy to see $\phi_{\theta,h} = 0$ for any $\theta \in \Theta_I^o(P, \Phi)$ and any $h \in \mathbb{R}^{d_\theta}$; this is due to the "observational equivalence" $\Phi(\theta_i, P) = \Phi(\theta_j, P) = 0$ for $\theta_i, \theta_j \in \Theta_I(\Phi, P)$ with $\theta_i \neq \theta_j$, whose equivalence in this case comes from the moment restriction. Assumption 4.9 (iv) serves a similar role as in Assumption 4.9 (ii) but with respect to $P \in \mathcal{P}$.

I now set up the approximation scheme $\mathcal{P}_\gamma \subset \mathcal{P}$ via multinomial distributions. Suppose that $\{X_i\}_{i\in[n]}$ is a random sample with $X_i \sim P_{\delta_0}$ on $(\mathcal{X}, \mathcal{B}_\mathcal{X})$, where P_{δ_0} is a multinomial distribution; that is, $\mathrm{supp}(P_{\delta_0}) = \{x_s\}_{s\in[d_s]} \subset \mathcal{X}$ with $d_s < \infty$. In this case, the parameter of P_{δ_0} is $\delta_0 := (\delta_0(1), \cdots, \delta_0(d_s)) \in \Delta_{d_s-1} \subset \mathbb{R}^{d_s}$ with $\sum_{s\in[d_s]} \delta_0(s) = 1$, where $\delta_0(s) := P_{\delta_0}(x_s)$ for any $s \in [d_s]$ and Δ_{d_s-1} is the unit simplex of dimension d_s. To avoid triviality, assume that $\delta_0 \in \Delta^o_{d_s-1}$.

The above set up yields a parametric model $X \sim P_{\delta_0}$ with parameters $\delta_0 \in \Delta^o_{d_s-1} \subset \mathbb{R}^{d_s}$ and the identified set of interest $\Theta_I(\Phi, P_{\delta_0})$ defined via moment restrictions (4), i.e.,

$$\Theta_I(\Phi, P_{\delta_0}) = \left\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : \Phi(\theta, P_{\delta_0}) = q\left(\sum\nolimits_{s\in[d_s]} m(\theta, x_s)\cdot\delta_0(s)\right) \le 0\right\}.$$

From this parametric model P_{δ_0} I derive the local Fisher information $I_{\theta,h}$ for $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$ that depends on direction $h \in \mathbb{R}^{d_\theta}$. The following Lemma 4.10 makes this procedure possible by establishing the model $P_\gamma \in \mathcal{P}_\gamma$ with a parameter $\gamma = (\theta^t, \xi^t)^t$ which preserves the information in P_{δ_0}. Here the parameters of interest are represented as the first d_θ-projection of γ, i.e. $\theta = \pi_{d_\theta} \circ \gamma$.

Lemma 4.10 (Auxiliary parametric model P_γ)**.** *Suppose that Assumption 3.3 and 4.9 hold with $X \sim P = P_{\delta_0}$ with $d_s \gg d_m$. Then there exist an open set $\Gamma^o \subset \mathbb{R}^{d_\gamma}$ with $d_\gamma = d_s - (d_q - d_\theta) - 1$ and a family of models $\{P_\gamma : \gamma \in \Gamma^o \subset \mathbb{R}^{d_\gamma}\}$ such that the following properties hold:*

(a). *$supp(P_\gamma) = \{x_s\}_{s\in[d_s]}$ for any $\gamma \in \Gamma^o \subset \mathbb{R}^{d_\gamma}$,*

(b). *there exists $\Gamma_I(\Phi, P_{\delta_0}) \subset \Gamma^o$ such that (i) $\Theta_I(\Phi, P_{\delta_0}) = \pi_{d_\theta} \circ \Gamma_I(\Phi, P_{\delta_0})$ and (ii) for any $\gamma_i, \gamma_j \in \Gamma_I(\Phi, P_{\delta_0})$ with $\gamma_i^t := (\theta_i^t, \xi_i^t)$ and $\gamma_j^t := (\theta_j^t, \xi_j^t)$, it holds that $P_{\gamma_i} = P_{\gamma_j} = P_{\delta_0}$,*

(c). *for any $h_\gamma^t = (h^t, h_\xi^t) \in \mathbb{R}^{d_\gamma}$ there exists $\partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0}) \subset \partial\Gamma_I(\Phi, P_{\delta_0})$ such that $\partial\Theta_{I,h}(\Phi, P_{\delta_0}) = \pi_{d_\theta} \circ \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ and that there exists a non-degenerate $d_\gamma \times d_\gamma$ matrix $I_{\gamma,h_\gamma} \neq 0$ for any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$. Its first $d_\theta \times d_\theta$-matrix represents*

$$I_{\theta,h} = \phi_{\theta,h}^t \cdot \left[\mathbb{E}_{\delta_0}[(\partial/\partial\delta)\Phi(\theta, P_\delta)|_{\delta=\delta_0}]^{\otimes 2}\right]^{-1} \cdot \phi_{\theta,h}, \tag{14}$$

where $\mathbb{E}_{\delta_0} = \mathbb{E}_{P_{\delta_0}}$.

Proof. See Appendix B.4. □

Lemma 4.10 shows that, from the pair (Φ, P_{δ_0}) of the moment restrictions Φ and the parametric model P_{δ_0}, one can construct the corresponding model P_γ such that directional local Fisher information $I_{\theta,h}$ on $\partial\Theta_{I,h}(\Phi, P_{\delta_0})$ is deducible. I now apply the result in section 4.2, Corollary 4.6 in particular, to show that this local Fisher information determines the lower bound of set estimators, as precisely stated in the following Proposition 4.11.

Proposition 4.11 (Local asymptotic minimax bound for Θ_I with $X \sim P_{\delta_0}$)**.** *Suppose that Assumption 3.3 and 4.9 hold with $X_i \overset{i.i.d.}{\sim} P_{\delta_0}$. Then for any estimators $\tilde{T}_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta \times \mathcal{K}_\Xi$ satisfying the condition (ii) in Theorem 4.4 and for any subconvex loss function*

$l : \mathbb{R} \mapsto \mathbb{R}$, *it holds that*

$$\sup_{D_\gamma \in \mathcal{D}_\gamma} \liminf_{n\to\infty} \sup_{h_\gamma \in D_\gamma} \sup_{\lambda_\gamma \in \Lambda_\gamma} \mathbb{E}_{P^n_{h_\gamma}} \left[l \circ \left(\sqrt{n} \cdot d_H(\tilde{T}_n, \partial\Gamma_I \oplus h_\gamma/\sqrt{n}) \right) \right] \geq \sup_{h_\gamma \in \mathbb{R}^{d_\gamma}} \sup_{\gamma \in \partial\Gamma_{I,h_\gamma}} \int_{\mathbb{R}^{d_\gamma}} l \circ \|x\| \, dN(0, I^\dagger_{\gamma,h_\gamma})(x),$$

where Λ_γ *is the family of all Borel probability measures on* $\partial\Gamma_I$, $dP_{h_\gamma} = p_{h_\gamma} d\sharp$ *with* $p_{h_\gamma} = p_{\partial\Gamma_{I,h_\gamma \oplus h_\gamma/\sqrt{n}}}$ *is as in (11),* I_{γ,h_γ} *is as in Lemma 4.10 and* $\mathcal{D}_\gamma := \{D_\gamma \subset \mathbb{R}^{d_\gamma} : 0 < |D_\gamma| < \infty\}$ *is a directed set with the partial order* $\subset$. *In particular, for any estimator* $T_n = \pi_{d_\theta} \circ \tilde{T}_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$ *it holds that*

$$\sup_{D_\gamma \in \mathcal{D}_\gamma} \liminf_{n\to\infty} \sup_{h_\gamma \in D_\gamma} \sup_{\lambda \in \Lambda} \mathbb{E}_{P^n_{h_\gamma}} \left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I \oplus (\pi_{d_\theta} \circ h_\gamma)/\sqrt{n}) \right) \right] \geq \sup_{h \in \mathbb{R}^{d_\theta}} \sup_{\theta \in \partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l \circ \|x\| \, dN(0, I^\dagger_{\theta,h})(x),$$

where $I_{\theta,h}$ *is as in (14).*

Proof. See Appendix B.4. □

The difference between Proposition 4.11 and Corollary 4.6 in section 4.2 is the source of set identification; the former deals with the case where the identified set is defined via a moment restriction while via a likelihood function in the latter.

Proposition 4.11 is based on the parametric model $X \sim P_{\delta_0}$ with $\delta_0 \in \Delta^o_{d_s-1}$ ($d_s < \infty$). Now I extend this result to the nonparametric counterpart P while maintaining a moment restriction Φ. To this end, let $(\mathcal{P}, \tau)$ be the space of probability measure $\mathcal{P}$ on $\mathcal{X}$ equipped with the same topology τ as in Chamberlain (1987). Specifically, given $P_0 \in \mathcal{P}$ the topology τ is induced by the basic neighborhoods of P_0 defined as

$$\mathcal{N}_{P_0} := \left\{ P \in \mathcal{P} : \left| \int_{\mathcal{X}} f_j(x) \left(dP(x) - dP_0(x) \right) \right| < \epsilon_j, \ j = 1, \cdots, k \right\},$$

where $\epsilon_j > 0$, $k \in \mathbb{N}$ is some positive integer, and $f_j : \mathcal{X} \mapsto \mathbb{R}$ are measurable function such that $f \in L^1(P_0)$[16]. With these notions, my main result is framed as the minimax criteria as in section 4.2, where the worst risk is taken within this neighborhood $\mathcal{N}_{P_0}$ with Assumption 4.9 satisfied. The formal result is stated as follows.

Theorem 4.12 (Local asymptotic minimax bound for Θ_I: a general case)**.** *Suppose that Assumption 3.3 and 4.9 hold with* $\Theta_I(\Phi, P_0)$ *being the set of interest. Let* $\mathcal{P}_0 \subset \mathcal{N}_{P_0}(\epsilon)$ *a class such that* (Φ, P) *satisfies Assumption 4.9 for any* $P \in \mathcal{P}_0$. *Then for any estimators* $T_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$ *satisfying the condition (ii) in Theorem 4.4 and for any subconvex loss*

[16] This topology is different from the weak topology, in which the metric is defined via continuous and bounded map $f : \mathcal{X} \mapsto \mathbb{R}$.

function $l : \mathbb{R} \mapsto \mathbb{R}$*, it holds that*

$$\liminf_{n\to\infty} \sup_{P\in\mathcal{P}_0} \sup_{\lambda\in\Lambda} \mathbb{E}_P \left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I(\Phi, P)) \right) \right] \geq \sup_{h\in\mathbb{R}^{d_\theta}} \sup_{\theta\in\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l \circ \|x\| \, dN(0, I^{\dagger}_{\theta,h})(x),$$

where $I_{\theta,h}$ *is as in (14).*

Proof. See Appendix B.4. □

Theorem 4.12 provides generic efficiency bound of set estimators for identified set in terms of the Hausdorff metric. The result is a generalization of Corollary 4.6 and Proposition 4.11 in that the data is free from parametric assumptions. A crucial aspect is that this efficiency bound is valid regardless of the parameter being point or set identified.

Remark 4.13. Semiparametric efficiency of GMM estimator under point identification, treated in Chamberlain (1987), can be seen as a special case of my result. Specifically, let $q(t) = t$, $\Theta_I(\Phi, P_0) = \{\theta_0\}$ and $\Phi(\theta, P)$ fully differentiable. Then it follows that $\phi_{\theta_0,h} = \mathbb{E}_0 \nabla m(\theta_0, X)$ and $\mathbb{E}_0[(\partial/\partial P)\Phi(\theta_0, P)]^{\otimes 2} = \mathbb{E}_0 m(\theta_0, X)^{\otimes 2}$, where $\mathbb{E}_0 = \mathbb{E}_{P_0}$. This indicates

$$I_{\theta_0,h} = [\mathbb{E}_0 \nabla m(\theta_0, X)]^t \cdot \left[\mathbb{E}_0 m(\theta_0, X)^{\otimes 2}\right]^{-1} \cdot [\mathbb{E}_0 \nabla m(\theta_0, X)],$$

which is independent of $h \in \mathbb{R}^{d_\theta}$ due to the full differentiability. The quantity $I^{-1}_{\theta_0,h}$ is the classical efficiency bound for the point identified parameter θ_0, that is achieved by the GMM estimator with the optimal weighting matrix $\left[n^{-1} \sum_{i\in[n]} m(\cdot, X_i)^{\otimes 2}\right]^{-1}$.

5. Limiting Distribution of Set Estimator from Criterion Function

I here consider an estimator T_n which I expect to achieve the efficiency bound derived in section 4. As in section 4, I remain agnostic about the characteristic of identified sets except for Assumption 3.3 for the purpose of gaining adaptiveness. The analysis in section 4, in particular in section 4.3, indicates that T_n could be defined as the extremum estimator of a criterion function of GMM-type $Q : \Theta \mapsto \mathbb{R}_+$ where Θ_I is defined as the zero-level set of Q. In fact, this type of criterion function has been studied in the literature (e.g. Chernozhukov et al. (2007) and Romano and Shaikh (2010)). The main result on this section shows that the extremum-type set estimators from this criterion function are indeed efficient under certain conditions in the sense of the Hausdorff metric d_H.

In section 5.1, I first set up criterion function, define set estimator, and show its consistency and rate of convergence. Based on this result, in section 5.2 I derive its limiting distribution in terms of the Hausdorff metric d_H; I compare it to the bound in section 4.

5.1. **Setting and basic properties of set estimator $\widehat{\Theta}_I$.** In section 5.1 and 5.2, I split the type of moment equality and inequality and express $\Phi = (\Phi^t_{\mathcal{J}_1}, \Phi^t_{\mathcal{J}_2})^t$, where $\Phi_{\mathcal{J}_1}(\theta, P) = (\Phi_j(\theta, P))_{j\in\mathcal{J}_1}$ and $\Phi_{\mathcal{J}_2}(\theta, P) = (\Phi_j(\theta, P))_{j\in\mathcal{J}_2}$ represent the class of moment equality and inequality type restrictions respectively. By definition, $|\mathcal{J}_1| + |\mathcal{J}_2| = d_q$. I then express

$$\Theta_I = \Theta_I(\Phi, P) := \left\{\theta \in \Theta^o \subset \mathbb{R}^{d_\theta} : \Phi_{\mathcal{J}_1}(\theta, P) = 0 \in \mathbb{R}^{|\mathcal{J}_1|}, \Phi_{\mathcal{J}_2}(\theta, P) \leq 0 \in \mathbb{R}^{|\mathcal{J}_2|}\right\}.$$

This is just a re-expression of the same object Θ_I (letting $\Phi_j(\theta, P) = |\Phi_j(\theta, P)|$ for any $j \in \mathcal{J}_1$ leads to the original (4)); I work the above expression for convenience.

From the model and the identified set expressed above, define the criterion function $Q : \Theta \mapsto \mathbb{R}_+$ such that

$$Q(\theta) = \left\|\Phi^t_v(\theta, P) \cdot W^{1/2}\right\|, \quad \theta \in \Theta, \tag{15}$$

where $W \in \mathbb{R}^{d_q \times d_q}$ is some positive definite matrix and

$$\Phi^t_v(\theta, P) = \left(\Phi^t_{\mathcal{J}_1}(\theta, P),\ [\Phi^t_{\mathcal{J}_2}(\theta, P)]_+\right) = \left(\left(\Phi^t_j(\theta, P)\right)_{j\in\mathcal{J}_1},\ \left([\Phi^t_j(\theta, P)]_+\right)_{j\in\mathcal{J}_2}\right),$$

with $[\Phi_j(\theta, P)]_+ := (\Phi_j(\theta, P) \vee 0)$ for any $j \in \mathcal{J}_2$. Clearly, $\Theta_I = \{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : \Phi_{\mathcal{J}_1}(\theta, P) = 0, \Phi_{\mathcal{J}_2}(\theta, P) \leq 0\} = \{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : Q(\theta) \leq 0\}$.

The set estimator, denoted by $\widehat{\Theta}_I(\tau)$ with the threshold level $\tau \in \mathbb{R}_+$ which could be data-dependent, is a lower level set of the sample analogue of (15), i.e.,

$$\widehat{\Theta}_I(\tau) = \left\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : a_n \cdot Q_n(\theta) \leq \tau\right\}, \tag{16}$$

where $\{a_n\}_{n\in\mathbb{N}}$ is a normalizing sequence and

$$Q_n(\theta) = \left\|\Phi^t_v(\theta, \mathbb{P}_n) \cdot W_n^{1/2}\right\|, \quad \theta \in \Theta, \tag{17}$$

where W_n the sample weighting matrix and

$$\Phi^t_v(\theta, \mathbb{P}_n) = \left(\Phi^t_{\mathcal{J}_1}(\theta, \mathbb{P}_n),\ [\Phi^t_{\mathcal{J}_2}(\theta, \mathbb{P}_n)]_+\right) = \left(\left(\Phi^t_j(\theta, \mathbb{P}_n)\right)_{j\in\mathcal{J}_1},\ \left([\Phi^t_j(\theta, \mathbb{P}_n)]_+\right)_{j\in\mathcal{J}_2}\right),$$

with $\mathbb{P}_n$ an empirical measure on $(\mathcal{X}, \mathcal{B}_\mathcal{X})$. Regarding the choice of τ, a key quantity is

$$c_n := \inf_{\theta'\in\Theta} Q_n(\theta'). \tag{18}$$

The setting above is standard. The vector $\Phi_v(\theta, P)$ represents deviation from constraint, which treats $\Phi_{\mathcal{J}_1}(\theta, P)$ and $\Phi_{\mathcal{J}_2}(\theta, P)$ differently. The quantity c_n is important in terms of self-adjustment of the threshold level which determines the estimator $\widehat{\Theta}_I$; in moment equality with over identified point it holds that $c_n > 0$. Note that this type of classic criterion function potentially appears to kill the information of identified set in a sense that differentiability does not hold and Hadamard directional derivatives of $\theta \in \Theta_I$ vanish (with respect to some direction). Note that, even though this criterion function is not fully differentiable or some direction of Hadamard directional derivative vanishes, the key in deriving limiting distribution is differentiability of Φ; this point is discussed in detail in section 5.2.

The consistency and rate of convergence of $\widehat{\Theta}_I$ can be derived via the approach in Chernozhukov et al. (2007). The next assumptions are used to analyze the consistency, the convergence rate and the limiting distribution of $\widehat{\Theta}_I$.

Assumption 5.1. (i). The functions $Q : \Theta \mapsto \mathbb{R}_+$ and $Q_n : \Theta \times \mathcal{X} \mapsto \mathbb{R}_+$ are Borel measurable. (ii). The class of moment functions $\{m(\theta, \cdot)\}_{\theta \in \Theta}$ has a P-Donsker property; uniformly over $\theta \in \Theta$ it holds that

$$\mathbb{G}_n[m(\theta, \cdot)] := \sqrt{n} \left(\mathbb{P}_n m(\theta, \cdot) - P m(\theta, \cdot) \right) \rightsquigarrow \xi(\theta), \quad \text{as } n \to \infty, \tag{19}$$

where ξ is the Gaussian field on Θ such that $\mathcal{L}(\xi(\theta)) = N(0, \Sigma(\theta))$ with $\Sigma(\theta) = \mathbb{E}_P m(\theta, \cdot)^{\otimes 2}$. (iii). $W_n(\theta) = W(\theta) + o_P(1)$ uniformly over $\theta \in \Theta$, where $W(\theta)$ is positive definite and continuous for any $\theta \in \Theta$.

Assumption 5.2 (The "degeneracy" condition in Chernozhukov et al. (2007).)**.** (i). There is a sequence of subsets $\{\Theta_n\}_{n \in \mathbb{N}} \subset \Theta$, which could be data dependent, such that $Q_n(\theta) - \inf_{\theta' \in \Theta} Q_n(\theta') = 0$ for each $\theta \in \Theta_n$ and for each $n \in \mathbb{N}$, and that $d_H(\Theta_n, \Theta_I) \leq \epsilon_n$ for some $\epsilon_n = o_P(1)$. (ii). The sequence ϵ_n is of stochastic order $a_n^{-1/\gamma}$ for some $\gamma > 0$.

Assumption 5.1 is a set of standard regularity conditions. Assumption 5.2 is satisfied in point identified models and many moment inequality models as shown in Chernozhukov et al. (2007)[17]. There are two quantities a_n and γ. First, recall that a_n appears as the scaling factor of Q_n in (17); in the present context, the order $a_n = \sqrt{n}$ is appropriate given the P-Donsker condition in Assumption 5.1 and the construction of Q in (15). Second, the quantity $\gamma > 0$ in Assumption 5.2 (ii) is related to the one in "Polynomial minorant" condition, Condition C.2 of Chernozhukov et al. (2007). I show in the proof of the convergence of $\widehat{\Theta}_I$ that Assumption 4.9 on Φ combined with Q in (15) implies Condition C.2 of Chernozhukov et al. (2007) with $\gamma = 1$.

Under these assumptions, the the set estimator $\widehat{\Theta}_I(\tau)$ converges to Θ_I in the Hausdorff metric; the results are stated formally as follows.

Proposition 5.3 (Consistency and convergence rate of $\widehat{\Theta}_I$)**.** *Suppose that Assumptions 4.9 hold with $\Theta_I = \Theta_I(\Phi, P)$ defined in (4). Let $\tau \in \mathbb{R}_+$ be such that $\tau \geq \sup_{\theta \in \Theta_I} \sqrt{n} \cdot Q_n(\theta)$ with probability approaching 1 and $\tau = o_P(\sqrt{n})$. Then $d_H(\widehat{\Theta}_I, \Theta_I) = o_P(1)$; more precisely,*

$$d_H(\widehat{\Theta}_I, \Theta_I) = O_P\left((1 \vee \tau)/\sqrt{n} \right).$$

If, in addition, Assumption 5.2 holds, then one can set $\tau = \sqrt{n} \cdot c_n$ and obtain

$$d_H(\widehat{\Theta}_I, \Theta_I) = O_P\left(1/\sqrt{n} \right).$$

Proof. See Appendix B.5. □

[17] Yildiz (2012) instead put conditions on moment functions; see Kaido, Molinari, et al. (2022) for their relationships in terms of constraint qualifications.

5.2. **Limiting distribution of $\widehat{\Theta}_I$.** Now I focus on the estimator $\widehat{\Theta}_I = \widehat{\Theta}_I(\tau)$ with $\tau = \sqrt{n} \cdot c_n$ and investigate its limiting distribution measured in the Hausdorff metric. In this section, I pay attention to the moment inequality restrictions $|\mathcal{J}_2| = d_q$ as this is the prototypical case in the literature. The key device underlying my analysis is the stochastic field whose supremum represents the Hausdorff metric between the estimator $\widehat{\Theta}_I$ and Θ_I, i.e.,

$$Z_n(x) := \sqrt{n} \cdot \left(d(x, \widehat{\Theta}_I) - d(x, \Theta_I) \right). \tag{20}$$

From the property of d_H (e.g. Rockafellar and Wets (1998) Chapter 4), it is known that $\sup_{x \in \mathbb{R}^{d_\theta}} |Z_n(x)| = \sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I)$. This stochastic field is not equicontinuous; the similar issue occurs in Chernozhukov et al. (2007) as well when they consider QLR statistic $\sup_{\theta' \in \Theta_I} a_n \cdot Q_n(\theta')$ for the purpose of constructing asymptotically valid confidence set. Hence care is needed in handling Z_n, and I employ the weak supremum convergence combined with finite set approximation as in Chernozhukov et al. (2007) and Kaido (2016).

In deriving limiting distribution of $\widehat{\Theta}_I$, I add the following assumptions.

Assumption 5.4. (i). $\Phi(\cdot, P) : \Theta \mapsto \mathbb{R}^{d_q}$ is C^2 on $\Theta \subset \mathbb{R}^{d_\theta}$. (ii). There exists positive constant $c > 0$ such that, for any $\theta \in \partial\Theta_I$,

$$d\left(0,\ \text{conv}\left\{\phi_j(\theta) : j \in \mathcal{A}(\theta)\right\}\right) \geq c > 0,$$

where $\phi_j(\theta) = \nabla \Phi_j(\theta, P)$ for any $j \in \mathcal{J}_2$ and

$$\mathcal{A}(\theta) = \{j \in \mathcal{J}_2 : \Phi_j(\theta) = 0\}. \tag{21}$$

(iii). $\{\phi_j(\theta, P)\}_{j \in \mathcal{A}(\theta)}$ are linearly independent at any $\theta \in \partial\Theta_I$.

Assumption 5.4 (i) says that Φ is Fréchet differentiable, which makes my argument simple in this section. Note that this is a condition on $\Phi(\cdot, P)$ and it does not exclude sets like polyhedrons. The set $\mathcal{A}(\theta)$ captures all binding moment restrictions. Assumption 5.4 (ii) and (iii) are related to the Mangasarian-Fromovitz Constraint Qualification (MFCQ) and the Linear Independence Constraint Qualification (LICQ) respectively, which are often used in the literature in partially identified models; see Kaido, Molinari, et al. (2022) for the detailed discussion.

Remark 5.5. It is possible to consider the other cases such as $|\mathcal{J}_1| = d_q$ or $|\mathcal{J}_1| + |\mathcal{J}_2| = d_q$. The first case nests the point identified situation, while the second case corresponds to the one, where for example a parameter is point identified via moment equalities and the moment inequalities put additional restriction. Each has its own difficulty. A difficulty in the first case, for example, arises when the parameter is under identified; under identified non-linear IV case corresponds to this situation as mentioned in Chernozhukov et al. (2007). In this case, even $\sqrt{n}$ consistency is not established in the criterion function based approach. A difficulty in the second case occurs when a parameter is uniquely identified via moment equalities and moment inequalities binds at the same time; this situation requires specific boundary-corrected asymptotic analysis as in Andrews (1999). The detail investigation of these cases is left for future research.

My main result here shows that the (scaled) Hausdorff-based distribution of $\widehat{\Theta}_I$, which is centered at the target set Θ_I, weakly converges to the supremum of a Gaussian field on $\partial\Theta_I$. The formal statement is as follows.

Theorem 5.6 (Limiting distribution of $\widehat{\Theta}_I$ in the Hausdorff metric.)**.** *Suppose that Assumptions 5.1, 5.2, and 5.4 hold. Then for the set estimator $\widehat{\Theta}_I(\tau)$ with $\tau = \sqrt{n} \cdot c_n$ with $c_n = \inf_{\theta' \in \Theta} Q_n(\theta')$ it holds that*

$$\begin{aligned} &\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I) \\ &\quad \rightsquigarrow \sup_{\theta \in \partial\Theta_I} \sup \left\{ |\alpha^t D_{\mathcal{A}(\theta)}(\theta)| : \alpha \geq 0, \left\| \alpha^t G_{\mathcal{A}(\theta)}(\theta) \right\| = 1 \right\}, \quad \text{as } n \to \infty, \end{aligned} \tag{22}$$

where $\mathcal{D}_{\mathcal{A}(\theta)}(\theta)$ and $G_{\mathcal{A}(\theta)}(\theta)$ are the sets of rows in $D(\theta) = \nabla\Phi(\theta, P)(\xi(\theta)) \in \mathbb{R}^{d_\theta}$ with $\xi(\theta)$ weak limit in (19) and $G(\theta) = \nabla\Phi(\theta, P) \in \mathbb{R}^{|\mathcal{J}_2| \times d_\theta}$ respectively, i.e.

$$\begin{aligned} D_{\mathcal{A}(\theta)}(\theta) &= (D_j(\theta))_{j \in \mathcal{A}(\theta)} = (\nabla\Phi_j(\theta, P)(\xi(\theta)))_{j \in \mathcal{A}(\theta)}, \\ G_{\mathcal{A}(\theta)}(\theta) &= (G_j(\theta))_{j \in \mathcal{A}(\theta)} = (\nabla\Phi_j(\theta, P))_{j \in \mathcal{A}(\theta)}. \end{aligned} \tag{23}$$

Proof. See Appendix B.6. □

The result in (22) shows a generic limiting distribution of $\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I)$ based on the criterion function. The key quantities are $D_{\mathcal{A}(\theta)}(\theta)$ and $G_{\mathcal{A}(\theta)}(\theta)$ on $\partial\Theta_I$. For any active or binding constraint $j \in \mathcal{A}(\theta)$, the component $D_j(\theta) = \nabla\Phi_j(\theta, P)(\xi(\theta))$ is a functional of Gaussian $\xi(\theta)$ which comes from the Donsker condition in Assumption 5.1; this roughly captures how the estimator $\widehat{\Theta}_I$ approaches to $\theta \in \partial\Theta_I$. Because of Fréchet differentiability of Φ, this $D_j(\theta)$ is (unconditionally) Gaussian. Regarding this local information, there could be multiple direction in which $\widehat{\Theta}_I$ approaches to $\theta \in \partial\Theta_I$, and this information of potential direction is contained in $G_{\mathcal{A}(\theta)}(\theta)$. For example, if θ is on edge or vertex of $\partial\Theta_I$, multiple restrictions bind ($|\mathcal{A}(\theta)| > 1$) while one direction matters if θ is on smooth part of $\partial\Theta_I$ because one restriction is active in this case ($|\mathcal{A}(\theta)| = 1$). The result tells that the limiting distribution of $\widehat{\Theta}_I$ in d_H is dominated at the local point $\theta \in \partial\Theta_I$ with harder direction to approach.

5.3. **Hausdorff-ball confidence set.** The result in Section 5.2 suggests a bootstrap-based confidence set, which is similar to the one for the convex identified set (Beresteanu and Molinari, 2008; Kaido, 2016) and differs from the approach taken in much of the literature, where confidence sets are constructed as level sets of the criterion function (e.g., Chernozhukov et al., 2007; Bugni, 2010; Romano and Shaikh, 2010; Chen et al., 2018). Motivated by the result in Section 5.2, I here briefly illustrate how to compute the bootstrap-based confidence set. As in Section 5.2, I focus on the case of moment inequalities. The validity of this bootstrap procedure is left for future research.

The key observation is that the set estimator $\widehat{\Theta}_I$ has the following relation

$$\Theta_I \subset (\widehat{\Theta}_I)^r \quad \Leftrightarrow \quad d_H^{\rightarrow}(\Theta_I, \widehat{\Theta}_I) \leq r,$$

for $r > 0$, where $d_H^{\rightarrow}(\Theta_I, \widehat{\Theta}_I) = \sup_{\theta \in \Theta_I} d(\theta, \widehat{\Theta}_I)$ is the directed Hausdorff distance. Note that $(\widehat{\Theta}_I)^r$ is the closed r-envelope of $\widehat{\Theta}_I$; see Appendix A.2. To construct a valid

confidence set, the problem is to find an $r > 0$ with suitable coverage property. This quantity can be derived from

$$C_n(\alpha) := (\widehat{\Theta}_I)^{r_n^{\rightarrow}(\alpha)}, \quad r_n^{\rightarrow}(\alpha) = n^{-1/2} \cdot \zeta_n^{\rightarrow}(1-\alpha),$$

where $\alpha \in (0,1)$ and $\zeta_n^{\rightarrow}(1-\alpha)$ is the $(1-\alpha)$ quantile of the conditional distribution, given the data, of

$$D_n^{\rightarrow *} := \sqrt{n} \cdot d_H^{\rightarrow}(\widehat{\Theta}_I, \widehat{\Theta}_I^*),$$

with $\widehat{\Theta}_I^*$ denoting the bootstrap set estimator specified below. By the definition of the envelope, $\Theta_I \subset C_n(\alpha)$ if and only if $\sqrt{n} \cdot d_H^{\rightarrow}(\Theta_I, \widehat{\Theta}_I) \leq \zeta_n^{\rightarrow}(1-\alpha)$. Hence, if the conditional distribution of $D_n^{\rightarrow *}$ consistently estimates the limiting distribution of $\sqrt{n} \cdot d_H^{\rightarrow}(\Theta_I, \widehat{\Theta}_I)$, and the latter is continuous at its $(1-\alpha)$ quantile, then

$$\lim_{n\to\infty} P\{\Theta_I \subset C_n(\alpha)\} = 1 - \alpha,$$

so that $C_n(\alpha)$ has asymptotic coverage $1-\alpha$. A formal proof of the validity of the bootstrap is beyond the scope of this paper.

For the construction of $\widehat{\Theta}_I^*$, I consider the standard procedure. Let $e_1, \cdots, e_n$ be i.i.d. with mean zero and unit variance and independent of the data (e.g., $e_i \sim N(0,1)$), and let

$$\mathbb{P}_n^* m(\theta) := \mathbb{P}_n m(\theta) + \frac{1}{n}\sum_{i=1}^n e_i \cdot \{m(\theta, X_i) - \mathbb{P}_n m(\theta)\},$$
$$Q_n^*(\theta) := \left\| \Phi_v^t(\theta, \mathbb{P}_n^*) \cdot W_n^{1/2} \right\|, \quad c_n^* := \inf_{\theta' \in \Theta} Q_n^*(\theta').$$

The bootstrap set estimator $\widehat{\Theta}_I^*$ is defined as

$$\widehat{\Theta}_I^* := \{\theta \in \Theta : Q_n^*(\theta) \leq c_n^*\}.$$

The algorithm for computing the confidence set $C_n(\alpha) = (\widehat{\Theta}_I)^{r_n^{\rightarrow}(\alpha)}$ is illustrated in Algorithm 1.

Remark 5.7 (Two-sided version). The confidence set $C_n(\alpha)$ is one-sided: it controls only how far Θ_I can extend beyond $\widehat{\Theta}_I$. A two-sided version is obtained by replacing the directed distance $d_H^{\rightarrow}$ with the Hausdorff distance d_H throughout. Let $\zeta_n(1-\alpha)$ be the $(1-\alpha)$ quantile of the conditional distribution, given the data, of

$$D_n^* := \sqrt{n} \cdot d_H(\widehat{\Theta}_I, \widehat{\Theta}_I^*),$$

and let $r_n(\alpha) := n^{-1/2} \cdot \zeta_n(1-\alpha)$. Since $d_H(\Theta_I, \widehat{\Theta}_I) \leq r$ if and only if both $\Theta_I \subset (\widehat{\Theta}_I)^r$ and $\widehat{\Theta}_I \subset (\Theta_I)^r$, the same argument as above gives, under the analogous conditions,

$$\lim_{n\to\infty} P\{d_H(\Theta_I, \widehat{\Theta}_I) \leq r_n(\alpha)\} = 1 - \alpha.$$

That is, with asymptotic probability $1-\alpha$, the identified set lies in the Hausdorff ball of radius $r_n(\alpha)$ centred at $\widehat{\Theta}_I$: Θ_I is contained in $(\widehat{\Theta}_I)^{r_n(\alpha)}$, and in addition no point of $\widehat{\Theta}_I$ is farther than $r_n(\alpha)$ from Θ_I. Because $d_H \geq d_H^{\rightarrow}$, the radius satisfies $r_n(\alpha) \geq r_n^{\rightarrow}(\alpha)$;

this is the price of the additional statement. In Algorithm 1, only the bootstrap statistic changes, to

$$D_n^{(b)} := \sqrt{n} \cdot \max\left\{ \max_{\theta_g \in \widehat{\Theta}_{I,s}} \delta_g^{(b)},\ \max_{\theta_g \in \widehat{\Theta}_{I,s}^{(b)}} \hat{\delta}_g \right\},$$

with ζ_n and r_n in place of $\zeta_n^{\rightarrow}$ and $r_n^{\rightarrow}$.

Algorithm 1 One-sided Hausdorff-ball confidence set (grid implementation)

Grid and precomputation

Form the grid $\Theta_s := \Theta \cap \left(\theta_s + s\,\mathbb{Z}^{d_\theta}\right) = \{\theta_1, \cdots, \theta_G\}$.

For $g = 1, \cdots, G$ and $i = 1, \cdots, n$: $\bar{m}_g := \mathbb{P}_n m(\theta_g)$ and $a_{gi} := m(\theta_g, X_i) - \bar{m}_g$.

Set estimate

$Q_g := \left\| \Phi_v^t(\theta_g, \mathbb{P}_n) \cdot W_n^{1/2} \right\|$ for all g, where $\Phi(\theta_g, \mathbb{P}_n) = q(\bar{m}_g)$, and $\hat{c}_n := \min_g Q_g$.

$\widehat{\Theta}_{I,s} := \{\theta_g :\ Q_g \le \hat{c}_n\}$, and $\hat{\delta}_g := d(\theta_g, \widehat{\Theta}_{I,s})$ for all g.

Bootstrap

for $b = 1, \cdots, B$ **do**

  draw $e_1^{(b)}, \cdots, e_n^{(b)}$ i.i.d. from the multiplier law (e.g., $N(0,1)$)

  $\bar{m}_g^{(b)} := \bar{m}_g + n^{-1} \sum_{i=1}^n e_i^{(b)} a_{gi}$ and $Q_g^{(b)} := \left\| \Phi_v^t(\theta_g, \mathbb{P}_n^{(b)}) \cdot W_n^{1/2} \right\|$ for all g, where $\Phi(\theta_g, \mathbb{P}_n^{(b)}) = q(\bar{m}_g^{(b)})$

  $\hat{c}_n^{(b)} := \min_g Q_g^{(b)}$ and $\widehat{\Theta}_{I,s}^{(b)} := \{\theta_g :\ Q_g^{(b)} \le \hat{c}_n^{(b)}\}$

  $\delta_g^{(b)} := d\left(\theta_g, \widehat{\Theta}_{I,s}^{(b)}\right)$ for all g

  $D_n^{\rightarrow(b)} := \sqrt{n}\, \max_{\theta_g \in \widehat{\Theta}_{I,s}} \delta_g^{(b)}$

Critical value and output

$\zeta_n^{\rightarrow} := D_n^{\rightarrow(k)}$, the k-th smallest of $D_n^{\rightarrow(1)}, \cdots, D_n^{\rightarrow(B)}$, with $k := \lceil (B+1)(1-\alpha) \rceil$; set $r_n^{\rightarrow} := \zeta_n^{\rightarrow} / \sqrt{n}$.

$C_{n,s}(\alpha) := \{\theta_g \in \Theta_s :\ \hat{\delta}_g \le r_n^{\rightarrow}\}$.

6. Finite sample properties

6.1. **Data-generating process.** I here evaluate the finite-sample performance of the set estimator $\widehat{\Theta}_I$ and the Hausdorff-ball confidence set $C_n(\alpha)$ in Section 5.3, using the optimal top tax rate example in Section 2.

Model. For convenience, I briefly recap the model in Section 2. Taxpayers with income above a threshold z^* face a constant marginal rate on income above z^* and misperceive the tax schedule as in Farhi and Gabaix (2020): a taxpayer with income z responds to a weighted average of the true marginal rate, with attention weight $\omega \in [0,1]$, and

of a weighted average of the marginal rates at other income levels. In the revenue-maximizing case without income effects, the optimal top rate is

$$\bar{\tau} = \frac{1}{1 + \bar{\zeta}_{c,r} \cdot \pi \cdot \kappa(A)}, \quad \kappa(A) = \omega + (1-\omega)A, \tag{24}$$

where $\bar{\zeta}_{c,r}$ is the elasticity of a taxpayer who perceives the schedule correctly, $\pi = z_m/(z_m - z^*)$ is the Pareto parameter of the upper tail with $z_m = \mathbb{E}_P[Z \mid Z \geq z^*]$ and Z reported income, and A represents the shape of misperception. At $\kappa = 1$, i.e., $\omega = 1$ or $A = 1$, (24) is the formula of Saez (2001). The basic conditions maintained through this exercise are as follows. The attention weight $\omega = \omega_0$ and the elasticity $\bar{\zeta}_{c,r}$ are assumed to be known, the misperception shape lies in a maintained range $A \in [A_L, A_U]$, and the tail parameter is partially identified because incomes are topcoded at a known cap $z_c > z^*$. Under the setting, the identified set of our interest for the parameter $\theta = (\bar{\tau}, A)^t$ is expressed as (see Proposition 3.6 for detail)

$$\Theta_I = \left\{ (\bar{\tau}, A)^t \in \Theta : \; A_L \leq A \leq A_U, \; \frac{1}{1 + \bar{\zeta}_{c,r}\pi_U \kappa(A)} \leq \bar{\tau} \leq \frac{1}{1 + \bar{\zeta}_{c,r}\pi_L \kappa(A)} \right\}, \tag{25}$$

where, with $S = P(Z \geq z^*)$ and the censoring moments

$$\begin{aligned} M_L &= \mathbb{E}_P\left[(Z \wedge z_c) \cdot \mathbb{1}\{Z \geq z^*\}\right], \\ M_U &= \mathbb{E}_P\left[(Z \cdot \mathbb{1}\{Z < z_c\} + \bar{z} \cdot \mathbb{1}\{Z \geq z_c\}) \cdot \mathbb{1}\{Z \geq z^*\}\right], \end{aligned}$$

the bounds on the tail parameter are expressed

$$\pi_L = \frac{M_U/S}{M_U/S - z^*}, \quad \pi_U = \frac{M_L/S}{M_L/S - z^*}.$$

Note that $\bar{z}$ is the maintained upper bound on the mean income above the cap in Assumption 3.5(ii).

Below I explain the detail of the data generating process for ingredients of the identified set in (25).

Population. I normalize the bracket threshold to $z^* = 1$ and let reported income in the top bracket follow an exact Pareto law with tail index π_0, i.e.,

$$P(Z > t) = t^{-\pi_0}, \quad t \geq 1, \quad \pi_0 = 1.5, \tag{26}$$

where the value $\pi_0 = 1.5$ is consistent with the stability of the Pareto parameter above roughly \$300,000 documented in Diamond and Saez (2011). Since every object in Θ_I depends on the income distribution only through moments conditional on $Z \geq z^*$, I simulate the top-bracket sample directly; the sampling design is therefore $Z_1, \cdots, Z_n$ i.i.d. from (26), so that $S = P(Z \geq z^*) = 1$ in this sample[18].

Censoring. Recall that reported income is topcoded at a known cap z_c. Accordingly, the econometrician observes

$$(Z_i \wedge z_c, \; D_i), \quad D_i = \mathbb{1}\{Z_i \geq z_c\}, \quad i \in [n].$$

[18]With a full-population sample, S is estimated as well and the only change is an additional moment, which is strongly correlated with the two moments M_L and M_U.

The cap z_c is set to the $(1-p_c)$ quantile of (26), i.e., $z_c = p_c^{-1/\pi_0}$, where p_c is the share of the bracket that is censored. I use $p_c = 0.05$, which gives $z_c = 7.37$, and set $\bar{z} = 50$. Assumption 3.5(ii) holds, since under (26) the mean income above the cap is $\pi_0 z_c/(\pi_0 - 1) = 22.1$; the interval for π below contains π_0 precisely because of this.

Implied bounds on the tail parameter. Under (26) the censoring moments are available in closed form. With the density $f(z) = \pi_0 z^{-\pi_0 - 1}$, it follows that

$$M_L = \mathbb{E}_P[Z \wedge z_c] = \frac{\pi_0}{\pi_0 - 1}\left(1 - z_c^{-(\pi_0 - 1)}\right) + z_c \cdot p_c,$$
$$M_U = \mathbb{E}_P\left[Z \cdot \mathbb{1}\{Z < z_c\} + \bar{z} \cdot \mathbb{1}\{Z \geq z_c\}\right] = M_L + (\bar{z} - z_c) \cdot p_c.$$

At $p_c = 0.05$ these equal $M_L = 2.2632$ and $M_U = 4.3948$. Hence the expressions for π_L and π_U imply

$$\pi \in [\pi_L, \pi_U] = [1.2946,\ 1.7916], \tag{27}$$

which contains the true value $\pi_0 = 1.5$.

Behavioral parameters. I consider two values of the attention weight, $\omega_0 \in \{0.2,\ 0.45\}$, and set the range of the misperception shape to $[A_L, A_U] = [0.55, 1.3]$. The two values are close to the range of estimates of attention to the French income tax in Lardeux (2023), who finds that tax filers assign a weight of 0.22 to 0.44 to the effective tax schedule, depending on the estimation method. The range also contains the estimate of attention to sales taxes in Taubinsky and Rees-Jones (2018), 0.25. The lower endpoint A_L is the value implied by ironing, $A = 1/\pi$, at the largest admissible tail parameter $\pi_U = 1.79$; Rees-Jones and Taubinsky (2020) find that taxpayers commonly iron. The upper endpoint allows for moderate over-perception of the top rate, for which I am not aware of a direct estimate. The true misperception shape need not be specified, since Θ_I depends only on the range.

Sources of the calibration. The tail index $\pi_0 = 1.5$ and the elasticity $\bar{\zeta}_{c,r} = 0.25$ are the values used in Diamond and Saez (2011); the elasticity $\bar{\zeta}_{c,r} = 0.57$ is the estimate for taxable income at the top of the distribution in Gruber and Saez (2002), which Diamond and Saez (2011) regard as a conservative upper bound.

Designs. I consider four designs, depending on the values of $\omega_0 \in \{0.2, 0.45\}$ and $\bar{\zeta}_{c,r} = \{0.25, 0.57\}$; for other parameters, all of the designs share $\pi_0 = 1.5$, $p_c = 0.05$, $\bar{z} = 50$ and $[A_L, A_U] = [0.55, 1.3]$. Figure 2 displays Θ_I in the four designs. The section at $A = 1$ is the interval implied by Saez (2001) at $e = \bar{\zeta}_{c,r}$ and does not depend on ω_0; points with $A < 1$ lie above it and points with $A > 1$ below it. A lower attention weight makes the optimal rate more sensitive to A and the upper boundary more curved.

For each design, I consider the sample sizes $n \in \{500,\ 1{,}000,\ 5{,}000\}$ with 1,000 Monte Carlo replications and 499 bootstrap or subsampling draws per replication.

6.2. **Results.** As set estimators and confidence sets, here I compare different methods described below.

(i) *Hausdorff ball.* The confidence set $C_n(\alpha)$ of Section 5.3: the set estimator $\widehat{\Theta}_I$ enlarged by the radius $r_n^{\rightarrow}(\alpha) = n^{-1/2}\zeta_n^{\rightarrow}(1-\alpha)$, where $\zeta_n^{\rightarrow}(1-\alpha)$ is the

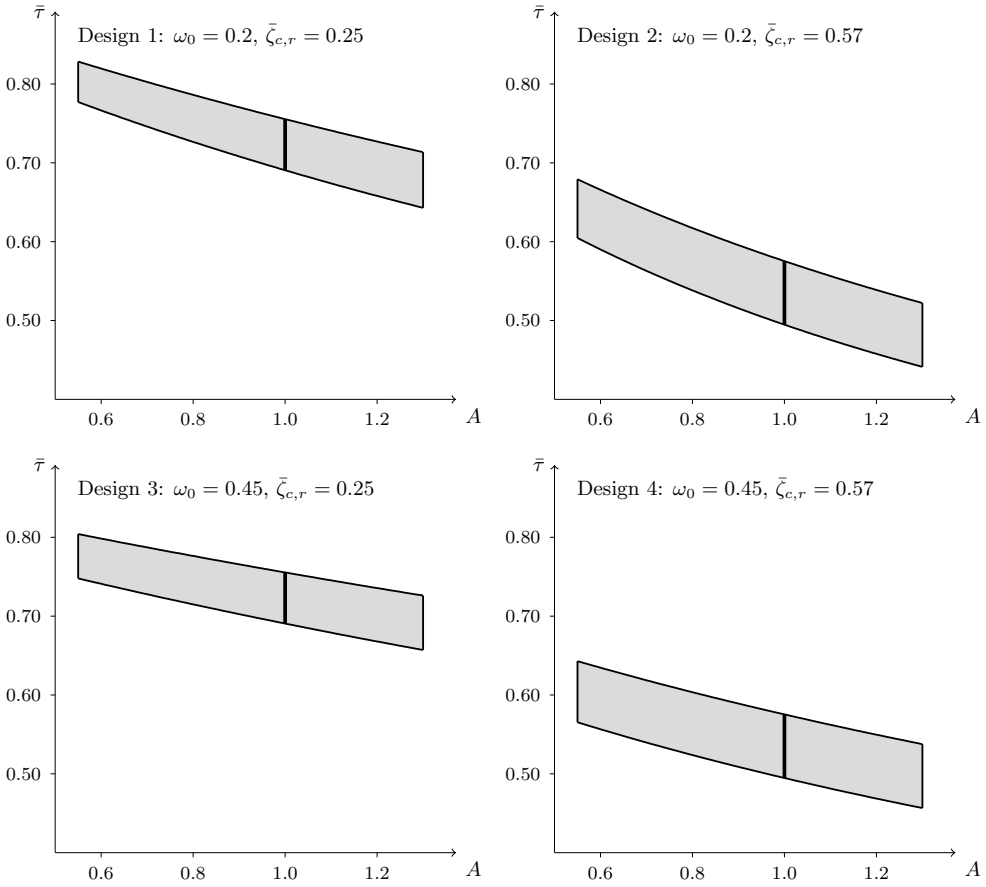


FIGURE 2. The population identified set (25) in the four designs. The thick vertical segment is the section at $A = 1$.

$(1-\alpha)$ quantile of the multiplier bootstrap distribution of the one-sided distance $\sqrt{n} \cdot d_H^{\rightarrow}$, as in Algorithm 1. In this section $C_n(\alpha)$ is intersected with the known range $A_L \leq A \leq A_U$ (The other four methods impose this range as the non-stochastic constraints $\Phi_3, \Phi_4 \leq 0$).

(ii) *Support function.* The Wald-type confidence set of Kaido (2016), which builds on the support-function approach of Beresteanu and Molinari (2008) and Kaido and Santos (2014). I use the subsampling version as implemented in Kaido (2016).[19]

(iii) *Subsampling.* The confidence set of Chernozhukov et al. (2007): a level set of the criterion function, with the critical value obtained by subsampling the supremum of the criterion over the estimated set. I use a single iteration started from $\widehat{\Theta}_I$, as their Condition C.3 permits.

[19]The upper face of $\text{conv}(\Theta_I)$ is a straight segment whose slope depends on P, which is excluded in Kaido and Santos (2014) (Remark 3.2 and Assumption 4.1) and makes a bootstrap of the support-function statistic doubtful; subsampling requires only a continuous limit distribution.

(iv) *Bootstrap.* The bootstrap of Bugni (2010), which approximates the distribution of the same supremum by retaining only the moment inequalities that are close to binding, with the threshold $\tau_n = \ln \ln n$.

(v) *Monte Carlo confidence set.* Procedure 2 of Chen et al. (2018). The moment inequalities are written as moment equalities with slackness parameters, quasi-posterior draws are generated by their sequential Monte Carlo sampler with 10,000 draws, and the confidence set consists of the values of θ whose profiled quasi-likelihood ratio lies below the $(1-\alpha)$ quantile of its draws.

For the two subsampling methods, (ii) and (iii), the subsample size is $b = \lfloor 0.2n \rfloor$, which follows the rule $b_n \propto n$ of Bugni (2010). All resampling methods use $B = 499$ draws, and their critical values are the $\lceil (B+1)(1-\alpha) \rceil$-th order statistic of these draws.

Set estimator. Table 1 reports the median Hausdorff distance of $\widehat{\Theta}_I$ and of the support-function estimator $\mathrm{conv}(\widehat{\Theta}_I)$ to Θ_I. The distances of $\widehat{\Theta}_I$ are smaller than thoes of $\mathrm{conv}(\widehat{\Theta}_I)$, which furthermore falls much more slowly as the sample size increases. Its ratio to the distance of $\widehat{\Theta}_I$ therefore grows with n, up to 5.00 in design 2 at $n = 5{,}000$. Figure 3 compares the distribution of $\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I)$ with the limiting distribution of Theorem 5.6. At $n = 1{,}000$ the two are almost indistinguishable in all four designs.

				$\widehat{\Theta}_I$			Support function			Ratio		
Design	ω_0	$\bar{\zeta}_{c,r}$	$n =$	500	1,000	5,000	500	1,000	5,000	500	1,000	5,000
1	0.2	0.25		0.55	0.39	0.18	0.70	0.58	0.43	1.27	1.47	2.43
2	0.2	0.57		0.66	0.46	0.20	1.16	1.05	1.00	1.75	2.27	5.00
3	0.45	0.25		0.55	0.40	0.18	0.61	0.46	0.26	1.11	1.14	1.47
4	0.45	0.57		0.67	0.46	0.20	0.83	0.65	0.48	1.25	1.43	2.42

TABLE 1. Median Hausdorff distance to Θ_I of the set estimators, times 100.

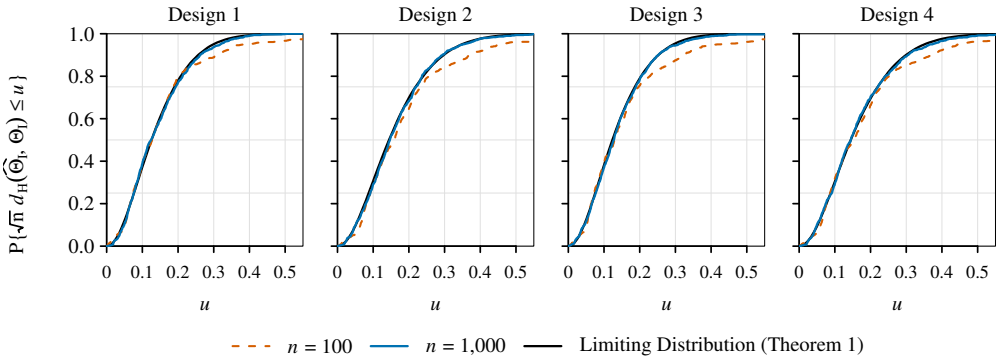


FIGURE 3. Distribution function of $\sqrt{n}\, d_H(\widehat{\Theta}_I, \Theta_I)$ (at $n = 100$ and $n = 1{,}000$) and the limiting distribution of Theorem 5.6.

Coverage and Hausdorff distance. Panel A and B of Table 2 report coverage and median Hausdorff distance to Θ_I at the nominal level 0.90. The Hausdorff ball covers Θ_I with

frequency 0.908–0.932 at $n = 500$ and 0.893–0.906 at $n = 5{,}000$. The bootstrap of Bugni (2010) and the Monte Carlo confidence set of Chen et al. (2018) under-cover slightly at $n = 500$, with 0.871–0.907, and are close to the nominal level at $n = 5{,}000$, with 0.891–0.900. The subsampling confidence set of Chernozhukov et al. (2007) has coverage that falls with n, to 0.863–0.871 at $n = 5{,}000$. With b/n held fixed, this is in line with the finding of Bugni (2010). The support-function confidence set is conservative, with coverage 0.96–0.99, as expected for a set that targets $\text{conv}(\Theta_I)$. For Hausdorff distance, methods except the support function have similar distance at $n = 5,000$. The similar results can be observed for the case of the nominal level 0.95 in Table 3.

				Panel A. Coverage			**Panel B.** $100\times$ **median** d_H		
Design	ω_0	$\bar{\zeta}_{c,r}$	Method $n =$	500	1,000	5,000	500	1,000	5,000
1	0.2	0.25	Hausdorff ball $C_n(\alpha)$	0.932	0.909	0.906	1.49	1.04	0.45
			Support function	0.985	0.969	0.963	2.78	2.38	1.92
			Chernozhukov et al. (2007), subsampling	0.927	0.876	0.863	1.35	0.94	0.41
			Bugni (2010), bootstrap	0.905	0.888	0.897	1.30	0.95	0.44
			Chen et al. (2018)	0.907	0.889	0.898	1.32	0.96	0.44
2	0.2	0.57	Hausdorff ball $C_n(\alpha)$	0.908	0.924	0.902	1.73	1.24	0.52
			Support function	0.968	0.978	0.964	3.22	2.77	2.22
			Chernozhukov et al. (2007), subsampling	0.907	0.898	0.871	1.57	1.09	0.47
			Bugni (2010), bootstrap	0.878	0.903	0.898	1.51	1.12	0.50
			Chen et al. (2018)	0.882	0.906	0.897	1.52	1.12	0.50
3	0.45	0.25	Hausdorff ball $C_n(\alpha)$	0.915	0.911	0.902	1.50	1.03	0.45
			Support function	0.977	0.968	0.963	2.57	2.15	1.65
			Chernozhukov et al. (2007), subsampling	0.913	0.881	0.863	1.38	0.92	0.41
			Bugni (2010), bootstrap	0.871	0.890	0.900	1.31	0.93	0.43
			Chen et al. (2018)	0.880	0.896	0.900	1.33	0.94	0.43
4	0.45	0.57	Hausdorff ball $C_n(\alpha)$	0.915	0.905	0.893	1.78	1.21	0.52
			Support function	0.972	0.962	0.970	2.80	2.23	1.60
			Chernozhukov et al. (2007), subsampling	0.907	0.880	0.870	1.61	1.08	0.48
			Bugni (2010), bootstrap	0.886	0.881	0.891	1.55	1.10	0.50
			Chen et al. (2018)	0.891	0.887	0.897	1.57	1.11	0.51

TABLE 2. Coverage of Θ_I by the 90% confidence sets (Panel A) and their median Hausdorff distance to Θ_I, times 100 (Panel B).

				Panel A. Coverage			**Panel B.** $100\times$ **median** d_H		
Design	ω_0	$\bar{\zeta}_{c,r}$	Method $n=$	500	1,000	5,000	500	1,000	5,000
			Hausdorff ball $C_n(\alpha)$	0.981	0.963	0.953	1.76	1.22	0.52
			Support function	0.994	0.989	0.987	3.06	2.53	1.97
1	0.2	0.25	Chernozhukov et al. (2007), subsampling	0.969	0.949	0.933	1.60	1.08	0.47
			Bugni (2010), bootstrap	0.950	0.936	0.946	1.47	1.08	0.49
			Chen et al. (2018)	0.951	0.938	0.947	1.48	1.09	0.50
			Hausdorff ball $C_n(\alpha)$	0.965	0.978	0.956	2.10	1.45	0.61
			Support function	0.987	0.991	0.985	3.52	2.94	2.28
2	0.2	0.57	Chernozhukov et al. (2007), subsampling	0.958	0.961	0.929	1.89	1.28	0.54
			Bugni (2010), bootstrap	0.932	0.955	0.948	1.70	1.26	0.57
			Chen et al. (2018)	0.939	0.954	0.950	1.72	1.26	0.57
			Hausdorff ball $C_n(\alpha)$	0.974	0.962	0.952	1.77	1.20	0.52
			Support function	0.993	0.986	0.980	2.83	2.29	1.71
3	0.45	0.25	Chernozhukov et al. (2007), subsampling	0.971	0.941	0.921	1.64	1.07	0.46
			Bugni (2010), bootstrap	0.924	0.939	0.947	1.48	1.05	0.49
			Chen et al. (2018)	0.938	0.944	0.945	1.51	1.06	0.49
			Hausdorff ball $C_n(\alpha)$	0.966	0.958	0.963	2.13	1.43	0.61
			Support function	0.992	0.984	0.990	3.13	2.41	1.66
4	0.45	0.57	Chernozhukov et al. (2007), subsampling	0.964	0.934	0.936	1.92	1.26	0.55
			Bugni (2010), bootstrap	0.930	0.929	0.952	1.74	1.23	0.57
			Chen et al. (2018)	0.935	0.933	0.956	1.75	1.24	0.57

TABLE 3. Coverage of Θ_I by the 95% confidence sets (Panel A) and their median Hausdorff distance to Θ_I, times 100 (Panel B).

7. Discussion

This paper studies the asymptotic efficiency of set estimation for a general class of identified sets. I frist derive the asymptotic minimax efficiency bound for set estimators under risk functions based on the Hausdorff metric. The analysis relies on a limit experiment framework tailored to the local structure of the identified set, and the bound recovers the classical efficiency bound as a special case when the parameter is point identified. I then show that a set estimator based on a criterion function is efficient under mild conditions. To establish this, I derive the limiting distribution of the set estimator in the Hausdorff metric without globally characterizing the identified set through its support functions, which allows the identified set to be non-convex. This limiting distribution motivates a bootstrap-based confidence set for the identified set and I examine its finite sample performance. I illustrate the framework with an application to optimal taxation with behavioral agents.

Appendices

A. Some primers

A.1. **Notation.** For any $r \in [1, \infty]$, $\|.\|_{L^r}$ denotes the L^r-norm. For any vector $a \in \mathbb{R}^d$, $\|.\|$ denotes the Euclidean norm. For positive sequences $\{a_n\}_{n\in\mathbb{N}}$ and $\{b_n\}_{n\in\mathbb{N}}$, let $a_n \lesssim b_n$ if $\limsup_n a_n/b_n < \infty$ and $a_n \asymp b_n$ if $a_n \lesssim b_n$ and $b_n \lesssim a_n$. For any set A, $|A|$ represents the cardinality of A. For a set A in a topological space, $\overline{A}$, A^o, and ∂A represent the closure, the interior, and the boundary of A respectively. For a metric space (V, d_V), $B_v(\epsilon)$ is an open ball centered at $v \in V$ with radius $\epsilon > 0$. For a and b in $\mathbb{R}$, $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$. For a vector $u \in \mathbb{R}^d$, let $u^{\otimes 2} = uu^T$ denote the outer product. Given a square matrix A, $A > 0$ and $A < 0$ indicate that A is positive definite (p.d.) and negative definite (n.d.) respectively; positive semi-definite (p.s.d.) and negative semi-definite (n.s.d.) correspond to weak inequalities. For any random variable X, its law is denoted by $\mathcal{L}(X)$. Given a probability measure P and a measurable function f, let $\mathbb{E}_P f = \int f dP$. For any function f, $\mathrm{Im}(f)$ represents the image of f. For sets A and B, $A \oplus B$ represents their Minkowski sum; see Appendix A.2 for the definition.

A.2. **Random set theory.** Throughout this section, $\mathcal{F}, \mathcal{G}$ and $\mathcal{K}$ represent the family of non-empty closed, open and compact sets in $\mathbb{R}^{d_\theta}$ respectively.

Definition A.1 (Basic set operations)**.** For any $A, B \subset \mathbb{R}^{d_\theta}$ and a scalar α,

$$A \oplus B := \{a + b : \ a \in A, b \in B\}, \quad \alpha A := \{\alpha a : \ a \in A\}.$$

The operation $A \oplus B$ is called the *Minkowski sum* of sets A and B.

Definition A.2 (Hausdorff metric d_H)**.** For any $A, B \in \mathcal{K}$,

$$d_H(A, B) := \max\left\{\sup_{a\in A}\inf_{b\in B}\|a - b\|, \ \sup_{b\in B}\inf_{a\in A}\|a - b\|\right\}. \tag{A.1}$$

The Hausdorff metric d_H admits alternative representations (Molchanov (2017) Proposition D.8). The first one connects d_H to ϵ-envelop of sets; for a set A, its ϵ-envelop denoted by A^ϵ is the set

$$A^\epsilon := \{x \in \mathbb{R}^{d_\theta} : \ \|x - A\| \le \epsilon\}. \tag{A.2}$$

Then d_H is equivalent to

$$d_H(A, B) = \max\left\{\rho_H(A, B), \ \rho_H(B, A)\right\}, \quad A, B \in \mathcal{K}, \tag{A.3}$$

where $\rho_H(A, B) := \inf\{\epsilon > 0 : \ A \subset B^\epsilon\}$. The second one takes the form

$$d_H(A, B) = \sup_{x\in\mathbb{R}^{d_\theta}} \left|\|x - A\| - \|x - B\|\right|. \tag{A.4}$$

Note that this formulation works in general for $A, B \in \mathcal{F}$ as well with the possibility that $d_H(A, B) = \infty$ (Rockafellar and Wets (1998) Chapter 4).

Definition A.3 (Myopic topology on $\mathcal{K}$)**.** The *myopic* topology on $\mathcal{K}$ has a sub-base consisting of

$$\begin{aligned} \mathcal{K}^F &= \{K \in \mathcal{K} : K \cap F = \emptyset\}, \quad F \in \mathcal{F}, \\ \mathcal{K}_G &= \{K \in \mathcal{K} : K \cap G \neq \emptyset\}, \quad G \in \mathcal{G}. \end{aligned} \tag{A.5}$$

Under the condition that the underlying metric space is separable, which corresponds to the present case $\mathbb{R}^{d_\theta}$, the following two types of sets are open in the topology induced by the Hausdorff metric: the sets $\{K \in \mathcal{K} : K \subset G\}$ and $\{K \in \mathcal{K} : K \cap G \neq \emptyset\}$ for $G \in \mathcal{G}$ (e.g. Theorem D.6. in Molchanov (2017)). In particular, this implies that the Borel σ-algebra $\mathcal{B}_\mathcal{K}$ generated by the Hausdorff metric is equivalent to the one generated by the Myopic topology.

Definition A.4 (Random compact set)**.** A map $T : \mathcal{X} \mapsto \mathcal{K}$ defined on a probability space $(\mathcal{X}, \mathcal{B}_\mathcal{X}, P)$ is called a *random compact set* if

$$\{x \in \mathcal{X} : \ T(x) \cap K \neq \emptyset\} \in \mathcal{B}_\mathcal{X} \quad \text{for any } K \in \mathcal{K}.$$

More generally, a map $T : \mathcal{X} \mapsto \mathcal{F}(\mathcal{G})$ is called a *random closed (open) set*. For the present purpose, it suffices to focus on random compact sets. Note that the above definition is equivalent to defining a random compact set T as a measurable map from $(\mathcal{X}, \mathcal{B}_\mathcal{X}, P)$ to $(\mathcal{K}, \mathcal{B}_\mathcal{K})$, where $\mathcal{B}_\mathcal{K}$ is the Borel σ-field induced by the Hausdorff metric (see Molchanov (2017) section 1.3.2 and Appendix D for the detail).

Definition A.5 (Containment functional)**.** Given a random compact set T, the functional $C_T : \mathcal{K} \mapsto [0, 1]$ such that

$$C_T(K) = \mathbb{P}(T \subset K), \quad K \in \mathcal{K},$$

is called the *containment functional*.

B. Theoretical background

B.1. **Proofs for Section 3.3.**

Proof. Proposition 3.6. I start from (a). First note that since $z_c > z^*$, every $Q \in \mathcal{Q}(P)$ satisfies $Q(Z \geq z^*) = S$. On the event $\{D = 0\}$ we have $Z = \tilde{Z}$ while on the event $\{D = 1\}$ we have $Z \geq z_c$ and $\tilde{Z} = z_c \geq z^*$. This implies $\mathbb{E}_Q[Z \cdot \mathbb{1}\{Z \geq z_c\}] = P(D = 1) \cdot \mathbb{E}_Q[Z \,|\, Z \geq z_c] \in [z_c \cdot P(D = 1), \bar{z} \cdot P(D = 1)]$. Hence

$$\mathbb{E}_Q\left[Z \cdot \mathbb{1}\{Z \geq z^*\}\right] = \mathbb{E}_P\left[\tilde{Z}(1 - D) \cdot \mathbb{1}\{\tilde{Z} \geq z^*\}\right] + \mathbb{E}_Q\left[Z \cdot \mathbb{1}\{Z \geq z_c\}\right] \in [M_L, M_U],$$

so that $z_{m,Q} \in [M_L/S, M_U/S]$. Conversely, for $\lambda \in [0, 1]$, let Q_λ be such that it agrees with P on $\{Z < z_c\}$ and that, conditional on $\{Z \geq z_c\}$, it puts mass $1 - \lambda$ at z_c and λ at $\bar{z}$. Then $Q_\lambda \in \mathcal{Q}(P)$, since $\mathbb{E}_{Q_\lambda}[Z \,|\, Z \geq z_c] = z_c + \lambda(\bar{z} - z_c) \leq \bar{z}$, and $z_{m,Q_\lambda} = (M_L + \lambda(M_U - M_L))/S$, which attains every point of $[M_L/S, M_U/S]$. By Assumption 3.5(iv), $M_L/S > z^*$, and by Assumption 3.5(ii) and (iv), $M_U > M_L$ with $M_U/S < \infty$. The map $z_m \mapsto z_m/(z_m - z^*)$ is continuous and strictly decreasing on (z^*, ∞) with values in $(1, \infty)$, so it maps $[M_L/S, M_U/S]$ onto $[\pi_L, \pi_U]$ with $1 < \pi_L < \pi_U < \infty$. This shows the result in (a).

For (b), note that the result in (a) implies that $\theta \in \Theta_I$ if and only if $A_L \leq A \leq A_U$ and $\pi_L \leq \pi(\theta) \leq \pi_U$. Since $M_U - z^*S > 0$ and $M_L - z^*S > 0$, multiplying through by these quantities gives

$$\begin{aligned}
\pi(\theta) \geq \pi_L &\iff \pi(\theta)(M_U - z^*S) \geq M_U \iff \Phi_1(\theta, P) \leq 0, \\
\pi(\theta) \leq \pi_U &\iff \pi(\theta)(M_L - z^*S) \leq M_L \iff \Phi_2(\theta, P) \leq 0,
\end{aligned}$$

and $A_L \leq A \leq A_U$ if and only if $\Phi_3, \Phi_4 \leq 0$. Finally, $\pi(\theta)$ is strictly decreasing in $\bar{\tau}$ for each A, and $\pi(\theta) = \pi$ if and only if $\bar{\tau} = 1/(1 + \bar{\zeta}_{c,r}\pi\kappa(A))$, so $\pi_L \leq \pi(\theta) \leq \pi_U$ is the pair of inequalities on $\bar{\tau}$ in (3). □

B.2. **Proofs for Section 4.1.**

B.2.1. *Proofs for the main text.*

Proof. Lemma 4.2 (a). Below I consider the cases $\Theta_I \in \mathcal{K}_i$ with $i = 1, 2, 3$ separately and prove the statements of Lemma 4.2 (a-i) and (a-ii) for each case.

Case 1: $\Theta_I \in \mathcal{K}_1$. In this case, the identified set can be expressed as $\Theta_I = \{\theta_1, \cdots, \theta_J\} \subset \Theta^o$ with $J < \infty$ and $\theta_i \neq \theta_j$ for any $i \neq j$. Take $\partial\Theta_{I,h} = \Theta_I$ for any $h \in \mathbb{R}^{d_\theta} \setminus \{0\}$. Since $H_{\theta_j} = \mathbb{R}^{d_\theta} \setminus \{\theta_j\}$ in this case as $T(\theta_j) = \{0\}$, Assumption 4.1 shows $\dot{\ell}_{\theta_j,h} \neq 0$ for any $j \in [J]$, implying $I_{\theta_j,h} \neq 0$. Since the continuity of $I_{\theta_j,h}$ is trivial for any $j \in [J]$, Lemma 4.2 (a-i) holds. Lemma 4.2 (a-ii) is satisfied with $H = \mathbb{R}^{d_\theta} \setminus \{0\}$.

Case 2: $\Theta_I \in \mathcal{K}_2$. In this case, the identified set can be expressed as $\Theta_I = \cup_{j=i}^{J}[\theta_{a_j}, \theta_{b_j}] \subset \mathbb{R}$ with $J < \infty$ and $\theta_{a_j} < \theta_{b_j} < \theta_{a_{j+1}} < \theta_{b_{j+1}}$ for any $j = 1, \cdots, J-1$. Note that in this case any (direction) $h \in \mathbb{R}$ can be uniquely characterized via the multiple of some positive scalar with the unit vectors $h_1 = -1$ and $h_2 = 1$; thus it suffices to consider h_1 and h_2. Take $\partial\Theta_{I,h_1} = \{\theta_{a_j}\}_{j\in[J]}$. For any $\theta_{a_j} \in \partial\Theta_{I,h_1}$, Assumption 4.1 combined with the condition (i) in Lemma 4.2 implies $\dot{\ell}_{\theta_{a_j},h_1} \neq 0$ for $j \in [J]$. Hence $I_{\theta_{a_j},h_1} \neq 0$ for $j \in [J]$; the similar logic applies to $\partial\Theta_{I,h_2}$. Hence non-degeneracy in Lemma 4.2 (a-i) follows; the continuity is trivial. Lemma 4.2 (a-ii) holds with $H = \{h_1, h_2\}$.

Case 3: $\Theta_I \in \mathcal{K}_3$. In this case, each member in $\mathcal{K}_3$ is a compact smooth manifold with corners as defined in Section 3.2. Pick $\theta_1 \in \partial\Theta_I$. By Assumption 3.3 and the condition (i) in Lemma 4.2, there exist $h - 1 \in \mathbb{R}^{d_\theta}$ such that $\dot{\ell}_{\theta_1,h_1} h_1 \neq 0$. For this h_1 let $S_k^{h_1}$ be a partition of $\partial\Theta_I$ and let $C_{h_1} := \{\theta \in \partial\Theta_I : \dot{\ell}_{\theta,h_1} h_1 \neq 0\}$; note that $C_{h_1} \neq \emptyset$ as $\theta_1 \in C_{h_1}$. Note also that $C_{h_1} = \bigcup_{k=1}^{K}(C_{h_1} \cap S_k^{h_1})$. Now take a compact set $\partial\Theta_{I,h_1}$ such that $\text{Leb}(C_{h_1} \setminus \partial\Theta_{I,h_1}) < \epsilon$ for arbitrary $\epsilon > 0$ and that $\partial\Theta_{I,h_1} \cap S_k \subset C_{h_1} \cap S_k$ for any k. On each $\partial\Theta_{I,h_1} \cap S_k$, the map $\theta \mapsto \dot{\ell}_{\theta,h_1}$ is continuous (with respect to the operator norm) by the condition (ii) of Lemma 4.2 and $\dot{\ell}_{\theta,h_1} h_1 \neq 0$ for any $\theta \in \partial\Theta_{I,h_1} \cap S_k$ as $\partial\Theta_{I,h_1} \cap S_k \subset C_{h_1} \cap S_k$. Since the choice of θ_1 is arbitrary, this indicates that for any $\theta \in \partial\Theta_I$ one can construct $\partial\Theta_{I,h}$ such that $\theta \in \partial\Theta_{I,h}$ and that the map $\theta \to \dot{\ell}_{\theta,h}$ is continuous in $\theta \in \partial\Theta_{I,h}$. This shows the results of Lemma 4.2 (a-i) and (a-ii) by noting that the continuity of $I_{\theta,h}$ follows from that of $\dot{\ell}_{\theta,h}$.

Proof. Lemma 4.2 (b). For each $h \in \mathbb{R}^{d_\theta}$, the set $\partial\Theta_{I,h} \subset \partial\Theta_I$ is compact and $I_{\theta,h} \neq 0$ holds uniformly over $\theta \in \partial\Theta_{I,h}$. For h and any $\theta \in \partial\Theta_{I,h}$, it follows that (e.g. Theorem 7.2 in van der Vaart (1998))

$$\log \prod_{i=1}^{n} \left(\frac{p_{\theta+h/\sqrt{n}}}{p_\theta} \right)(X_i) = \frac{1}{\sqrt{n}} h^t \sum_{i=1}^{n} \dot{\ell}_{\theta,h}^t(X_i) - \frac{1}{2} h^t I_{\theta,h} h + o_{p_0}(1)$$
$$\rightsquigarrow N(-\frac{1}{2} h^t I_{\theta,h} h, h^t I_{\theta,h} h) \quad \text{as } n \to \infty.$$

Moreover $I_{\theta,h} \neq 0$ is continuous in $\theta \in \partial\Theta_{I,h}$. Now the uniform weak convergence of $\{p^n_{\theta+h/\sqrt{n}}/p^n_\theta\}$ on $\partial\Theta_{I,h}$ follows from a version of Proposition 2.1.2 in Bickel et al. (1998)

without using non-singularity of $I_{\theta,h}$, which holds exactly by following the same steps in Appendix A.9. of Bickel et al. (1998). □

Proof. Theorem 4.4. Since $p_\theta = p_0$ μ-a.s. for any $\theta \in \partial\Theta_{I,h} \subset \Theta_I$, it holds that

$$\begin{aligned}
\mathcal{L}_{p_0}\left(\frac{p^n_{\partial\Theta_{I,h}\oplus h/\sqrt{n}}}{p^n_{\partial\Theta_{I,h}}}\right) &\overset{(a)}{=} \mathcal{L}_{p_0}\left(\frac{\int_{\partial\Theta_{I,h}} p^n_{\theta+h/\sqrt{n}}\, d\lambda_h(\theta)}{p^n_0}\right) \\
&= \mathcal{L}_{p_0}\left(\int_{\partial\Theta_{I,h}} \left(\frac{p^n_{\theta+h/\sqrt{n}}}{p^n_0}\right)\, d\lambda_h(\theta)\right) \\
&\overset{(b)}{=} \mathcal{L}_{p_0}\left(\int_{\partial\Theta_{I,h}} \left(\frac{p^n_{\theta+h/\sqrt{n}}}{p^n_\theta}\right)\, d\lambda_h(\theta)\right) \\
&\overset{(c)}{\rightsquigarrow} \mathcal{L}\left(\int_{\partial\Theta_{I,h}} \frac{dN(I_{\theta,h}h, I_{\theta,h})}{dN(0, I_{\theta,h})}\, d\lambda_h(\theta)\right).
\end{aligned}$$

Note that (a) follows from $p_\theta = p_0$ μ-a.s. for any $\theta \in \partial\Theta_{I,h} \subset \Theta_I$ and $\int_{\partial\Theta_{I,h}} p_\theta\, d\lambda_h(\theta) = \int_{\partial\Theta_{I,h}} p_0\, d\lambda_h(\theta) = p_0$, where recall that λ_h is a probability measure on $\partial\Theta_{I,h}$. Part (b) again follows from $p_\theta = p_0$ μ-a.s. for any $\theta \in \partial\Theta_{I,h} \subset \Theta_I$. The weak convergence in (c) follows from the result on uniform weak convergence in Lemma 4.2 combined with the continuous mapping theorem (note that $\lambda(\partial\Theta_{I,h}) = 1 < \infty$ and that the integral operation $\lambda_h : \mathcal{C}(\partial\Theta_{I,h}) \to \mathbb{R}$ such that $\lambda_h(f) = \int_{\partial\Theta_{I,h}} f\, d\lambda_h$ for any $f \in \mathcal{C}(\partial\Theta_{I,h})$ is continuous with respect to $\|\cdot\|_\infty$). In particular, this shows that the experiment $\{p^n_{\partial\Theta_{I,h}\oplus h/\sqrt{n}} : h \in \mathbb{R}^{d_\theta}\}$ converges to the Gaussian mixture one $\{G_h = \int_{\partial\Theta_{I,h}} N(h, I^\dagger_{\theta,h})\, d\lambda_h(\theta) : h \in \mathbb{R}^{d_\theta}\}$.

The definition of local d_H-neighborhood of $\partial\Theta_{I,h}$ admits the set-valued sequential expression

$$\begin{aligned}
\mathcal{N}(\partial\Theta_{I,h}) = \{\{A_n\}_{n\in\mathbb{N}} \subset \mathcal{K} :\, &\exists A \in \mathcal{K} \text{ s.t. } \lim_{n\to\infty} \sqrt{n}\cdot d_H(A_n, \partial\Theta_{I,h}\oplus A/\sqrt{n}) = 0 \\
&\text{and } \exists C < \infty \text{ s.t. } \lim_{n\to\infty} \sqrt{n}\cdot d_H(A_n, \partial\Theta_{I,h}) \le C\}.
\end{aligned} \tag{B.1}$$

Define $d_H^{\sqrt{n}} : \mathcal{N}(\partial\Theta_{I,h}) \times \mathcal{N}(\partial\Theta_{I,h}) \to \mathbb{R}_+$ as

$$d_H^{\sqrt{n}}(\{A_n\}_{n\in\mathbb{N}}, \{A'_n\}_{n\in\mathbb{N}}) = \lim_{n\to\infty} \sqrt{n}\cdot d_H(A_n, A'_n), \tag{B.2}$$

for any $\{A_n\}_{n\in\mathbb{N}}, \{A'_n\}_{n\in\mathbb{N}} \in \mathcal{N}(\partial\Theta_{I,h})$. This $d_H^{\sqrt{n}}$ is a well-defined metric. Let $\{K_q\}_{q\in\mathbb{Q}} \subset \mathcal{K}$ be a topological base with respect to d_H; since $(\mathcal{K}, d_H)$ is second countable, at most the countable number of sets in a base. For any $\{A_n\}_{n\in\mathbb{N}} \in \mathcal{N}(\partial\Theta_{I,h})$, let $A \in \mathcal{K}$ satisfy $\sqrt{n}\cdot d_H(A_n, \partial\Theta_{I,h}\oplus A/\sqrt{n}) < \epsilon/2$ for any $n > n_1$. Take $\{B_n\}_{n\in\mathbb{N}}$ such that $B_n = \cup_{q\in Q\subset\mathbb{Q}}(K_q/\sqrt{n})$ and $\sqrt{n}\cdot d_H(B_n, \partial\Theta_{I,h}\oplus A/\sqrt{n}) < \epsilon/2$ for large enough $n > n_2$. Then $\sqrt{n}\cdot d_H(A_n, B_n) \le \sqrt{n}\cdot d_H(A_n, \partial\Theta_{I,h}\oplus A/\sqrt{n}) + \sqrt{n}\cdot d_H(\partial\Theta_{I,h}\oplus A/\sqrt{n}, B_n) < \epsilon$ for any $n > n_1 \vee n_2$, indicating $d_H^{\sqrt{n}}(\{A_n\}_{n\in\mathbb{N}}, \{B_n\}_{n\in\mathbb{N}}) = 0$. This shows that the space $(\mathcal{N}(\partial\Theta_{I,h}), d_H^{\sqrt{n}})$ is separable.

I next show that the space $(\mathcal{N}(\partial\Theta_{I,h}), d_H^{\sqrt{n}})$ is complete. Let $\{A_n^{(k)}\}_{n,k\in\mathbb{N}} \subset \mathcal{N}(\partial\Theta_{I,h})$ be a Caushy sequence in $k \in \mathbb{N}$, i.e. for any $\epsilon > 0$ there exists $k_0 \in \mathbb{N}$ such that $d_H^{\sqrt{n}}(\{A_n^{(k)}\}_{n\in\mathbb{N}}, \{A_n^{(k')}\}_{n\in\mathbb{N}}) < \epsilon$ for any $k, k' > k_0$. For such $k, k' > k_0$ denote by $A^{(k)}$, $A^{(k')} \in \mathcal{K}$ the sets satisfying $\lim_{n\to\infty} \sqrt{n} \cdot d_H(A_n^{(k)}, \partial\Theta_{I,h} \oplus A^{(k)}/\sqrt{n}) = 0$ and $\lim_{n\to\infty} \sqrt{n} \cdot d_H(A_n^{(k')}, \partial\Theta_{I,h} \oplus A^{(k')}/\sqrt{n}) = 0$ respectively. By the triangle inequality, the set-valued sequence $\{\partial\Theta_{I,h} \oplus A^{(k)}/\sqrt{n}\}_{k\in\mathbb{N}}$ is Cauchy with respect to $d_H^{\sqrt{n}}$. By completeness of $(\mathcal{K}, d_H)$, for any $\epsilon > 0$ there exists $A^* \in \mathcal{K}$ such that $d_H^{\sqrt{n}}(\partial\Theta_{I,h} \oplus A^{(k)}/\sqrt{n}, \partial\Theta_{I,h} \oplus A^*/\sqrt{n}) < \epsilon$ for any $k > k_0$. Now take $\{A_n^*\}_{n\in\mathbb{N}}$ such that $A_n^* = \partial\Theta_{I,h} \oplus A^*/\sqrt{n}$ for any $n \in \mathbb{N}$. It is trivial that $\lim_{n\to\infty} \sqrt{n} \cdot d_H(A_n^*, \partial\Theta_{I,h} \oplus A^*/\sqrt{n}) = 0$ and that

$$\begin{aligned}\lim_{n\to\infty} \sqrt{n} \cdot d_H(A_n^*, \partial\Theta_{I,h}) &= \lim_{n\to\infty} \sqrt{n} \cdot d_H(\partial\Theta_{I,h} \oplus A^*/\sqrt{n}, \partial\Theta_{I,h}) \\ &\leq \sup\{\|a\| : a \in A^*\} < C\end{aligned}$$

for some $C < \infty$ as $A^* \in \mathcal{K}$ is compact; this shows $\{A_n^*\}_{n\in\mathbb{N}} \in \mathcal{N}(\partial\Theta_{I,h})$ and the completeness of the space $(\mathcal{N}(\partial\Theta_{I,h}), d_H^{\sqrt{n}})$ follows. The analysis above indicates that the space $(\mathcal{N}(\partial\Theta_{I,h}), d_H^{\sqrt{n}})$ is Polish.

The condition (ii) in Theorem 4.4 implies that the law $\mathcal{L}(T_n^h)$ stays on a Borel measurable space $(\mathcal{N}(\partial\Theta_{I,h}), \mathcal{B}_{\mathcal{N}})$ with the Borel σ-field $\mathcal{B}_{\mathcal{N}}$ induced by $d_H^{\sqrt{n}}$. Note that $\{\partial\Theta_{I,h} \oplus h/\sqrt{n}\}_{n\in\mathbb{N}} \in \mathcal{N}(\partial\Theta_{I,h})$. As a probability measure on $(\mathbb{R}_+, \mathcal{B}_{\mathbb{R}_+})$ the asymptotic law $S_h \overset{a}{\sim} \sqrt{n} \cdot d_H(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n})$ is tight due to the Ulam's theorem. Since the map $d_H^{\sqrt{n}}(\cdot, \{\partial\Theta_{I,h} \oplus h/\sqrt{n}\}_{n\in\mathbb{N}}) : \mathcal{N}(\partial\Theta_{I,h}) \mapsto \mathbb{R}_+$ is uniformly continuous, the tightness of the asymptotic law S_h on $(\mathbb{R}_+, \mathcal{B}_{\mathbb{R}_+})$ indicates the asymptotic tightness of $\mathcal{L}(T_n^h)$ on $(\mathcal{N}(\partial\Theta_{I,h}), d_H^{\sqrt{n}})$. It then follows from the Prokhorov's theorem that there exists a subsequence $\{n_k\}_{k\in\mathbb{N}} \subset \{n\}_{n\in\mathbb{N}}$ such that $\{\mathcal{L}(T_{n_k}^h)\}_{k\in\mathbb{N}}$ weakly converges to some probability measure $\mathcal{L}_{\mathcal{N}}^h$ on $(\mathcal{N}(\partial\Theta_{I,h}), \mathcal{B}_{\mathcal{N}})$, i.e. $\mathcal{L}(T_{n_k}^h) \rightsquigarrow \mathcal{L}_{\mathcal{N}}^h$, and that $\mathcal{L}(\sqrt{n_k} \cdot d_H(T_{n_k}^h, \partial\Theta_{I,h} \oplus h/\sqrt{n_k})) \rightsquigarrow S_h$.

Now Lemma B.3 applied to the space $(\mathcal{N}(\partial\Theta_{I,h}), \mathcal{B}_{\mathcal{N}})$ and Theorem 3.1 in van der Vaart (1991) show that there exists a randomized statisic $T^h : \mathcal{X} \times \mathcal{U} \mapsto (\mathcal{N}(\partial\Theta_{I,h}), \mathcal{B}_{\mathcal{N}})$ with $X \sim G_h$ and $U \perp\!\!\!\perp X$ such that the limiting law $\mathcal{L}_{\mathcal{N}}^h \overset{a}{\sim} T_{n_k}^h$ on $(\mathcal{N}(\partial\Theta_{I,h}), \mathcal{B}_{\mathcal{N}})$ is equivalent to the law $\mathcal{L}(T^h(X, U))$. Due to the continuity of $d_H^{\sqrt{n}}$ and by the continuous mapping theorem, it holds that

$$\begin{aligned}\mathcal{L}\left(\sqrt{n} \cdot d_H(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n})\right) &\overset{a}{=} \mathcal{L}\left(\sqrt{n_k} \cdot d_H(T_{n_k}^h, \partial\Theta_{I,h} \oplus h/\sqrt{n_k})\right) \\ &\overset{a}{=} \mathcal{L}\left(d_H^{\sqrt{n_k}}(T^h, \{\partial\Theta_{I,h} \oplus h/\sqrt{n_k}\}_{k\in\mathbb{N}})\right).\end{aligned} \tag{B.3}$$

Finally note that for any $\{A_n\}_{n\in\mathbb{N}}, \{B_n\}_{n\in\mathbb{N}} \in \mathcal{N}(\partial\Theta_{I,h})$ it holds that

$$\begin{aligned}d_H^{\sqrt{n}}(\{A_n\}_{n\in\mathbb{N}}, \{B_n\}_{n\in\mathbb{N}}) &= \lim_{n\to\infty} \sqrt{n} \cdot d_H(A_n, B_n) \\ &= \lim_{n\to\infty} \sqrt{n} \cdot d_H(\partial\Theta_{I,h} \oplus A/\sqrt{n}, \partial\Theta_{I,h} \oplus B/\sqrt{n}) \\ &= d_H(\partial\Theta_{I,h} \oplus A, \partial\Theta_{I,h} \oplus B),\end{aligned}$$

where A and B are such that $\lim_{n\to\infty}\sqrt{n}\cdot d_H(A_n, \partial\Theta_{I,h}\oplus A/\sqrt{n}) = 0$ and $\lim_{n\to\infty}\sqrt{n}\cdot d_H(B_n, \Theta_{I,h}\oplus B/\sqrt{n}) = 0$ respectively. Note that $B_{n_k} = \partial\Theta_{I,h}\oplus h/\sqrt{n_k}$ has $B = \{h\}$. Since the law $\mathcal{L}(T^h)$ on $(\mathcal{N}(\partial\Theta_{I,h}), \mathcal{B}_{\mathcal{N}}))$ uniquely determines the law on $\{\partial\Theta_{I,h}\oplus K : K\in\mathcal{K}$ and $\exists\{A_n\}_{n\in\mathbb{N}}\in\mathcal{N}(\partial\Theta_{I,h})$ s.t. $\lim_{n\to\infty}\sqrt{n}\cdot d_H(A_n, \partial\Theta_{I,h}\oplus K/\sqrt{n}) = 0\}$, having $A_{n_k} = T^h$ and $B_{n_k} = \partial\Theta_{I,h}\oplus h/\sqrt{n_k}$ in the above relation gives

$$\mathcal{L}\left(d_H^{\sqrt{n_k}}(T^h, \{\partial\Theta_{I,h}\oplus h/\sqrt{n_k}\}_{k\in\mathbb{N}})\right) = \mathcal{L}\left(d_H(T^h, \partial\Theta_{I,h}\oplus h)\right). \tag{B.4}$$

Combining (B.3) and (B.4) provides the result. □

B.2.2. *Auxiliary lemmas.* I here state specific notions and results used for the main results in section 4.1. Notice that the discussion and results below are all stated in terms of the space $(\mathcal{K}_\Theta, \mathcal{B}_{\mathcal{K}_\Theta})$, but these are applicable to the Polish space $(\mathcal{K}, \mathcal{B}_{\mathcal{K}})$ or $(\mathcal{N}(\partial\Theta_{I,h}), \mathcal{B}_{\mathcal{N}})$ in (B.1) and (B.2) as well.

Definition B.1 (Markov kernel)**.** A *Markov kernel* $\delta : \mathcal{K}_\Theta\times\mathcal{X}\mapsto\mathbb{R}$ defined in $(\mathcal{K}_\Theta\times\mathcal{X}, \mathcal{B}_{\mathcal{K}_\Theta}\otimes\mathcal{B}_{\mathcal{X}})$ is such that

(i). for any $x\in\mathcal{X}$, $\delta(.,x)$ is a probability measure on $(\mathcal{K}_\Theta, \mathcal{B}_{\mathcal{K}_\Theta})$,
(ii). for any $A\in\mathcal{B}_{\mathcal{K}_\Theta}$, the map $\delta(A,.) : \mathcal{X}\mapsto\mathbb{R}$ is measurable.

The condition (i) in Definition B.1 requires that, given $x\in\mathcal{X}$, the functional $\delta(.,x)$ on $(\mathcal{K}_\Theta, \mathcal{B}_{\mathcal{K}_\Theta})$ is σ-additive. Not all functional defined on a family of sets satisfy this requirement; a canonical example is the capacity functional[20]. This condition is, however, satisfied by the containment functional in Definition A.5; the containment functional additionally characterizes a law of a random compact set X as shown below.

Lemma B.2 (Unique characterization of random compact set T via C_T)**.** *Let $T : \mathcal{X}\mapsto\mathcal{K}_\Theta$ is a random compact set. Then its law on $(\mathcal{K}_\Theta, \mathcal{B}_{\mathcal{K}_\Theta})$ is uniquely determined by the containment functional C_T.*

Proof. Since the underlying space is $\mathbb{R}^{d_\theta}$ and $\mathbb{P}(T\in\mathcal{K}_\Theta) = 1$, the present setting satisfies the conditions of Theorem 1.3.12 in Molchanov (2017). □

My analysis builds on this result and connects a random set-valued statistic T in Theorem 4.4 to the Markov kernel $\delta : \mathcal{K}_\Theta\times\mathcal{X}\mapsto[0,1]$, which is viewed as the containment functional conditionally on $x\in\mathcal{X}$. This connection is formally stated as the following.

Lemma B.3 (Set-indexed Markov kernel δ)**.** *The following relations hold for a random statistic T and a Markov kernel δ:*

[20] Given a random closed set X, the functional $T_X : \mathcal{K}\mapsto[0,1]$ such that

$$T_X(K) = \mathbb{P}(X\cap K\neq\emptyset), \quad K\in\mathcal{K},$$

is called the *capacity functional* of X. The capacity functional is σ-subadditive.

(i). *For any Markov kernel* $\delta : \mathcal{K}_\Theta \times \mathcal{X} \mapsto \mathbb{R}$, *there exists a measurable map* $T : (\mathcal{X} \times [0,1], \mathcal{B}_\mathcal{X} \otimes \mathcal{B}_{[0,1]}) \mapsto (\mathcal{K}_\Theta, \mathcal{B}_{\mathcal{K}_\Theta})$ *such that*

$$\delta(A, x) = \mathbb{P}(T(x, U) \in A), \quad \textit{for any } A \in \mathcal{B}_{\mathcal{K}_\Theta}, \tag{B.5}$$

where U *is a random variable taking values in* $([0,1], \mathcal{B}_{[0,1]})$.

(ii). *Given a map* T *above, the equation (B.5) defines a Markov kernel* $\delta : \mathcal{X} \times \mathcal{K}_\Theta \mapsto \mathbb{R}$.

Proof. This is the result in Lemma 9.1 in van der Vaart (1991) with $(\mathcal{K}_\Theta, \mathcal{B}_{\mathcal{K}_\Theta})$ the target space, noting that the metric space $(\mathcal{K}_\Theta, d_H)$ is complete and separable. □

B.3. **Proofs for Section 4.2.**

B.3.1. *Proofs for the main text.*

Proof. Theorem 4.5. First notice that $\partial\Theta_{I,h} \subset \partial\Theta_I$ for any $h \in \mathbb{R}^{d_\theta}$ implies

$$d_H(T_n, \partial\Theta_I) \geq \sup_{h\in\mathbb{R}^{d_\theta}} d_H(T_n^h, \partial\Theta_{I,h}). \tag{B.6}$$

I show this by contradiction. Suppose the statement is not true; then there exist some $v \in \mathbb{R}^{d_\theta}$ such that $d_H(T_n, \partial\Theta_I) < d_H(T_n^v, \partial\Theta_{I,v})$. This implies

$$\begin{aligned} d_H(T_n, \partial\Theta_I) &= \max\{\rho_H(T_n, \partial\Theta_I),\ \rho_H(\partial\Theta_I, T_n)\} \\ &< \max\{\rho_H(T_n^v, \partial\Theta_{I,v}),\ \rho_H(\partial\Theta_{I,v}, T_n^v)\} = d_H(T_n^v, \partial\Theta_{I,v}), \end{aligned}$$

as in (A.3). This cannot occur as $\rho_H(T_n^v, \partial\Theta_{I,v}) \leq \rho_H(T_n, \partial\Theta_I)$ and $\rho_H(\partial\Theta_{I,v}, T_n^v) \leq \rho_H(\partial\Theta_I, T_n)$ due to $\partial\Theta_{I,v} \subset \partial\Theta_I$ and $\mathrm{Im}(T_n^v) \subset \mathrm{Im}(T_n)$, both of which hold for any $v \in \mathbb{R}^{d_\theta}$. Since $\partial\Theta_{I,h} \oplus h/\sqrt{n} \subset \partial\Theta_I \oplus h/\sqrt{n}$ for any $h \in \mathbb{R}^{d_\theta}$ and any $n \in \mathbb{N}$, the inequality (B.6) holds as well with the shifted sets $\partial\Theta_I \oplus h/\sqrt{n}$ and $\partial\Theta_{I,h} \oplus h/\sqrt{n}$ for any $n \in \mathbb{N}$. Thus the left hand side of the expression in Theorem 4.5 is bounded below by

$$\begin{aligned} &\sup_{D\in\mathcal{D}} \liminf_{n\to\infty} \sup_{h\in D} \mathbb{E}_{P_h^n}\left[l \circ \left(\sqrt{n}\cdot d_H(T_n, \partial\Theta_I \oplus h/\sqrt{n})\right)\right] \\ &\quad \geq \sup_{D\in\mathcal{D}} \liminf_{n\to\infty} \sup_{h\in D} \mathbb{E}_{P_h^n}\left[l \circ \left(\sqrt{n}\cdot d_H(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n})\right)\right] =: R_s(l). \end{aligned} \tag{B.7}$$

Now for a given subconvex loss function $l : \mathbb{R} \to \mathbb{R}$, let

$$l_m(t) = 2^{-m} \sum_{i=1}^{m\cdot 2^m} (1 - \mathbb{1}_{C_{im}}(t)) \quad m \in \mathbb{Z}_+, \tag{B.8}$$

where $C_{im} := \{s \in \mathbb{R} : l(s) \leq i \cdot 2^{-m}\}$. The sequence of functions $\{l_m\}_{m\in\mathbb{Z}_+}$ satisfies $0 \leq l_m \leq l$, $l_m \to l$ as $m \to \infty$, and that l_m is bounded, subconvex for each $m \in \mathbb{Z}_+$, and continuous Leb-a.e.

Regarding $R_s(l)$ defined in (B.7), note that there exist two subsequences $\{n'(D)\} \subset \{n(D)\} \subset \mathbb{N}$ depending on D such that

(B.9)

$$\begin{aligned} R_s(l) &\geq R_s(l_m) \\ &= \sup_{D\in\mathcal{D}} \liminf_{n\to\infty} \sup_{h\in D} \mathbb{E}_{P_h^n}\left[l_m \circ \left(\sqrt{n}\cdot d_H(T_n^h, \partial\Theta_{I,h} \oplus h/\sqrt{n})\right)\right] \\ &= \sup_{D\in\mathcal{D}} \lim_{n(D)\to\infty} \sup_{h\in D} \mathbb{E}_{P_h^{n(D)}}\left[l_m \circ \left(\sqrt{n(D)}\cdot d_H(T_{n(D)}^h, \partial\Theta_{I,h} \oplus h/\sqrt{n(D)})\right)\right] \\ &\overset{(a)}{\geq} \sup_{D\in\mathcal{D}} \liminf_{n'(D)\to\infty} \sup_{h\in D} \mathbb{E}_{P_h^{n'(D)}}\left[l_m \circ \left(\sqrt{n'(D)}\cdot d_H(T_{n'(D)}^h, \partial\Theta_{I,h} \oplus h/\sqrt{n'(D)})\right)\right] \\ &\overset{(b)}{\geq} \sup_{D\in\mathcal{D}} \sup_{h\in D} \mathbb{E}_{G_h\otimes\mathcal{L}(U)}\left[l_m \circ \left(d_H(T^h, \partial\Theta_{I,h} \oplus h)\right)\right] \\ &\overset{(c)}{=} \sup_{h\in\mathbb{R}^{d_\theta}} \mathbb{E}_{G_h}\left[l_m \circ \left(d_H(T^h, \partial\Theta_{I,h} \oplus h)\right)\right] \end{aligned}$$

where (a) follows by the existence of the subsubsequence $\{n'(D)\} \subset \{n(D)\}$ that assures the weak convergence of Theorem 4.4 and (b) follows from Theorem 4.4 and Portmanteau theorem, noting that l_m is bounded and continuous Leb-a.e., and (c) is due to the construction of $\mathcal{D}$ and the independence $G_h \perp\!\!\!\perp \mathcal{L}(U)$.

Now I show

$$\mathbb{E}_{G_h}\left[l_m \circ \left(d_H(T^h, \partial\Theta_{I,h} \oplus h)\right)\right] \geq \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^d_\theta} l_m \circ \|x\| \, dN(0, I^\dagger_{\theta,h})(x) \, d\lambda_h(\theta). \tag{B.10}$$

Recall $G_h = \int_{\partial\Theta_{I,h}} N(h, I^\dagger_{\theta,h}) \, d\lambda_h(\theta)$. Let $X \sim G_h$ and $X_{\theta,h} \sim N(h, I^\dagger_{\theta,h})$; then $X = \int_{\partial\Theta_{I,h}} X_{\theta,h} \, d\lambda_h(\theta)$. Note that $\partial\Theta_{I,h} \oplus h = \{\theta + h : \theta \in \partial\Theta_{I,h}\}$. For $T^h(X) = \partial\Theta_{I,h} \oplus X = \{\theta + X_{\theta,h} : \theta \in \partial\Theta_{I,h}\}$ it follows that

$$\begin{aligned} d_H\left(T^h(X), \partial\Theta_{I,h} \oplus h\right) &= d_H\left(\partial\Theta_{I,h} \oplus X, \partial\Theta_{I,h} \oplus h\right) \\ &\geq \int_{\partial\Theta_{I,h}} \|(\theta + X_{\theta,h}) - (\theta + h)\| \, d\lambda_h(\theta), \end{aligned}$$

by the property of d_H combined with λ_h being a probability measure. This observation, together with Fubini's theorem and $X_{\theta,h} - h \sim N(0, I^\dagger_{\theta,h})$, indicates

$$\mathbb{E}_{G_h}\left[d_H(\partial\Theta_{I,h} \oplus X, \partial\Theta_{I,h} \oplus h)\right] \geq \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} \|x\| \, dN(0, I^\dagger_{\theta,h})(x) \, d\lambda_h(\theta).$$

As in (B.8), one can construct a sequence $\{l_{m,k}\}_{k\in\mathbb{Z}_+}$ such that $0 \leq l_{m,k} \leq l_m$, $l_{m,k} \to l_m$ as $k \to \infty$, and that $l_{m,k}$ is piecewise constant, bounded, subconvex for each $k \in \mathbb{Z}_+$, and continuous Leb-a.e. For such $\{l_{m,k}\}_{k\in\mathbb{Z}_+}$, it holds that

$$\mathbb{E}_{G_h}\left[l_{m,k} \circ d_H\left(\partial\Theta_{I,h} \oplus X, \partial\Theta_{I,h} \oplus h\right)\right] \geq \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l_{m,k} \circ \|x\| \, dN(0, I^\dagger_{\theta,k})(x) \, d\lambda_h(\theta).$$

Accordingly, letting $k \to \infty$ and applying the monotone convergence theorem to the above inequality show

$$\mathbb{E}_{G_h}\left[l_m \circ (d_H(\partial\Theta_{I,h} \oplus X, \partial\Theta_{I,h} \oplus h))\right] \geq \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l_m \circ \|x\| \, dN(0, I^\dagger_{\theta,h})(x) \, d\lambda_h(\theta).$$

Together with (B.9), it follows that

$$R_s(l) \geq R_s(l_m) \geq \sup_{h\in\mathbb{R}^{d_\theta}} \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l_m \circ \|x\| \, dN(0, I^\dagger_{\theta,h})(x) \, d\lambda_h(\theta),$$

for any $m \in \mathbb{Z}_+$. With (B.7), the final result follows by taking $m \to \infty$ and using the monotone convergence theorem, combined with

$$\begin{aligned} &\lim_{m\to\infty} \sup_{h\in\mathbb{R}^{d_\theta}} \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l_m \circ \|x\| \, dN(0, I^\dagger_{\theta,h})(x) \, d\lambda_h(\theta) \\ &\qquad = \sup_{h\in\mathbb{R}^{d_\theta}} \int_{\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l \circ \|x\| \, dN(0, I^\dagger_{\theta,h})(x) \, d\lambda_h(\theta), \end{aligned}$$

which again holds from the monotone convergence theorem. □

Proof. Corollary 4.6. In (B.7) in the proof of Theorem 4.5, it holds that

$$\begin{aligned}
&\sup_{D\in\mathcal{D}} \liminf_{n\to\infty} \sup_{h\in D} \sup_{\lambda\in\Lambda} \mathbb{E}_{P_h^n}\left[l\circ\left(\sqrt{n}\cdot d_H(T_n,\partial\Theta_I\oplus h/\sqrt{n})\right)\right] \\
&\quad\geq \sup_{D\in\mathcal{D}} \liminf_{n\to\infty} \sup_{h\in D} \sup_{\lambda_h\in\Lambda_h} \mathbb{E}_{P_h^n}\left[l\circ\left(\sqrt{n}\cdot d_H(T_n,\partial\Theta_I\oplus h/\sqrt{n})\right)\right] \\
&\quad\geq \sup_{D\in\mathcal{D}} \liminf_{n\to\infty} \sup_{h\in D} \sup_{\lambda_h\in\Lambda_h} \mathbb{E}_{P_h^n}\left[l\circ\left(\sqrt{n}\cdot d_H(T_n^h,\partial\Theta_{I,h}\oplus h/\sqrt{n})\right)\right] =: R_s^*(l),
\end{aligned}$$

where the first inequality is due to the fact that $\Lambda_h \subset \Lambda$. Recall that the analysis in (B.9) depends on a fixed $\lambda_h \in \Lambda_h$. By making its dependence on λ_h explicit by denoting $R_s(l,\lambda_h)$ and $R_s(l_m,\lambda_h)$ in (B.9), it holds that

$$R_s^*(l) \geq R_s(l,\lambda_h) \geq R_s(l_m,\lambda_h) \geq \sup_{h\in\mathbb{R}^{d_\theta}} \mathbb{E}_{G_h}\left[l_m\circ\left(T^h,\partial\Theta_{I,h}\oplus h\right)\right],$$

where the first inequality follows from the construction of $R_s^*(l)$ and the last two are from the proof in Theorem 4.5. Note that from (B.10) in the proof of Theorem 4.5 it holds that

$$\sup_{\lambda_h\in\Lambda_h} \mathbb{E}_{G_h}\left[l_m\circ\left(d_H(T^h,\partial\Theta_{I,h}\oplus h)\right)\right] \geq \sup_{\theta\in\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l_m\circ\|x\|\, dN(0,I_{\theta,h}^\dagger)(x),$$

which follows from the fact that λ is a probability measure. Combining the argument yields

$$R_s^*(l) \geq \sup_{h\in\mathbb{R}^{d_\theta}} \sup_{\theta\in\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l_m\circ\|x\|\, dN(0,I_{\theta,h}^\dagger)(x).$$

Taking $m\to\infty$ and applying the monotone convergence theorem shows the result. □

B.4. **Proofs for Section 4.3.**

B.4.1. *Proofs for the main text.*

Proof. Lemma 4.10. Fix $h \in \mathbb{R}^{d_\theta}$ and let $N_h = \{\theta \in \partial\Theta_I(\Phi, P_{\delta_0}) : h \in H_\theta\}$. Note that $N_h \neq \emptyset$ by Assumption 4.9 (ii). Take a non-empty compact set $\partial\Theta_{I,h}(\Phi, P_{\delta_0}) \subset N_h$.

Consider the quantity $(\partial/\partial\delta^t)\Phi|_{\delta=\delta_0} = \nabla q(\mathbb{E}_{\delta_0} m)\cdot M$, where $M := (m(\cdot, x_1), \cdots, m(\cdot, x_s))$ is the $d_m \times d_s$-matrix and the quantity is understood as the pathwise derivative restricted to any path $\{(1-\eta_n)\delta_0 + \eta_n \delta_n\}_{n\in\mathbb{N}} \subset H_{P_{\delta_0},\theta}$ for $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$ in the sense of Assumption 4.9 (iv). Note that Assumption 4.9 (iv) indicates that $\text{rank}(\nabla q(\mathbb{E}_{\delta_0} m)\cdot M) = d_q$. Note also that Assumption 4.9 (iv) implies $\mathbb{E}_{\delta_0}(\partial/\partial\delta^t)\Phi|_{\delta=\delta_0} = (\partial/\partial\delta^t)\Phi|_{\delta=\delta_0}\cdot\delta_0 = \nabla q(\mathbb{E}_{\delta_0} m)\cdot\mathbb{E}_{\delta_0} m = 0 \in \mathbb{R}^{d_q}$ for any $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$ and P_{δ_0} (note that the quantity $\nabla q(\mathbb{E}_\delta m)\cdot M$ depends on $\theta \in \Theta$ and on $\delta \in \Delta^o_{d_s-1}$).

For any $d_1 \times d_2$-matrix A, denote by $\text{span}(\text{row}[A])$ the linear space spanned by $\text{row}[A] := \{A_1, \cdots, A_{d_1}\}$ where $A_i = (A_{i,1}, \cdots, A_{i,d_2})^t \in \mathbb{R}^{d_2}$ with $i \in [d_1]$ is the i-th row of A. I now show $\iota \notin \text{span}(\text{row}[\nabla q(\mathbb{E}_{\delta_0} m)\cdot M])$ for any $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$ and any P_{δ_0} under Assumption 4.9 (iv) (again note that the quantity $\nabla q(\mathbb{E}_\delta m)\cdot M$ depends on $\theta \in \Theta$ and on $\delta \in \Delta^o_{d_s-1}$) , i.e. ι is not in the row space of $\nabla q(\mathbb{E}_{\delta_0} m)\cdot M$, where $\iota = (1, \cdots, 1)^t \in \mathbb{R}^{d_s}$. Suppose this is not true; there then exists a non-zero vector $z \in \mathbb{R}^{d_q}$ such that $z^t \cdot \nabla q(\mathbb{E}_{\delta_0} m)\cdot M = \iota^t$. Multiplying both sides by δ_0 from the right yields $z^t \cdot \nabla q(\mathbb{E}_{\delta_0} m)\cdot\mathbb{E}_{\delta_0} m = \iota^t\delta_0 = 1$, which contradicts the fact that Assumption 4.9 (iv) implies $(\partial/\delta^t)\Phi|_{\delta=\delta_0}\cdot\delta_0 = 0$ for any $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$. This observation implies that there exists a $d_s - (d_q+1) \times d_s$-matrix Ξ_θ, which depends on $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$, such that $\text{rank}(\Xi_\theta) = d_s - (d_q+1)$ and $\text{row}[\Xi_\theta] \not\subset \text{span}(\text{row}[\nabla q(\mathbb{E}_{\delta_0} m)\cdot M],\ \iota)$ for any $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$.

Define $\xi_0 := \Xi_\theta\delta_0 \in \mathbb{R}^{d_\xi}$ with $d_\xi := \dim(\xi) = d_s - (d_q+1)$. Let $U_\theta \subset \Theta \subset \mathbb{R}^{d_\theta}$, $U_\delta \subset \mathbb{R}^{d_s}$ and $U_\xi \subset \mathbb{R}^{d_\xi}$ be open neighborhoods of $\Theta_I(\Phi, P_{\delta_0}), \delta_0$ and ξ_0 respectively. Define maps $g_1 : U_\theta \times U_\delta \to \mathbb{R}^{d_q}$, $g_2 : U_\delta \mapsto \mathbb{R}$ and $g_3 : U_\theta \times U_\xi \times U_\delta \mapsto \mathbb{R}^{d_\xi}$ as

$$
\begin{aligned}
&g_1(\theta, \delta) = q\left(\int_{\mathcal{X}} m(\theta, x)\, dP_\delta(x)\right) = q\left(\sum\nolimits_{s\in[d_s]} m(\theta, x_s)\cdot\delta(s)\right), \\
&g_2(\delta) = \iota^t\delta - 1, \\
&g_3(\theta, \xi, \delta) = \xi - \Xi(\theta)\delta,
\end{aligned}
\tag{B.11}
$$

where $\Xi : \mathbb{R}^{d_\theta} \to \mathbb{R}^{d_\xi \times d_s}$ is a map such that $\Xi(\theta) = \Xi_{\theta^*}$ with $\theta^* = \arg\min\{\|\theta - \theta'\| : \theta' \in \partial\Theta_I(\Phi, P_{\delta_0})\}$. Let $g^t = (g_1^t, g_2^t, g_3^t) : \mathbb{R}^{d_\gamma} \times \mathbb{R}^{d_s} \mapsto \mathbb{R}^{d_s}$, where $\gamma^t = (\theta^t, \xi^t) \in \mathbb{R}^{d_\gamma}$. By the construction, it holds that $d_\gamma = d_\theta + d_\xi$.

Now define

$$
\partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0}) := \left\{\gamma \in U_\theta \times U_\xi \subset \mathbb{R}^{d_\gamma} : g(\gamma, \delta_0) = 0,\ \pi_{d_\theta}\circ\gamma \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})\right\},
\tag{B.12}
$$

where $h_\gamma := (h^t, h_\xi^t)^t \in \mathbb{R}^{d_\gamma}$. Note that $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ implies $\theta = \pi_{d_\theta}\circ\gamma \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$; indeed for any $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$ there exists Ξ_θ by the analysis above, indicating the existence of ξ such that $\xi = \Xi_\theta\delta_0$ and hence $g_3(\theta, \xi, \delta_0) = 0$ for any $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$. This yields the relation $\partial\Theta_{I,h}(\Phi, P_{\delta_0}) = \pi_{d_\theta}\circ\partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$. Finally

let

$$\partial\Gamma_I(\Phi, P_{\delta_0}) = \bigcup\nolimits_{h_\gamma \in \mathbb{R}^{d_\gamma}} \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0}). \tag{B.13}$$

Clearly for any $\gamma \in \partial\Gamma_I(\Phi, P_{\delta_0})$ it holds that $\pi_{d_\theta} \circ \gamma \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$; on the other hand, by Assumption 4.9 (ii) for any $\theta \in \partial\Theta_I(\Phi, P_{\delta_0})$ there exists $h_* \in \mathbb{R}^{d_\theta}$ such that $\theta \in \partial\Theta_{I,h_*}(\Phi, P_{\delta_0})$, implying the existence of $h_{\gamma *} = (h_*^t, h_{\xi *}^t)^t \in \mathbb{R}^{d_\gamma}$ with $\partial\Theta_{I,h_*}(\Phi, P_{\delta_0}) = \pi_{d_\theta} \circ \partial\Gamma_{I,h_{\gamma *}}(\Phi, P_{\delta_0})$. This establishes the relation $\pi_{d_\theta} \circ \partial\Gamma_I(\Phi, P_{\delta_0}) = \partial\Theta_I(\Phi, P_{\delta_0})$.

Now for $\delta_0 \in U_\delta$ and any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$, let $\nabla_{h_\gamma} g_i$ and $\nabla_\delta g_i$ represent the directional derivatives with respect to $h_\gamma \in \mathbb{R}^{d_\gamma}$ and the restricted pathwise derivatives, respectively, for $i = 1, 2, 3$. Note that the existence of the standard derivatives implies those of the directional and the restricted pathwise derivatives; thus $\nabla_{h_\gamma} g_i$ and $\nabla_\delta g_i$ are well defined for any $i = 1, 2, 3$. One then obtains

$$\nabla_{h_\gamma} g(\gamma, \delta_0) = \begin{pmatrix} \nabla_{h_\gamma} g_1(\theta, \delta_0) \\ \nabla_{h_\gamma} g_2(\delta_0) \\ \nabla_{h_\gamma} g_3(\theta, \xi, \delta_0) \end{pmatrix} = \begin{pmatrix} \phi_{\theta,h} & 0_{d_q \times d_\xi} \\ 0_{1 \times d_\theta} & 0_{1 \times d_\xi} \\ 0_{d_\xi \times d_\theta} & \mathbf{I}_{d_\xi \times d_\xi} \end{pmatrix} \in \mathbb{R}^{d_s \times d_\gamma},$$

$$\nabla_\delta g(\gamma, \delta_0) = \begin{pmatrix} \nabla_\delta g_1(\theta, \delta_0) \\ \nabla_\delta g_2(\delta_0) \\ \nabla_\delta g_3(\theta, \xi, \delta_0) \end{pmatrix} = \begin{pmatrix} (\partial/\partial\delta)\Phi(\theta, P_\delta)|_{\delta=\delta_0} \\ \iota^t \\ -\Xi_\theta \end{pmatrix} \in \mathbb{R}^{d_s \times d_s},$$

for any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$. Note that $\nabla_\delta g(\gamma, \delta_0)$ is invertible for any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ by the construction of Ξ_θ.

It is clear that $g(\gamma, \delta_0) = 0 \in \mathbb{R}^{d_s}$ for $\delta_0 \in U_\delta$ and any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$. I now apply an implicit function theorem via directional derivatives in Lemma B.8 to conclude that for any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ there exists a map $\upsilon : \mathbb{R}^{d_\gamma} \to \mathbb{R}^{d_s}$ such that $\delta_0 = \upsilon(\gamma)$ and that it has directional derivative represented by

$$\nabla_{h_\gamma} \upsilon(\gamma) = -\left[\nabla_\delta g(\gamma, \delta)\right]^{-1} \cdot \nabla_{h_\gamma} g(\gamma, \delta). \tag{B.14}$$

The local Fisher information of $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ with respect to the direction $h_\gamma \in \mathbb{R}^{d_\gamma}$ is expressed as the $d_\gamma \times d_\gamma$-matrix

$$\begin{aligned} I_{\gamma,h_\gamma} &= \mathbb{E}_{\delta_0}\left[\nabla_{h_\gamma} \upsilon(\gamma)\right]^{\otimes 2} \\ &= \left[\nabla_{h_\gamma} g(\gamma, \delta_0)\right]^t \cdot \left[\mathbb{E}_{\delta_0}[\nabla_\delta g(\gamma, \delta_0)]^{\otimes 2}\right]^{-1} \cdot \left[\nabla_{h_\gamma} g(\gamma, \delta_0)\right], \end{aligned}$$

which is non-singular. From I_{γ,h_γ} one obtains

$$\begin{aligned} I_{\theta,h} &= \left(\mathbf{I}_{d_\theta \times d_\theta}, 0_{d_\theta \times d_\xi}\right) \cdot I_{\gamma,h_\gamma} \cdot \left(\mathbf{I}_{d_\theta \times d_\theta}, 0_{d_\theta \times d_\xi}\right)^t \\ &= \phi_{\theta,h}^t \cdot \left[\mathbb{E}_{\delta_0}[(\partial/\partial\delta)\Phi(\theta, P_\delta)|_{\delta=\delta_0}]^{\otimes 2}\right]^{-1} \cdot \phi_{\theta,h} \end{aligned}$$

for any $\theta \in \partial\Theta_{I,h}(\Phi, P_{\delta_0})$ and $h = \pi_{d_\theta} \circ h_\gamma$.

What remain to prove are Lemma 4.10 (a) and (b). For any $\gamma \in \partial\Gamma_I(\Phi, P_{\delta_0})$, set a probability measure P_γ on $\{x_s\}_{s \in [d_s]} \subset (\mathcal{X}, \mathcal{B}_\mathcal{X})$ such that $P_\gamma(x_s) = \upsilon_s(\gamma)$ for any $s \in [d_s]$, where υ_s is understood as a component of $\upsilon = (\upsilon_1, \cdots \upsilon_{d_s})$. Note that, by the argument above, for any $\gamma \in \partial\Gamma_I(\Phi, P_{\delta_0})$ it holds that $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ for some $h_\gamma \in \mathbb{R}^{d_\gamma}$ and a map $\upsilon : \mathbb{R}^{d_\gamma} \to \mathbb{R}^{d_s}$, whose functional form may depend on each $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$, such that $\upsilon(\gamma) = \delta_0$. This implies that for any $\gamma_i, \gamma_j \in \partial\Gamma_I(\Phi, P_{\delta_0})$

it holds that $P_{\gamma_i}(x_s) = \delta_0(s) = P_{\gamma_j}(x_s)$. Note also that both $g(\gamma, \delta_0) = 0$ and $P_{\delta_0} = P_\gamma$ hold for $\gamma \in \mathbb{R}^{d_\gamma}$ with $\pi_{\mathrm{d}_\theta} \circ \gamma \in \Theta_I^o(\Phi, P_{\delta_0})$. Hence

$$\Gamma_I(\Phi, P_{\delta_0}) = \{\gamma \in \mathbb{R}^{d_\gamma} : \pi_{d_\theta} \circ \gamma \in \Theta_I(\Phi, P_{\delta_0})\}$$

is well-defined with $\partial\Gamma_I(\Phi, P_{\delta_0}) \subset \Gamma_I(\Phi, P_{\delta_0})$. This shows Lemma 4.10 (b), except for the construction of Γ^o that also needs to satisfy Lemma 4.10 (a). To this end, let

$$H_\gamma := \{h_\gamma \in \mathbb{R}^{d_\gamma} : \pi_{d_\theta} \circ h_\gamma \in H_\theta\}. \tag{B.15}$$

Note that for any $h_\gamma \in \mathbb{R}^{d_\gamma}$ and any $\gamma \in \partial\Theta_{I,h_\gamma}(\Phi, P_{\delta_0})$, continuity of $\gamma \mapsto P_\gamma$ implies the existence of $B_\gamma(\epsilon_\gamma)$ such that $\mathrm{supp}(P_{\gamma'}) = \{x_s\}_{s\in[d_s]}$ for any $\gamma' \in B_\gamma(\epsilon_\gamma) \cap H_\gamma$. Letting

$$\Gamma^o = \left(\bigcup\nolimits_{\gamma\in\partial\Gamma_I(\Phi,P_{\delta_0})} B_\gamma(\epsilon_\gamma)\right) \cup \Gamma_I(\Phi, P_{\delta_0})$$

then indicates $\Gamma_I(\Phi, P_{\delta_0}) \subset \Gamma^o$ and also proves Lemma 4.10 (a). □

Proof. Proposition 4.11. To derive the asymptotic lower bound, I apply Corollary 4.6 to the parametric model $\{P_\gamma : \gamma \in \Gamma^o \subset \mathbb{R}^{d_\gamma}\}$ obtained in Lemma 4.10. To this end, I show that the conditions required in Theorem 4.5 are all satisfied. Assumption 3.3 is clear. Hence I verify below Assumption 4.1 and the conditions (i) (ii) in Lemma 4.2.

I start from Assumption 4.1. Recall the parametric model $\{P_\gamma : \gamma \in \Gamma^o \subset \mathbb{R}^{d_\gamma}\}$ constructed in Lemma 4.10, where $P_\gamma(x_s) = \upsilon_s(\gamma)$ for any $s \in [d_s]$. Consider a map $\gamma \mapsto \sqrt{P_\gamma}(x_s) = \sqrt{\upsilon_s(\gamma)}$ for any $s \in [d_s]$. For any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$, recall H_γ in (B.15). Then for any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ and $h_\gamma \in H_\gamma$ it holds that

$$\nabla_{h_\gamma}\sqrt{P_\gamma}(x_s) = \frac{1}{2}\left(\upsilon_s(\gamma)\right)^{-1/2} \cdot \nabla_{h_\gamma}\upsilon_s(\gamma), \quad s \in [d_s], \tag{B.16}$$

where $((\nabla_{h_\gamma}\upsilon_1(\gamma))^t, \cdots, (\nabla_{h_\gamma}\upsilon_{d_s}(\gamma))^t)^t = \nabla_{h_\gamma}\upsilon(\gamma)$ is as in (B.14); note that this chain rule is possible as the map $\gamma \mapsto \sqrt{P_\gamma}$ is the composite of the fully differentiable map and the directionally differentiable map in the sense of Hadamard (Proposition 3.6 in Shapiro (1990)). This Hadamard directional derivative (B.16) at $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ for $h_\gamma \in H_\gamma$ implies that, with $\mu = \sharp$,

$$\begin{aligned}
&\int_{\mathcal{X}} \left(\frac{\sqrt{P_{\gamma+t_n h_{\gamma,n}}} - \sqrt{P_\gamma}}{t_n} - \left(\nabla_{h_\gamma}\sqrt{P_\gamma}\right)\cdot h_\gamma\right)^2 d\mu \\
&\quad = \sum\nolimits_{s\in[d_s]} \left(\frac{\sqrt{P_{\gamma+t_n h_{\gamma,n}}}(x_s) - \sqrt{P_\gamma}(x_s)}{t_n} - \left(\nabla_{h_\gamma}\sqrt{P_\gamma}\right)(x_s)\cdot h_\gamma\right)^2 \\
&\quad = \sum\nolimits_{s\in[d_s]} \left(\frac{\sqrt{\upsilon_s(\gamma+t_n h_{\gamma,n})} - \sqrt{\upsilon_s(\gamma)}}{t_n} - \frac{1}{2}\left(\frac{\nabla_{h_\gamma}\upsilon_s(\gamma)}{\upsilon_s(\gamma)}\right)\cdot h_\gamma \cdot \sqrt{\upsilon_s(\gamma)}\right)^2 = o(1),
\end{aligned}$$

for any $\{h_{\gamma,n}\}_{n\in\mathbb{N}} \subset H_\gamma$ and any $\{t_n\}_{n\in\mathbb{N}}$ such that $h_{\gamma,n} \to h_\gamma \in H_\gamma$ and $t_n \to 0$ as $n \to \infty$. Hence Assumption 4.1 is satisfied with the directional score defined as $\dot{\ell}_{\gamma,h_\gamma} = ((\nabla_{h_\gamma}\upsilon_1(\gamma)/\upsilon_1(\gamma))^t, \cdots, (\nabla_{h_\gamma}\upsilon_{d_s}(\gamma)/\upsilon_{d_s}(\gamma))^t)^t$.

The condition (i) in Lemma 4.2 follows from the construction of H_γ above combined with Assumption 4.9 (ii). The condition (ii) is clear from Assumption 4.9 (iii), the construction of γ and $\dot{\ell}_{\gamma,h_\gamma} = ((\nabla_{h_\gamma}\upsilon_1(\gamma)/\upsilon_1(\gamma))^t, \cdots, (\nabla_{h_\gamma}\upsilon_{d_s}(\gamma)/\upsilon_{d_s}(\gamma))^t)^t$ with $\nabla_{h_\gamma}\upsilon(\gamma)$ in (B.14).

The result for $\partial\Theta_I = \pi_{d_\theta} \circ \partial\Gamma_I$ and $T_n = \pi_{d_\theta} \circ \tilde{T}_n$ is immediate from their projection structure. □

Proof. Theorem 4.12. Fix $P_0 \in \mathcal{P}$ and $\epsilon > 0$. It follows from the definition of $\mathcal{N}_{P_0}$ that $\left|\int_{\mathcal{X}} f\, dP - \int_{\mathcal{X}} f\, dP_0\right| < \epsilon$ for any $P \in \mathcal{N}_{P_0}$ and any $f : \mathcal{X} \mapsto \mathbb{R}^d$ with $\|f\|_{L^1(P_0)} < \infty$ and $d < \infty$. Since the map $\theta \mapsto m(\theta, \cdot)$ satisfies $\int_{\mathcal{X}} \|m(\theta, x)\|\, dP_0(x) < \infty$ for any θ in a neighborhood of $\Theta_I(\Phi, P_0)$, denoting by $\|f\|_{\text{op}}$ operator norm of a map f it holds that

$$\begin{aligned}\|\Phi(\theta, P) - \Phi(\theta, P_0)\| &= \left\| q\left(\int_{\mathcal{X}} m(\theta, x)\, dP(x)\right) - q\left(\int_{\mathcal{X}} m(\theta, x)\, dP_0(x)\right)\right\| \\ &\leq \left\|\frac{\partial}{\partial P}\Phi(\theta, P)|_{P=P_0}\right\|_{\text{op}} \cdot \int_{\mathcal{X}} \|m(\theta, x)\|\ (dP(x) - dP_0(x)) < \epsilon\end{aligned}$$

for any $P \in \mathcal{P}$ such that $\int_{\mathcal{X}} \|m(\theta, x)\|\ (dP(x) - dP_0(x)) < \epsilon/\|(\partial/\partial P)\Phi(\theta, P)|_{P=P_0}\|_{\text{op}}$.

By Lemma B.9 there exists a multinomial probability measure P_{δ_0} with $\text{supp}(P_{\delta_0}) \subset \mathcal{X}$ and $|\text{supp}(P_{\delta_0})| = d_s < \infty$ such that $|\int_{\mathcal{X}} f\ (dP_{\delta_0} - dP_0)| < \epsilon$ for any f with $\|f\|_{L^1(P_0)} < \infty$, i.e. $P_{\delta_0} \in \mathcal{N}_{P_0}$. The above analysis indicates $\|\Phi(\theta, P_{\delta_0}) - \Phi(\theta, P_0)\| < \epsilon$ for any θ in a neighborhood of $\Theta_I(\Phi, P_0)$. I now apply Lemma 4.10 to construct a family of parametric models $\{P_\gamma : \gamma \in \Gamma^o \subset \mathbb{R}^{d_\gamma}\}$ such that $P_\gamma = P_{\delta_0}$ for any $\gamma \in \Gamma_I(\Phi, P_{\delta_0}) \subset \Gamma^o$; it follows that for any γ' in a neighborhood of $\Gamma_I(\Phi, P_{\delta_0})$, $\gamma_0 \in \arg\min\{\|\gamma' - \gamma\| : \gamma \in \Gamma_I(\Phi, P_{\delta_0})\}$ and

$$\|\Phi(\theta, P_{\gamma'}) - \Phi(\theta, P_0)\| \leq \|\Phi(\theta, P_{\gamma'}) - \Phi(\theta, P_{\gamma_0})\| + \|\Phi(\theta, P_{\gamma_0}) - \Phi(\theta, P_0)\| < 2\epsilon,$$

as the first term is arbitrarily small in the same analysis as above and the second term is $\|\Phi(\theta, P_{\gamma_0}) - \Phi(\theta, P_0)\| = \|\Phi(\theta, P_{\delta_0}) - \Phi(\theta, P_0)\| < \epsilon$.

Now for $T_n = \pi_{d_\theta} \circ \tilde{T}_n : \mathcal{X}^n \mapsto \mathcal{K}_\Theta$ it holds that

$$\begin{aligned}&\sup_{P\in\mathcal{P}_0} \sup_{\lambda\in\Lambda} \mathbb{E}_P\left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I(\Phi, P))\right)\right] \\ &\qquad \geq \sup_{h_\gamma \in D_\gamma} \sup_{\lambda\in\Lambda} \mathbb{E}_{P_{h_\gamma}}\left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I \oplus (\pi_{d_\theta} \circ h_\gamma)/\sqrt{n})\right)\right]\end{aligned}$$

for large enough $n \in \mathbb{N}$, where the inequality follows from $\{P_\gamma : \gamma \in \Gamma^o \subset \mathbb{R}^{d_\gamma}\} \subset \mathcal{P}_0$ and $P_{h_\gamma} = P_{\partial\Gamma_{I,h_\gamma} \oplus h_\gamma/\sqrt{n}} = P_{\gamma'}$ for $\gamma' \in \Gamma^o \cap H_\gamma$. Since this inequality holds for any $D_\gamma \in \mathcal{D}_\gamma$, combining the inequality with Proposition 4.11 yields

$$\begin{aligned}&\liminf_{n\to\infty} \sup_{P\in\mathcal{P}_0} \sup_{\lambda\in\Lambda} \mathbb{E}_P\left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I(\Phi, P))\right)\right] \\ &\qquad \geq \sup_{D_\gamma\in\mathcal{D}_\gamma} \liminf_{n\to\infty} \sup_{h_\gamma\in D_\gamma} \sup_{\lambda\in\Lambda} \mathbb{E}_{P_{h_\gamma}}\left[l \circ \left(\sqrt{n} \cdot d_H(T_n, \partial\Theta_I \oplus (\pi_{d_\theta} \circ h_\gamma)/\sqrt{n})\right)\right] \\ &\qquad \geq \sup_{h\in\mathbb{R}^{d_\theta}} \sup_{\theta\in\partial\Theta_{I,h}} \int_{\mathbb{R}^{d_\theta}} l \circ \|x\|\, dN(0, I^\dagger_{\theta,h})(x),\end{aligned}$$

which is the desired result. □

B.4.2. *Auxiliary lemmas.* One of the intermediate lemmas used to establish the main results in section 4.3 is a general implicit function theorem via directional derivatives; see also Dontchev and Rockafellar (2009) for a textbook treatment. Below I modify the results in Robinson (1991) (section 3) to the present setting. A key difference is that the results in Robinson (1991) are based on the *Bouligand derivatives* (or the *B-derivatives*) while my argument relies on the Hadamard directional derivatives with a certain tangential set (e.g. $H_\theta \subset \mathbb{R}^{d_\theta}$ for the map $\theta \mapsto \Phi(\theta, \cdot)$ in the present context). See Shapiro (1990) for their relationship.

As in the standard implicit function theorem, a basic approach is to establish bijectivity of the targeted map. Let (V, d_V) and (W, d_W) be metric spaces. For any map $f : V \mapsto W$, let

$$\delta(f, V) := \inf_{v_1 \neq v_2 \in V} \frac{d_W\left(f(v_1), f(v_2)\right)}{d_V(v_1, v_2)}. \tag{B.17}$$

Notice that f is injective if and only if $\delta(f, V) > 0$.

Lemma B.4 (A modification of the Banach Perturbation Lemma)**.** *Let $(V, \|\cdot\|_V)$ be a Banach space, $(W, \|\cdot\|_W)$ a normed space, $U_V \subset V$ an open set, and $f : U_V \mapsto W$ maps such that $f(v_0) = w_0$ for $v_0 \in U_V$. Suppose that there exist closed sets $H_{v_0} \subset V$ and $H_{w_0} \subset W$, with $v_0 \in H_{v_0}$ and $w_0 \in H_{w_0}$ respectively, and a map $\tilde{f} : U_V \cap H_{v_0} \mapsto H_{w_0}$ such that*

(i). $\overline{B_\alpha(w_0)} \cap H_{w_0} \subset f(U_V \cap H_{v_0})$ *for some* $\alpha > 0$,
(ii). $0 \leq \eta < \delta := \delta(f, U_V \cap H_{v_0})$, *where η is the Lipschitz molulus of $\tilde{f}$ on $U_V \cap H_{v_0}$,*
(iii). $\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0} \subset U_V \cap H_{v_0}$,
(iv). $\kappa := (1 - \eta\delta^{-1})\alpha - \|\tilde{f}(v_0)\|_W > 0$.

Then it holds that

$$\overline{B_\kappa(w_0)} \cap H_{w_0} \subset (f + \tilde{f})\left(\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}\right), \quad \delta(f + \tilde{f}, U_V \cap H_{v_0}) \geq \delta - \eta > 0.$$

Proof. This follows from an extension of Lemma 3.1 in Robinson (1991) by restricting neighborhoods of v_0 and w_0 to those intersected via closed sets H_{v_0} and H_{w_0} respectively. For $w \in \overline{B_\kappa(w_0)} \cap H_{w_0}$ and $v \in \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$, let $T_w(v) = f^{-1}(w - \tilde{f}(v))$. Since

$$\begin{aligned}\|w - \tilde{f}(v) - w_0\|_W &\leq \|w - w_0\|_W + \|\tilde{f}(v) - \tilde{f}(v_0)\|_W + \|\tilde{f}(v_0)\|_W \\ &\leq \kappa + \eta(\delta^{-1}\alpha) + \|\tilde{f}(v_0)\|_W = \alpha,\end{aligned}$$

and $\tilde{f}(v) \in \overline{B_\alpha(w_0)} \cap H_{w_0}$ for any $v \in \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$, the set $T_w(v)$ is well-defined, non-empty as well as singleton (due to $\delta > 0$). Since

$$\|T_w(v) - v_0\|_V = \|f^{-1}(w - \tilde{f}(v)) - f^{-1}(w_0)\|_V \leq \delta^{-1}\alpha,$$

this further implies $T_w : \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0} \mapsto \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$. For $v_1, v_2 \in \overline{B_{\delta^{-1}\alpha}(v_0)}$, it holds that

$$\|T_w(v_1) - T_w(v_2)\|_V = \|f^{-1}(w - \tilde{f}(v_1)) - f^{-1}(w - \tilde{f}(v_2))\|_V \leq \delta^{-1}\eta\|v_1 - v_2\|_V.$$

This, combined with the assumption $\delta^{-1}\eta < 1$, implies that the map T_w is a strong contraction mapping. Note that the set $\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$ is complete as a closed subset of $(V, \|\cdot\|_V)$. Then the Banach fixed-point theorem (or the contraction mapping theorem) shows that T_w has a unique fixed point $v(w) \in \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$. The analysis above shows that for any $w \in \overline{B_\kappa(w_0)} \cap H_{w_0}$ it holds that $(f + \tilde{f})(v(w)) = w$, indicating the first result of Lemma B.4. Finally, it holds that

$$\begin{aligned}
\delta(f + \tilde{f}, U_V \cap H_{v_0}) &= \inf_{v_1 \neq v_2 \in U_V \cap H_{v_0}} \frac{\|[f(v_1) - f(v_2)] + [\tilde{f}(v_1) - \tilde{f}(v_2)]\|_W}{\|v_1 - v_2\|_V} \\
&\geq \delta(f, U_V \cap H_{v_0}) - \sup_{v_1 \neq v_2 \in U_V \cap H_{v_0}} \frac{\|\tilde{f}(v_1) - \tilde{f}(v_2)\|_W}{\|v_1 - v_2\|_V} \\
&\geq \delta - \eta > 0,
\end{aligned}$$

which establishes the second result of Lemma B.4. □

Now I state a result in Lemma B.5 which shows the existence of an implicitly defined map. To this end, I use the following notion. Let $f_1 : V \times W \mapsto Z$, where $(V, \|\cdot\|_V), (W, \|\cdot\|_W)$, and $(Z, \|\cdot\|_Z)$ are normed spaces, and let $f_2 : V \mapsto Z$. Then f_2 is said to *approximate* f_1 with respect to v at (v_0, w_0), denoted by $f_2 \overset{v}{\sim} f_1$, if

$$f_1(v, w_0) - f_2(v) = o(\|v - v_0\|_V).$$

f_2 is said to *strongly approximate* f_1 with respect to v at (v_0, w_0), denoted by $f_2 \overset{v}{\approx} f_1$ at (v_0, w_0), if for any $\epsilon > 0$ there exist neighborhoods N_{v_0} of v_0 and N_{w_0} of w_0 such that

$$\|[f_1(v, w) - f_2(v)] - [f_1(v', w) - f_2(v')]\|_Z \leq \epsilon \|v - v'\|_V,$$

for any $v, v' \in N_{v_0}$ and any $w \in N_{w_0}$. If the inequality holds for $v, v' \in N_{v_0} \cap H_V$ for some $H_V \subset V$, then $f_2 \overset{v}{\approx} f_1$ is said to hold with respect to $v \in H_V$ at (v_0, w_0).

Lemma B.5 (Existence of an implicit function)**.** *Let $(V, \|\cdot\|_V)$ be a Banach space, $(W, \|\cdot\|_W)$ and $(Z, \|\cdot\|_Z)$ normed spaces, $U_V \subset V$ and $U_W \subset W$ open sets, and $F : U_V \times U_W \mapsto Z$ such that $F(v_0, w_0) = 0$ for $v_0 \in U_V$ and $w_0 \in U_W$. Suppose that there exist closed and connected sets $H_{v_0} \subset V$, $H_{z_0} \subset Z$ and a closed cone $H_{w_0} \subset W$, with $v_0 \in H_{v_0}$, $w_0 \in H_{w_0}$ and $0 \in H_{z_0}$ respectively, and a map $f : U_V \cap H_{v_0} \mapsto Z$ such that*

(i). *$f \overset{v}{\approx} F$ with respect to $v \in U_V \cap H_{v_0}$ at (v_0, w_0),*
(ii). *for any $v \in U_V \cap H_{v_0}$, the map $w \mapsto F(v, w)$ is Lipschitz with modulus ϕ,*
(iii). *$0 \in H_{z_0} \subset f(U_V \cap H_{v_0})$,*
(iv). *$\delta(f, U_V \cap H_{v_0}) =: \delta > 0$.*

Then for any $\lambda > \delta^{-1}\phi$, there exist sets U_{v_0} and U_{w_0}, with $v_0 \in U_{v_0}$ and $w_0 \in U_{w_0}$ respectively, and a map $v : U_{w_0} \mapsto U_{v_0}$ such that

(a). *$v(w_0) = v_0$,*
(b). *the map v is Lipschitz with modulus λ,*
(c). *for any $w \in U_{w_0}$, $v(w)$ is the unique solution of $F(v, w) = 0$ on U_{v_0}.*

Proof. Fix $\lambda > \delta^{-1}\phi$. Pick $\alpha, \kappa, \eta > 0$ and a neighborhood $N_{w_0} \subset W$ with $w_0 \in N_{w_0}$ such that

(1). $\eta < \delta$ and $\phi(\delta - \eta)^{-1} \leq \lambda$,
(2). for any $v, v' \in \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$ and any $w \in N_{w_0}$, it holds that $\|[F(v,w) - f(v)] - [F(v',w) - f(v')]\|_Z \leq \eta \|v - v'\|_V$,
(3). $\overline{B_\alpha(0)} \cap H_{z_0} \subset f(U_V \cap H_{v_0})$,
(4). for any $w \in N_{w_0}$, it holds that $0 < \kappa \leq (1 - \eta\delta^{-1})\alpha - \|F(v_0, w)\|_Z$.

For any $w \in N_{w_0}$, let $\tilde{f}_w : \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0} \mapsto Z$ such that $\tilde{f}_w(v) = F(v,w) - f(v)$. The point (2) implies that $\tilde{f}_w$ is Lipschitz on $\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$ with modulus η. I now verify the conditions from (i) to (iv) in Lemma B.4 to apply its results to the map $(f + \tilde{f}_w)(\cdot) = F(\cdot, w)$ for $w \in N_w$; note that $\tilde{f}_w$ here corresponds to $\tilde{f}$ in Lemma B.4.

For Lemma B.4 (i), this immediately follows from the point in (3). Lemma B.4 (ii) is satisfied from the construction of $\tilde{f}_w$ with $w \in N_{w_0}$ and from the point (1), where note that $\tilde{f}_w$ is Lipschitz with modulus $\eta < \delta$. For Lemma B.4 (iii), the relation $\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0} \subset U_V \cap H_{v_0}$ follows from the observation that the point (3) and the condition (iv) in Lemma B.5 indicates $\overline{B_{\delta^{-1}\alpha}(v_0)} \subset U_V$. Lemma B.4 (iv) is clear from the point (4) because

$$(1 - \eta\delta^{-1})\alpha - \|\tilde{f}_w(v_0)\|_Z = (1 - \eta\delta^{-1})\alpha - \|F(v_0, w)\|_Z =: \kappa > 0,$$

where $f(v_0) = 0$ is used.

Hence it follows from Lemma B.4 that

$$\begin{aligned} &\delta(F(\cdot, w), \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}) \geq \delta - \eta > 0, \\ &\overline{B_\kappa(0)} \cap H_{z_0} \subset F(\cdot, w)(\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}). \end{aligned} \tag{B.18}$$

In particular, these results imply that for any $w \in N_{w_0}$ the map $v \mapsto F(v,w)$ is injective on $\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$ and surjective onto $\overline{B_\kappa(0)} \cap H_{z_0}$. Now for any $z \in \overline{B_\kappa(0)} \cap H_{z_0}$ and any $w \in N_{w_0}$ let $v_w(z)$ be the unique point in $\overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$ such that $F(v_w(z), w) = z$. Define $v : N_{w_0} \mapsto \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$ such that $v(w) = v_w(0)$ and take $U_{v_0} = \overline{B_{\delta^{-1}\alpha}(v_0)} \cap H_{v_0}$, $U_{w_0} = N_{w_0} \cap H_{w_0}$. The result (a) and (c) are clear. For the result (b), note that for any $w_1, w_2 \in U_{w_0}$,

$$\begin{aligned} (\delta - \eta)\|v(w_1) - v(w_2)\|_V &\leq \|F(v(w_1), w_1) - F(v(w_2), w_1)\|_Z \\ &= \|F(v(w_2), w_2) - F(v(w_2), w_1)\|_Z \\ &\leq \phi\|w_1 - w_2\|_W, \end{aligned}$$

where the first inequality is from (B.18), the second equality is from $F(v(w_1), w_1) = 0 = F(v(w_2), w_2)$, and the last inequality is from the Lipschitz condition in Lemma B.5 (ii). This shows the map $v : U_{w_0} \mapsto U_{v_0}$ is Lipschitz with modulus $\phi(\delta - \eta)^{-1} \leq \lambda$ and the result (b) follows. □

Lemma B.6 (Approximation of an implicit function)**.** *Let v_0, w_0, F and f be as in Lemma B.5. Suppose that a map $r : U_W \cap H_{w_0} \mapsto Z$ is continuous with $r(w_0) = 0$.*

Define $v_r(w)$ *for* $w \in U_W \cap H_{w_0}$ *by*

$$f(v_r(w)) + r(w) = 0.$$

If $r \overset{w}{\sim} F$ *at* (v_0, w_0)*, then* $v_r \sim v$ *for* $w \in U_W \cap H_{w_0}$ *at* w_0*.*

Proof. Fix $\epsilon > 0$. Since $f \overset{v}{\approx} F$ with respect to $v \in U_V \cap H_{v_0}$ and $r \overset{w}{\sim} F$, both of which hold at (v_0, w_0), there exist $N_{w_0} \subset W$ and $N_{v_0} \subset V$ with $w_0 \in N_{w_0}$ and $v_0 \in N_{v_0}$ such that, for any $y \in N_{w_0}$ there exist $v(w), v_r(w) \in N_{v_0}$ satisfying

$$\begin{aligned} &\|[F(v(w), w) - f(v(w))] - [F(v_0, w) - f(v_0)]\|_Z \leq \epsilon \|v(w) - v_0\|_V, \\ &\|F(v_0, w) - r(w)\|_Z \leq \epsilon \|w - w_0\|_W. \end{aligned} \tag{B.19}$$

It follows that

$$\begin{aligned} 0 &= F(v(w), w) - [f(v_r(w)) + r(w)] \\ &= [F(v(w), w) - f(v(w))] - [F(v_0, w) - f(v_0)] \\ &\quad + [f(v(w)) - f(v_r(w))] + [F(v_0, w) - r(w)]. \end{aligned}$$

By combining the equation above with (B.19), noting that $f^{-1} : Z \mapsto U_V \cap H_{v_0}$ is locally Lipschitz around $0 \in H_{z_0}$ with modulus δ^{-1}, one obtains

$$\begin{aligned} \|v(w) - v_r(w)\|_V &\leq \delta^{-1} \|f(v(w)) - f(v_r(w))\| \\ &\leq \delta^{-1} \left(\epsilon \|v(w) - v_0\|_V + \epsilon \|w - w_0\|_W\right) \\ &\leq \epsilon \delta^{-1} (\lambda + 1) \|w - w_0\|_W, \end{aligned}$$

which implies $v_r \sim v$. □

When there exists a set $H_V \subset V$ and a map $F : V \times W \mapsto Z$ has a partial (Hadamard) directional derivative with respect to v in $h_v \in H_V$ at (v_0, w_0), denoted by $\nabla_{h_v} F(v_0, w_0)$, $\nabla_{h_v} F(v_0, w_0)$ is then said to be *strong* in H_V if $F(v_0, w_0) + \nabla_{h_v} F(v_0, w_0)(v - v_0) \overset{v}{\approx} F(v, w)$ with respect to $v \in H_V$ at (v_0, w_0).

Lemma B.7 (Directional derivative of an implicit funciton)**.** *Let the same conditions as in Lemma B.5 hold except for* f*. Assume further that* F *has partial directional derivatives (in the sense of Hadamard) in* $h_v \in H_{v_0}$ *and in* $h_w \in H_{w_0}$ *at* (v_0, w_0) *with the following properties:*

(i). $\nabla_{h_v} F(v_0, w_0)$ *is strong in* H_{v_0}*,*
(ii). $0 \in H_{z_0} \subset \nabla_{h_v} F(v_0, w_0)(U_V \cap H_{v_0} - v_0)$*,*
(iii). $\delta(\nabla_{h_v} F(v_0, w_0), U_V \cap H_{v_0} - v_0) =: \delta > 0$*.*

Then for any $\lambda > \delta^{-1}\phi$*, there exist sets* U_{v_0} *and* U_{w_0}*, with* $v_0 \in U_{v_0}$ *and* $w_0 \in U_{w_0}$ *respectively, and a map* $v : U_{w_0} \mapsto U_{v_0}$ *such that Lemma B.5 hold. Moreover, the map* v *is Hadamard directionally differentiable at* w_0 *in* $h_w \in U_{w_0}$ *with*

$$\nabla_{h_w} v(w_0)(w) = -\left[\nabla_{h_v} F(v_0, w_0)\right]^{-1} \cdot \nabla_{h_w} F(v_0, w_0)(w)$$

for any $w \in U_{w_0}$*.*

Proof. Let $f(v) = \nabla_{h_v} F(v_0, w_0)(v - v_0)$ for $v \in U_V \cap H_{v_0}$. The condition (i) indicates $f \overset{v}{\approx} F$ with respect to $v \in U_V \cap H_{v_0}$ at (v_0, w_0). Thus, together with the conditions (ii) and (iii) in Lemma B.7, the conditions from (i) to (iv) in Lemma B.5 are all satisfied; accordingly the results of Lemma B.5 follows.

Now let $r(w) = \nabla_{h_w} F(v_0, w_0)(w - w_0)$ for any $h_w \in H_{w_0}$ and any $w \in U_W \cap H_{w_0}$; by the existence of partial Hadamard directional derivative, it holds that $r \overset{w}{\sim} F$ with respect to $w \in U_W \cap H_{w_0}$ at (v_0, w_0). Lemma B.6 then implies that $v_r \sim v$ for $w \in U_W \cap H_{w_0}$ at w_0, where v_r is defined through

$$f(v_r(w)) + r(w) = \nabla_{h_v} F(v_0, w_0)(v_r(w) - v_0) + \nabla_{h_w} F(v_0, w_0)(w - w_0) = 0.$$

As the map $f : U_V \cap H_{v_0} \mapsto H_{z_0}$ is bijective due to the condition (ii) and (iii) in Lemma B.7, the above equality determines $v_r(w)$ for any $w \in U_{w_0} \subset U_W \cap H_{w_0}$. Thus

$$v_r(w) - v_0 = -\left[\nabla_{h_v} F(v_0, w_0)\right]^{-1} \cdot \nabla_{h_w} F(v_0, w_0)(w - w_0).$$

Note that the map $w \mapsto \left[\nabla_{h_v} F(v_0, w_0)\right]^{-1} \cdot \nabla_{h_w} F(v_0, w_0)(w - w_0)$ in U_{w_0} is positively homogeneous of degree one. Note also that $v_r \sim v$ for $w \in U_{w_0}$ at w_0 implies

$$v(w_0) + \left[\nabla_{h_v} F(v_0, w_0)\right]^{-1} \cdot \nabla_{h_w} F(v_0, w_0)(w - w_0) \sim v(w)$$

at w_0. Thus the map $v : U_{w_0} \mapsto U_{v_0}$ is Hadamard directionally differentiable at w_0 with

$$\nabla_{h_w} v(w_0)(w') = -\left[\nabla_{h_v} F(v_0, w_0)\right]^{-1} \cdot \nabla_{h_w} F(v_0, w_0)(w'),$$

where $w' = \alpha(w - w_0) \in U_{w_0}$ for some $\alpha > 0$; note that such w' exists as $U_{w_0} = N_{w_0} \cap H_{w_0}$ where H_{w_0} is a closed cone and that the above equality is valid due to the map $\left[\nabla_{h_v} F(v_0, w_0)\right]^{-1} \cdot \nabla_{h_w} F(v_0, w_0)$ positively homogeneous of degree one. □

Lemma B.8 (Implicit function theorem via directional derivatives)**.** *Suppose that Assumptions 3.3 and 4.9 hold. For $g = (g_1^t, g_2^t, g_3^t) : \mathbb{R}^{d_\gamma} \times \mathbb{R}^{d_s} \to \mathbb{R}^{d_s}$ defined in (B.11), $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0}) \subset \mathbb{R}^{d_\gamma}$ as in (B.12) and $\delta_0 \in \mathbb{R}^{d_s}$, there exist sets U_γ, U_δ and a map $\upsilon : \mathbb{R}^{d_\gamma} \to \mathbb{R}^{d_s}$ such that*

(a). *$\upsilon(\gamma) = \delta_0$ for $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$,*
(b). *υ is Lipschitz on U_γ,*
(c). *for each $\gamma \in U_\gamma$, $\upsilon(\gamma)$ is the unique solution in U_δ of $g(\gamma, \delta) = 0$,*
(d). *υ admits the Hadamard directional derivative at $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ for $h_\gamma \in H_\gamma$, represented as*

$$\nabla_{h_\gamma} \upsilon(\gamma) = -\left[\nabla_\delta g(\gamma, \delta)\right]^{-1} \cdot \nabla_{h_\gamma} g(\gamma, \delta).$$

Proof. I apply Lemma B.7 to the map $g : \mathbb{R}^{d_\gamma} \times \mathbb{R}^{d_s} \to \mathbb{R}^{d_s}$; the properties from (a) to (d) in Lemma B.8 are then the direct consequences of Lemma B.7 and Lemma B.5. To this end, I verify the conditions in Lemma B.5 (except for f in Lemma B.5) and in Lemma B.7.

Conditions in Lemma B.5. Fix $h_\gamma \in \mathbb{R}^{d_\gamma}$. Note that the present context corresponds to $V = Z = \mathbb{R}^{d_s}$ and $W = \mathbb{R}^{d_\theta}$ with the Euclidean norms, and $F = g$. Let $U_V = U_\delta$ and $U_W = U_\theta \times U_\xi$ be open neighborhoods of δ_0 and $\partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$. Let H_{v_0} be a

compact and connected subset of Δ_{d_s-1} such that $\delta_0 \in H_{v_0}$ and $P_\delta \in H_{P_{\delta_0},\theta}$ for $\delta \in H_{v_0}$; $H_{w_0} = \gamma \oplus H_\gamma$ for any $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$, where H_γ is as defined in (B.15); H_{z_0} a closed and connected subset of $\{g(\gamma,\delta) : \gamma \in H_{w_0}, \delta \in H_{v_0}\} \subset \mathbb{R}^{d_s}$. Note that $g(\gamma, \delta_0) = 0$ for $\gamma \in H_{w_0}$ with $\gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})$ and $\delta_0 \in H_{v_0}$.

For the condition (ii) in Lemma B.5, i.e., the Lipschitz condition of the map $\gamma \mapsto g(\gamma,\delta)$ for any $\delta \in U_V \cap H_{v_0}$, this follows from the fact that $g(\gamma,\delta)$ has directional derivatives in $h_\gamma \in H_{w_0}$.

Conditions in Lemma B.7. Recall that the map $g(\gamma,\delta)$ has partial Hadamard directional derivatives at $(\gamma,\delta_0) \in \{(\gamma,\delta_0) : \gamma \in \partial\Gamma_{I,h_\gamma}(\Phi, P_{\delta_0})\}$, in $h_\gamma \in H_{w_0}$ and $\delta \in H_{v_0}$; these are denoted by $\nabla_{h_\gamma} g(\gamma,\delta_0)$ and by $\nabla_\delta g(\gamma,\delta_0)$ respectively.

For the condition (i) in Lemma B.7, note that Hadamard directional derivatives are sequentially continuous on $\delta \in H_{v_0}$ (e.g. Proposition 3.1 in Shapiro (1990)), which in turn implies the uniform continuity of $\nabla_\delta g(\gamma,\delta_0)$ in $\delta \in H_{v_0}$ as $H_{v_0} \subset \Delta_{d_s-1}$ is compact. This indicates that the first order approximation error $g(\gamma,\delta) - g(\gamma,\delta_0) - \nabla_\delta g(\gamma,\delta_0)(\delta - \delta_0)$ converges to zero uniformly over H_{v_0}, which shows $g \overset{\delta}{\approx} g(\gamma,\delta_0) + \nabla_\delta g(\gamma,\delta_0) + (\delta - \delta_0)$ and $\nabla_\delta g(\gamma,\delta_0)$ is strong in H_{v_0}.

The condition (ii) in Lemma B.7, i.e. $0 \in H_{z_0} \subset \nabla_\delta g(\gamma,\delta_0)(U_V \cap H_{v_0} - \delta_0)$ in the notation defined above, is trivial from the construction of H_{z_0}. The condition (iii) in Lemma B.7 follows immediately as $\nabla_\delta g(\gamma,\delta_0)$ is non-singular.

Hence I apply the results in Lemma B.7 to the map $g : \mathbb{R}^{d_\gamma} \times \mathbb{R}^{d_s} \to \mathbb{R}^{d_s}$ to conclude that there exists a map $\upsilon : \mathbb{R}^{d_\gamma} \mapsto \mathbb{R}^{d_s}$ satisfying the properties from (a) to (d) in Lemma B.8. □

Lemma B.9 (Lemma 3 in Chamberlain (1987))**.** *Let $f : \mathcal{X} \mapsto \mathbb{R}^d$ $(d < \infty)$ be a Borel measurable function and P a probability measure with support $\mathcal{X}_P \subset \mathcal{X}$. If $\|f\|_{L^1(P)} < \infty$, then there exists a probability measure P_δ whose support is a finite subset of $\mathcal{X}_P$ such that $\int_{\mathcal{X}} f\, dP = \int_{\mathcal{X}} f\, dP_\delta$.*

Proof. See Lemma 3 in Chamberlain (1987). □

B.5. **Proofs for Section 5.1.**

Proof. Proposition 5.3. The result follows by applications of Theorem 3.1 and 3.2 in Chernozhukov et al. (2007); thus it suffices to verify those conditions in Theorem 3.1 and 3.2.

Condition C.1 (a), (b) and (c) of Chernozhukov et al. (2007) are immediate from the present setting ($\Theta = [-M_\theta, M_\theta]^{d_\theta} \subset \mathbb{R}^{d_\theta}$ is compact; $\Theta_I = \{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : Q(\theta) = 0\}$ from the construction of Q in (15); (c) is the same as Assumption 5.1 (ii).). Condition C.1 (d) with $a_n = \sqrt{n}$ and (e) with $b_n = \sqrt{n}$ are direct consequences of Assumption 5.1 (ii), i.e. the class of moment functions $\{m(\theta, \cdot)\}_{\theta \in \Theta}$ being P-Donsker, combined with $\Phi(\theta, P) \leq 0$ for any $\theta \in \Theta_I$.

I next show that Condition C.2 ("Existence of a Polynomial Minorant") in Chernozhukov et al. (2007) is satisfied from Assumption 4.1. For completeness, I state their Condition C.2 here.

"Existence of a Polynomial Minorant": Condition C.2 in Chernozhukov et al. (2007). There exist positive constants (δ, κ, γ) such that for any $\epsilon \in (0, 1)$ there are $(\kappa_\epsilon, n_\epsilon)$ such that for all $n \geq n_\epsilon$,

$$Q_n(\theta) \geq \kappa \cdot [d(\theta, \Theta_I) \wedge \delta]^\gamma$$

uniformly on $\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : d(\theta, \Theta_I) \geq (\kappa_\epsilon / a_n)^{1/\gamma}\}$, with probability at least $1 - \epsilon$.

First Assumption 4.9 implies that there exist positive constants $C, \delta > 0$ such that

$$\|\Phi_v(\theta, P)\| \geq C \cdot [d(\theta, \Theta_I) \wedge \delta] \tag{B.20}$$

for any $\theta \in \Theta$. To see this result, take a positive constant C satisfying $0 < C < \inf_{h \in \mathbb{R}^{d_\theta}} \inf_{\theta \in \partial\Theta_{I,h}} \|\nabla_h \Phi(\theta, P)\|_{op}$, where $\|\cdot\|_{op}$ denotes the operator norm; this choice of C is possible as Assumption 4.9 indicates $0 < \inf_{h \in \mathbb{R}^{d_\theta}} \inf_{\theta \in \partial\Theta_{I,h}} \|\nabla_h \Phi(\theta, P)\|_{op}$. If $\theta \in \Theta_I$, then both sides of the equation in (B.20) is 0. If $\theta \notin \Theta_I$, let $\theta^* \in \partial\Theta_I$ be such that $d(\theta, \theta^*) = d(\theta, \Theta_I)$ and $h_\theta = \theta - \theta^*$. The Taylor expansion of Φ at θ^* via Hadamard directional derivative shows

$$\begin{aligned} \Phi(\theta, P) &= \Phi(\theta^*, P) + \nabla_{h_\theta} \Phi(\theta^*, P)(h_\theta) + o(\|h_\theta\|) \\ &= \nabla_{h_\theta} \Phi(\theta^*, P)(h_\theta) + o(\|h_\theta\|). \end{aligned}$$

Since $\|h_\theta\| = d(\theta, \Theta_I)$, this expansion indicates that for $\theta \notin \Theta_I$ it holds that $\|\Phi_v(\theta, P)\| \geq \|\nabla_{h_\theta} \Phi_v(\theta^*, P)(h_\theta)\| > C \cdot \|h_\theta\| = C \cdot d(\theta, \Theta_I)$ for small enough $C > 0$.

Note here that the P-Donsker property in Assumption 5.1 combined with Assumption 4.9 shows

$$\begin{aligned} \sqrt{n} \cdot [\Phi(\theta, \mathbb{P}_n) - \Phi(\theta, P)] &= \sqrt{n} \cdot [q(\mathbb{P}_n m(\theta, \cdot)) - q(P m(\theta, \cdot))] \\ &\rightsquigarrow \nabla q(P m(\theta, \cdot))(\xi(\theta)), \end{aligned} \tag{B.21}$$

which follows from the Delta method with Hadamard directional derivative. In particular, this implies $\|\sqrt{n} \cdot [\Phi_v(\theta, \mathbb{P}_n) - \Phi_v(\theta, P)]\| \leq \|\sqrt{n} \cdot [\Phi(\theta, \mathbb{P}_n) - \Phi(\theta, P)]\| = O_P(1)$.

From the construction of Q in (15), it follows that, with probability approaching 1,

$$\begin{aligned}
\sqrt{n}\cdot Q_n(\theta) &= \left\|\sqrt{n}\cdot[\Phi_v(\theta,\mathbb{P}_n)-\Phi_v(\theta,P)]^t\cdot W_n^{1/2}+\sqrt{n}\cdot\Phi_v^t(\theta,P)\cdot W_n^{1/2}\right\| \\
&\overset{(a)}{\geq} \mathrm{Eig}_1(W)^{1/2}\cdot\left\|\sqrt{n}\cdot[\Phi_v(\theta,\mathbb{P}_n)-\Phi_v(\theta,P)]^t+\sqrt{n}\cdot\Phi_v^t(\theta,P)\right\| \\
&\overset{(b)}{\geq} \mathrm{Eig}_1(W)^{1/2}\cdot\left|\left\|\sqrt{n}\cdot[\Phi_v(\theta,\mathbb{P}_n)-\Phi_v(\theta,P)]^t\right\|-\left\|\sqrt{n}\cdot\Phi_v^t(\theta,P)\right\|\right| \\
&\overset{(c)}{\geq} \mathrm{Eig}_1(W)^{1/2}\cdot\left|O_p(1)-\sqrt{n}\cdot C\cdot[d(\theta,\Theta_I)\wedge\delta]\right|,
\end{aligned}$$

where (a) follows from $\mathrm{Eig}_1(W)=\inf_{\theta\in\Theta}\mathrm{Eig}_1(W(\theta))>0$ ($\mathrm{Eig}_1(W)$represents the minimum eigenvalue of W) with Assumption 5.1 (iii), (b) follows from $\|a+b\|\geq|\|a\|-\|b\||$, and (c) follows from (B.20) and (B.21). Thus, for any $\epsilon>0$ there exist $(\kappa_\epsilon,n_\epsilon)$ large enough to satisfy

$$\sqrt{n}\cdot Q_n(\theta)\geq(\mathrm{Eig}_1(W)^{1/2}/2)\cdot\sqrt{n}\cdot C\cdot[d(\theta,\Theta_I)\wedge\delta],$$

uniformly over $\{\theta\in\Theta\subset\mathbb{R}^{d_\theta}:d(\theta,\Theta_I)\geq(\kappa_\epsilon/n)^{1/2}\}$. The above argument implies Condition C.2 is satisfied with $\gamma=1$.

Hence Condition C.1 and C.2 of Chernozhukov et al. (2007) are satisfied in the present setting and Theorem 3.1 holds. The result of Theorem 3.2 follows by adding Assumption 5.2 and by noting that $\tau=c_n=\inf_{\theta\in\Theta}\sqrt{n}\cdot Q_n(\theta)=O_p(1)$. □

B.6. **Proofs for Section 5.2.**

Proof. Theorem 5.6.
For any $\theta \in \Theta \subset \mathbb{R}^{d_\theta}$ recall the stochastic field

$$Z_n(\theta) := \sqrt{n} \cdot \left(d(\theta, \widehat{\Theta}_I) - d(\theta, \Theta_I) \right)$$

defined in (20). By the definition of d_H, it follows that $\sup_{\theta \in \mathbb{R}^{d_\theta}} |Z_n(\theta)| = \sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I)$. Lemma B.12 further implies that the domain of the supremum can be restricted to the $\sqrt{n}$-local neighborhood of the boundary $\partial\Theta_I$, $\mathcal{T}_n$ in (B.22). Thus with probability approaching 1 it holds that

$$\sqrt{n} \cdot \left(d_H(\widehat{\Theta}_I, \Theta_I) \right) = \sup_{x \in \mathcal{T}_n(M)} |Z_n(x)| = \sup_{(\theta,\psi) \in \partial\Theta_I \times \overline{B_M(0)}} |Z_n(\theta, \psi)| ,$$

where the second equality follows from Lemma B.14. This corresponds to the "generalized Chernoff regularity" (Condition S.1 in Chernozhukov et al. (2007)) in the interior parameter case (recall $\Theta_I \subset \Theta^o$).

The stochastic field $Z_n(\theta, \psi)$ is not equicontinuous with respect to their arguments; hence typical argument of uniform weak convergence on a function space $\mathcal{C}(\partial\Theta_I \times \overline{B_M(0)})$ or $\ell^\infty(\partial\Theta_I \times \overline{B_M(0)})$ does not work. Similar to Chernozhukov et al. (2007) and Kaido (2016), I instead rely on combining the weak supremum convergence with weak finite dimensional approximability. The key intermediate result is the following.

Proposition B.10. *Suppose that the same conditions as in Theorem 5.6 hold. Then for large enough $M > 0$ it holds that*

$$\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I) = \sup_{(\theta,\psi) \in \partial\Theta_I \times \mathbb{R}^{d_\theta}} |Z_n(\theta, \psi)| + o(1) \rightsquigarrow \sup_{(\theta,\psi) \in \partial\Theta_I \times \mathbb{R}^{d_\theta}} |Z(\theta, \psi)| ,$$

as $n \to \infty$, where $Z(\theta, \psi) = d(\psi, \Psi(\theta)) - d(\psi, T(\theta))$ as defined in Lemma B.15.

Proof. See Appendix B.6.2. □

This characterizes the limiting distribution of $\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I)$. Lemma B.16 further establishes the equivalence

$$\sup_{(\theta,\psi) \in \partial\Theta_I \times \mathbb{R}^{d_\theta}} |Z(\theta, \psi)| = \sup_{\theta \in \partial\Theta_I} V(\theta) = \sup_{\theta \in \partial\Theta_I} d_H(\Psi(\theta), T(\theta)),$$

where $\Psi(\theta)$ and $T(\theta)$ are the local limit of $\sqrt{n} \cdot (\widehat{\Theta}_I - \theta)$ and $\sqrt{n} \cdot (\Theta_I - \theta)$ at $\theta \in \partial\Theta_I$ as in Lemma B.13. Finally the local quantity $d_H(\Psi(\theta), T(\theta))$ at $\theta \in \partial\Theta_I$ is expressed via the supremum of the inner product of the unit normal vector and the combination of the Gaussian approximation error, i.e.

$$d_H(\Psi(\theta), T(\theta)) = \sup \left\{ \left| \alpha^t D_{\mathcal{A}(\theta)}(\theta) \right| : \alpha \geq 0, \ \left\| \alpha^t G_{\mathcal{A}(\theta)}(\theta) \right\| = 1 \right\} .$$

by Lemma B.16. Taking the supremum over $\theta \in \partial\Theta_I$ then shows the result. □

B.6.1. *Auxiliary lemmas.* Throughout this subsection, I assume $\mathcal{J}_1 = \emptyset$ as well as the same conditions in Theorem 5.6.

Lemma B.11. *For $\widehat{\Theta}_I$ and c_n, the following results hold:*

(i). $c_n = \inf_{\theta' \in \Theta} Q_n(\theta') = 0$*; hence* $\lim_{n\to\infty} \sqrt{n} \cdot c_n = 0$,

(ii). $\widehat{\Theta}_I(\tau)$ *with* $\tau = \sqrt{n} \cdot c_n$ *satisfies*

$$\widehat{\Theta}_I(\tau) = \left\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : Q_n(\theta) \leq c_n\right\} = \left\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : \max_{j \in \mathcal{J}_2} \Phi_j(\theta, \mathbb{P}_n) \leq 0\right\}.$$

Proof. For (i), note that for any $\theta \in \partial\Theta_I$ it holds that $d(0, \mathrm{conv}\{\phi_j(\theta) : j \in \mathcal{A}(\theta)\}) \geq c > 0$ by Assumption 5.4, implying that $0 \notin \mathrm{conv}\{\phi_j(\theta) : j \in \mathcal{A}(\theta)\}$. Now by Gordan's theorem (e.g., Borwein and Lewis (2006) Theorem 2.2.1), one of the following cases holds: (a) there exists $\psi \in \mathbb{R}^{d_\theta}$ such that $\langle \phi_j(\theta), \psi\rangle < 0$ for any $j \in \mathcal{A}(\theta)$, or (b) $0 \in \mathrm{conv}\{\phi_j(\theta) : j \in \mathcal{A}(\theta)\}$; but (b) does not hold, as shown above. For this ψ in (a), consider the Tayloer expansion around $\theta \in \partial\Theta_I$,

$$\Phi_j(\theta + t \cdot \psi, P) = \Phi_j(\theta, P) + t \cdot \langle \phi_j(\theta), \psi\rangle + o(t).$$

For $j \notin \mathcal{A}(\theta)$ it holds that $\Phi_j(\theta, P) < 0$; hence $t > 0$ can be taken small enough to satisfy $\Phi_j(\theta + t \cdot \psi, P) < 0$. For $j \in \mathcal{A}(\theta)$, it holds that $\Phi_j(\theta, P) = 0$ and $\langle \phi_j(\theta), \psi\rangle < 0$; hence $\Phi_j(\theta + t \cdot \psi, P) < 0$. Let $\theta^* = \theta + t \cdot \psi$ be such that $\Phi_j(\theta^*, P) < 0$ for any $j \in \mathcal{J}_2$. By continuity in t it holds that $\theta^* \in \Theta_I^o$.

For any such $\theta^* \in \Theta_I^o$ let $\delta := -\max_{j\in\mathcal{J}_2} \Phi_j(\theta^*, P)$; then $\delta > 0$ due to the slackness shown above. Note here that by the Donsker property of m with smooth Φ, it holds that $\sup_{\theta\in\Theta} \|\Phi(\theta, \mathbb{P}_n) - \Phi(\theta, P)\| = O_P(1/\sqrt{n})$. Thus, with probability approaching 1 it holds that

$$\begin{aligned}\max_{j\in\mathcal{J}_2} \Phi_j(\theta^*, \mathbb{P}_n) &\leq \max_{j\in\mathcal{J}_2} \Phi_j(\theta^*, P) + \left[\max_{j\in\mathcal{J}_2} \Phi_j(\theta^*, \mathbb{P}_n) - \max_{j\in\mathcal{J}_2} \Phi_j(\theta^*, P)\right] \\ &\leq -\delta + O_P(1/\sqrt{n}) < 0.\end{aligned}$$

This implies $\Phi_v(\theta^*, \mathbb{P}_n) = 0$ and $Q_n(\theta^*) = 0$, which shows the result (i).

For (ii), note that the result (i) implies that with probability approaching 1

$$\begin{aligned}\widehat{\Theta}_I(\tau) = \left\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : Q_n(\theta) \leq c_n\right\} &= \left\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : Q_n(\theta) = 0\right\} \\ &= \left\{\theta \in \Theta \subset \mathbb{R}^{d_\theta} : \Phi_v(\theta, \mathbb{P}_n) = 0\right\},\end{aligned}$$

where the second equality follows from $\mathcal{J}_1 = \emptyset$ and $W = W_n + o_P(1)$ being positive definite. The construction of Φ_v shows that the last set of the above is equivalent to the set in the result in (ii). □

Lemma B.12. *For $M > 0$ let*

$$\mathcal{T}_n(M) := \left\{\theta \in \mathbb{R}^{d_\theta} : d(\theta, \partial\Theta_I) \leq M/\sqrt{n}\right\}. \tag{B.22}$$

Under the condition that $d_H(\widehat{\Theta}_I, \Theta_I) = O_P(1/\sqrt{n})$*, one can take* M *large enough such that*

$$d_H(\widehat{\Theta}_I, \Theta_I) = \sup_{\theta \in \mathcal{T}_n(M)} \left| d(\theta, \widehat{\Theta}_I) - d(\theta, \Theta_I) \right|,$$

with probability approaching 1*.*

Proof. By the definition of d_H it holds that $d_H(\widehat{\Theta}_I, \Theta_I) = \sup_{\theta \in \mathbb{R}^{d_\theta}} \left| d(\theta, \widehat{\Theta}_I) - d(\theta, \Theta_I) \right|$, which implies the left hand side is bigger than the right one.

For the other direction, if $d_H(\widehat{\Theta}_I, \Theta_I) = 0$ with probability 1 the result is trivial; hence assume $d_H(\widehat{\Theta}_I, \Theta_I) > 0$ at least with some probability $\epsilon(n) > 0$ with $\epsilon(n) = O(1/\sqrt{n})$. First note that

$$P\left\{ \Theta_I^{-\delta_n} \subset \widehat{\Theta}_I \right\} \to 1,$$

where $\delta_n = M/\sqrt{n}$ and $\Theta_I^{-\delta_n}$ denotes the δ_n-contraction; $\Theta_I^{-\delta_n} = \{\theta \in \mathbb{R}^{d_\theta} : d\}$. To see this, note that for $\theta \in \Theta_I^{-\delta_n}$ Assumption 5.4 shows $\max_{j \in \mathcal{J}_2} \Phi_j(\theta, P) \leq -c \cdot \delta_n$. Since $\sup_{\theta \in \Theta} \|\Phi_n(\theta, \mathbb{P}_n) - \Phi(\theta, P)\| = O_P(1/\sqrt{n})$, it holds that

$$\begin{aligned} \max_{j \in \mathcal{J}_2} \Phi_j(\theta, \mathbb{P}_n) &\leq \max_{j \in \mathcal{J}_2} \Phi_j(\theta, P) + \left[\max_{j \in \mathcal{J}_2} \Phi_j(\theta, \mathbb{P}_n) - \max_{j \in \mathcal{J}_2} \Phi_j(\theta, P) \right] \\ &\leq -c \cdot \delta_n + O_P(1/\sqrt{n}) < 0 \end{aligned}$$

with probability approaching 1 for large M. Hence $\theta \in \widehat{\Theta}_I$ by Lemma B.11 and the result follows.

Now recall that the definition of d_H reads

$$d_H(\widehat{\Theta}_I, \Theta_I) = \max \left\{ \sup_{x \in \widehat{\Theta}_I} d(x, \Theta_I), \sup_{y \in \Theta_I} d(y, \widehat{\Theta}_I) \right\}.$$

Accordingly, two cases arise: (i) $d_H(\widehat{\Theta}_I, \Theta_I) = \sup_{x \in \widehat{\Theta}_I} d(x, \Theta_I)$ and (ii) $d_H(\widehat{\Theta}_I, \Theta_I) = \sup_{y \in \Theta_I} d(y, \widehat{\Theta}_I)$.

Suppose that the case (i) holds. Take $x^* \in \widehat{\Theta}_I$ such that $d(x^*, \Theta_I) = d_H(\widehat{\Theta}_I, \Theta_I)$. The restriction $d_H(\widehat{\Theta}_I, \Theta_I) > 0$ implies $x^* \notin \Theta_I$. Since

$$d(x^*, \partial\Theta_I) \leq d(x^*, \Theta_I) = d_H(\widehat{\Theta}_I, \Theta_I) \leq M/\sqrt{n},$$

it holds that $x^* \in \mathcal{T}_n$. Clearly $\left| d(x^*, \widehat{\Theta}_I) - d(x^*, \Theta_I) \right| = d_H(\widehat{\Theta}_I, \Theta_I)$ as $x^* \in \widehat{\Theta}_I$ and the desired result holds.

Suppose now that the case (ii) holds. Take $y^* \in \Theta_I$ such that $d(y^*, \widehat{\Theta}_I) = d_H(\widehat{\Theta}_I, \Theta_I)$. The restriction $d_H(\widehat{\Theta}_I, \Theta_I) > 0$ implies $y^* \notin \widehat{\Theta}_I$. Since $P\{\Theta_I^{-\delta_n} \subset \widehat{\Theta}_I\} \to 1$, it holds that $y^* \notin \Theta_I^{-\delta_n}$ for large $n \in \mathbb{N}$; but then $d(y^*, \partial\Theta_I) < \delta_n = M/\sqrt{n}$, implying $y^* \in \mathcal{T}_n$. Clearly $|d(y^*, \widehat{\Theta}_I) - d(y^*, \Theta_I)| = d_H(\widehat{\Theta}_I, \Theta_I)$ as $y^* \in \Theta_I$ and the desired result holds as well. □

Lemma B.13. *Suppose that Assumption 5.1 holds and that $\Phi(\theta, P)$ is fully differentiable at any $\theta \in \partial\Theta_I$. Then for any $M > 0$, uniformly for $\psi \in \overline{B_M(0)}$ it holds that*

$$\sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, \mathbb{P}_n) \rightsquigarrow \begin{cases} \langle \phi_j(\theta), \psi \rangle + D_j(\theta) & j \in \mathcal{A}(\theta), \\ -\infty & j \notin \mathcal{A}(\theta). \end{cases}$$

Hence with $c_n = 0$ with probability approaching 1 it holds that

$$\begin{aligned} \Psi_n(\theta) \cap \overline{B_M(0)} &:= \sqrt{n} \cdot (\widehat{\Theta}_I - \theta) \cap \overline{B_M(0)} \\ &\rightsquigarrow \Psi(\theta) \cap \overline{B_M(0)} = \left\{ \psi \in \overline{B_M(0)} : \langle \phi_j(\theta), \psi \rangle \leq -D_j(\theta),\ j \in \mathcal{A}(\theta) \right\} \\ &= \left\{ \psi \in \overline{B_M(0)} : G_{\mathcal{A}(\theta)}(\theta)\psi \leq -D_{\mathcal{A}(\theta)}(\theta) \right\}, \\ T_n(\theta) \cap \overline{B_M(0)} &:= \sqrt{n} \cdot (\Theta_I - \theta) \cap \overline{B_M(0)} \\ &\to T(\theta) \cap \overline{B_M(0)} = \{ \psi \in \overline{B_M(0)} : \langle \phi_j(\theta), \psi \rangle \leq 0,\ j \in \mathcal{A}(\theta) \} \\ &= \{ \psi \in \overline{B_M(0)} : G_{\mathcal{A}(\theta)}(\theta)\psi \leq 0 \}. \end{aligned} \tag{B.23}$$

Proof. Fix $\theta \in \partial\Theta_I$ and $M > 0$. Note that for any $\psi \in \overline{B_M(0)}$ it holds that

$$\begin{aligned} &\sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, \mathbb{P}_n) \\ &= \underbrace{\sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, P)}_{(a)} + \underbrace{\left[\sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, \mathbb{P}_n) - \sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, P) \right]}_{(b)}, \end{aligned}$$

for any $j \in \mathcal{J}_2$. For (a) it holds that

$$(a) = \sqrt{n} \cdot \Phi_j(\theta, P) + \langle \phi_j(\theta), \psi \rangle + \sqrt{n} \cdot o(\|\psi/\sqrt{n}\|),$$

while for (b) one obtains

$$\begin{aligned} (b) &= \sqrt{n} \cdot [\Phi_j(\theta, \mathbb{P}_n) - \Phi_j(\theta, P)] \\ &\quad + \sqrt{n} \cdot \left\{ \left[\Phi_j(\theta + \psi/\sqrt{n}, \mathbb{P}_n) - \Phi_j(\theta + \psi/\sqrt{n}, P) \right] - [\Phi_j(\theta, \mathbb{P}_n) - \Phi_j(\theta, P)] \right\} \\ &= \nabla\Phi_j(\xi(\theta)) + \sqrt{n} \cdot o_P(\|\psi/\sqrt{n}\|), \end{aligned}$$

where the last equality follows from the stochastic equicontinuity of the process $\Phi_j(\theta, \mathbb{P}_n) - \Phi(\theta, P)$ on Θ due to the Donsker property of m and smooth Φ. Combining the results and noting that $D_j(\theta) = \nabla\Phi_j(\xi(\theta))$, one obtains

$$\sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, \mathbb{P}_n) = \sqrt{n} \cdot \Phi_j(\theta, P) + \langle \phi_j(\theta), \psi \rangle + D_j(\theta) + o_P(1),$$

which holds uniformly for $\psi \in \overline{B_M(0)}$. If $j \in \mathcal{A}(\theta)$, $\Phi_j(\theta, P) = 0$ and the first term vanishes; if $j \notin \mathcal{A}$, then $\Phi_j(\theta, P) < 0$ and $\sqrt{n} \cdot \Phi_j(\theta, P) \to -\infty$. This gives the first result.

For the second result, recall that Lemma B.11 shows $\widehat{\Theta}_I = \{\theta \in \mathbb{R}^{d_\theta} : \max_{j \in \mathcal{J}_2} \Phi_j(\theta, \mathbb{P}_n) \leq 0\}$. From the construction $\Psi_n(\theta) = \sqrt{n} \cdot (\widehat{\Theta}_I - \theta)$ for $\theta \in \partial\Theta_I$, it follows that $\psi \in \Psi_n(\theta) \Leftrightarrow \theta + \psi/\sqrt{n} \in \widehat{\Theta}_I$. Hence

$$\begin{aligned} \Psi_n(\theta) &= \sqrt{n} \cdot (\widehat{\Theta}_I - \theta) \\ &= \left\{ \psi \in \mathbb{R}^{d_\theta} : \sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, \mathbb{P}_n) \leq 0, \text{ for all } j \in \mathcal{J}_2 \right\}. \end{aligned} \tag{B.24}$$

Now let $\psi_n \in \Psi_n(\theta) \cap \overline{B_M(0)}$ with $\psi_n \to \psi$. For $j \in \mathcal{A}(\theta)$, the first result combined with the uniform expansion of $\Phi_j(\theta + \psi/\sqrt{n}, \mathbb{P}_n)$ on $\overline{B_M(0)}$ above shows

$$0 \geq \sqrt{n} \cdot \Phi(\theta + \psi_n/\sqrt{n}, \mathbb{P}_n) \to \langle \phi_j(\theta), \psi \rangle + D_j(\theta).$$

Hence $\langle \phi_j(\theta), \psi \rangle \leq -D_j(\theta)$ for any $j \in \mathcal{A}(\theta)$, i.e. $\psi \in \Psi(\theta) \cap \overline{B_M(0)}$; this implies $\lim_{n\to\infty} \sqrt{n} \cdot (\widehat{\Theta}_I - \theta) \subset \Psi(\theta) \cap \overline{B_M(0)}$.

Next let $\psi \in \Psi(\theta) \cap \overline{B_M(0)}$. If $\langle \phi_j(\theta), \psi \rangle < -D_j(\theta)$ for any $j \in \mathcal{A}(\theta)$, then $\sqrt{n} \cdot \Phi(\theta + \psi/\sqrt{n}, \mathbb{P}_n) < 0$ for any $j \in \mathcal{J}_2$ and any large $n \in \mathbb{N}$; thus $\psi \in \Psi_n(\theta) \cap \overline{B_M(0)}$ by (B.24). The other case is $\psi \in \partial\Psi(\theta) \cap \overline{B_M(0)}$. As seen in the proof of Lemma B.11, Assumption 5.4 combined with the Gordan's theorem (e.g. Borwein and Lewis (2006) Theorem 2.2.1) implies that there exists $\psi \in \mathbb{R}^{d_\theta}$ such that $\langle \phi_j(\theta), \psi \rangle < 0$ for any $j \in \mathcal{A}(\theta)$, i.e. the tangent cone $T(\theta)$ has non-empty interior. Moreover Assumption 5.4 (iii) implies $G_{\mathcal{A}}(\theta)v = -D_{\mathcal{A}}(\theta)$ has a solution v. Therefore, $\Psi(\theta) = \{\psi : G_{\mathcal{A}}(\theta)\psi \leq -D_{\mathcal{A}}(\theta)\} = \{\psi : G_{\mathcal{A}}(\theta)(\psi - v) \leq 0\}$, which is a translation of $T(\theta)$. Hence take $\psi_n \in \Psi(\theta)^o \cap \overline{B_M(0)}$ such that $\psi_n \to \psi \in \partial\Psi(\theta) \cap \overline{B_M(0)}$ and that $\psi_n \in \Psi_n(\theta) \cap \overline{B_M(0)}$ for all large $n \in \mathbb{N}$; this construction gives $\psi \in \liminf_{n\to\infty} \Psi_n(\theta) \cap \overline{B_M(0)}$. Hence it has been shown $\Psi(\theta) \cap \overline{B_M(0)} \subset \lim_{n\to\infty} \sqrt{n} \cdot (\widehat{\Theta}_I - \theta) \cap \overline{B_M(0)}$.

The result on $T_n(\theta)$ and $T(\theta)$ follows in a similar manner. In particular, note that

$$\sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, P) = \sqrt{n} \cdot \Phi_j(\theta, P) + \langle \phi_j(\theta), \psi \rangle + o(1),$$

which holds uniformly for $\psi \in \overline{B_M(0)}$. Note also that, from the characterization $\Theta_I = \{\theta \in \mathbb{R}^{d_\theta} : \max_{j\in\mathcal{J}_2} \Phi_j(\theta, P) \leq 0\}$ and $T_n(\theta) = \sqrt{n} \cdot (\Theta_I - \theta)$ for $\theta \in \partial\Theta_I$, it follows that $\psi \in T_n(\theta) \Leftrightarrow \theta + \psi/\sqrt{n} \in \Theta_I$. Hence

$$\begin{aligned} T_n(\theta) &= \sqrt{n} \cdot (\Theta_I - \theta) \\ &= \left\{\psi \in \mathbb{R}^{d_\theta} : \sqrt{n} \cdot \Phi_j(\theta + \psi/\sqrt{n}, P) \leq 0, \text{ for all } j \in \mathcal{J}_2\right\}. \end{aligned}$$

The rest of the argument follows the same argument, where $\{\Psi_n, \Psi, D_{\mathcal{A}}, v\}$ is replaced by $\{T_n, T, 0 \in \mathbb{R}^{d_\theta}, 0 \in \mathbb{R}^{d_\theta}\}$. □

Lemma B.14. *Recall* $Z_n(x) := \sqrt{n} \cdot \left(d(x, \widehat{\Theta}_I) - d(x, \Theta_I)\right)$ *defined in (20). For the fixed compact local graph* $\partial\Theta_I \times \overline{B_M(0)}$, *define the local process of* Z_n *as*

$$Z_n(\theta, \psi) := Z_n(\theta + \psi/\sqrt{n}), \tag{B.25}$$

where $\Psi(\theta)$ *and* $T(\theta)$ *are as in Lemma B.13. Then it holds that* $Z_n(\theta, \psi) = d(\psi, \Psi_n(\theta)) - d(\psi, T_n(\theta))$, *where* $\Psi_n(\theta)$ *and* $T_n(\theta)$ *are are as in Lemma B.13. Moreover,*

$$\sup_{x\in\mathcal{T}_n(M)} |Z_n(x)| = \sup_{(\theta,\psi)\in\partial\Theta_I\times\overline{B_M(0)}} |Z_n(\theta, \psi)|.$$

Proof. First, for $x \in \mathcal{T}_n(M)$, let $\theta = \theta(x) \in \partial\Theta_I$ such that $d(x, \partial\Theta_I) = d(x, \theta)$, and define $\psi = \sqrt{n}(x - \theta)$; clearly $x = \theta + \psi/\sqrt{n}$. This map $x \mapsto (\theta, \psi)$ is bijective

from $\mathcal{T}_n(M)$ onto $\partial\Theta_I \times \overline{B_M(0)}$. Next, note also that since for any set A it holds that $d(x, A) = (1/\sqrt{n}) \cdot d(\psi, \sqrt{n}(A - \theta))$, one obtains

$$\begin{aligned} Z_n(x) &= Z_n(\theta + \psi/\sqrt{n}) \\ &= \sqrt{n} \cdot \left(d(\theta + \psi/\sqrt{n}, \widehat{\Theta}_I) - d(\theta + \psi/\sqrt{n}, \Theta_I) \right) \\ &= d(\psi, \Psi_n(\theta)) - d(\psi, T_n(\theta)). \end{aligned}$$

The first result then follows from $Z_n(\theta, \psi) := Z_n(\theta + \psi/\sqrt{n})$ in (B.25). Combining the arguments, one obtains

$$\begin{aligned} \sup_{x \in \mathcal{T}_n(M)} |Z_n(x)| &= \sup_{(\theta,\psi) \in \partial\Theta_I \times \overline{B_M(0)}} |d(\psi, \Psi_n(\theta)) - d(\psi, T_n(\theta))| \\ &= \sup_{(\theta,\psi) \in \partial\Theta_I \times \overline{B_M(0)}} |Z_n(\theta, \psi)|, \end{aligned}$$

which shows the second result. □

Lemma B.15. *Recall $Z_n(\theta, \psi) = d(\psi, \Psi_n(\theta)) - d(\psi, T_n(\theta))$ for $(\theta, \psi) \in \partial\Theta_I \times \overline{B_M(0)}$ as defined in Lemma B.14. For $Z_n(\theta, \psi)$ defined on the compact local graph $\partial\Theta_I \times \overline{B_M(0)}$ and define its limit counterpart on the limit graph $\partial\Theta_I \times \mathbb{R}^{d_\theta}$ as*

$$Z(\theta, \psi) := d(\psi, \Psi(\theta)) - d(\psi, T(\theta)), \tag{B.26}$$

where $\Psi(\theta)$ and $T(\theta)$ are as in Lemma B.13. Finally for any $\theta \in \partial\Theta_I$ define

$$V(\theta) := d_H(\Psi(\theta), T(\theta)). \tag{B.27}$$

Then it holds that

$$\sup_{(\theta,\psi) \in \partial\Theta_I \times \mathbb{R}^{d_\theta}} |Z(\theta, \psi)| = \sup_{\theta \in \partial\Theta_I} V(\theta).$$

Proof. Note that $\sup_{\psi \in \mathbb{R}^{d_\theta}} |Z(\theta, \psi)| = d_H(\Psi(\theta), T(\theta)) = V(\theta)$ for any $\theta \in \partial\Theta_I$ by the property of d_H; taking the supremum over $\theta \in \partial\Theta_I$ yields the result. □

Lemma B.16. *Let $\theta \in \partial\Theta_I$ and $v \in \mathbb{R}^{d_\theta}$ such that $G_{\mathcal{A}(\theta)}(\theta)v = -D_{\mathcal{A}(\theta)}(\theta)$ (v may not be unique as $v = v' + ker(G_{\mathcal{A}(\theta)}(\theta))$ with v' such solution). Then*

$$\begin{aligned} V(\theta) = d_H\left(\Psi(\theta), T(\theta)\right) &= \sup_{u \in N(\theta), \|u\|=1} |\langle u, v \rangle| \\ &= \sup\left\{ \left| \alpha^t D_{\mathcal{A}(\theta)}(\theta) \right| : \alpha \geq 0, \left\| \alpha^t G_{\mathcal{A}(\theta)}(\theta) \right\| = 1 \right\}. \end{aligned} \tag{B.28}$$

Moreover,

$$V(\theta) = \sup_{\psi \in \mathbb{R}^{d_\theta}} |d(\psi, \Psi(\theta)) - d(\psi, T(\theta))| = \max_{\psi \in \mathbb{R}^{d_\theta} : \psi \in \{0, v\}} |d(\psi, \Psi(\theta)) - d(\psi, T(\theta))|. \tag{B.29}$$

Proof. It follows from Lemma B.13 that $\Psi(\theta)$ and $T(\theta)$ are intersections of half spaces, implying that these sets are convex polyhedral cones (note that these sets $\Psi(\theta)$ and $T(\theta)$ capture local nature of Θ_I at $\theta \in \partial\Theta_I$ and this does not require convexity of Θ_I).

First note that $\Phi(\theta) = T(\theta) + v$ from Lemma B.13. Hence $\sup_{a\in\Psi(\theta)} d(a, T(\theta)) = \sup_{t\in T(\theta)} d(t + v, T(\theta))$. Since $d(t + v, T(\theta)) \leq d(t, T(\theta)) + d(v, T(\theta)) = d(v, T(\theta))$ for any $t \in T(\theta)$, it holds that $\sup_{a\in\Psi(\theta)} d(a, T(\theta)) = d(v, T(\theta))$. In a similar manner, it holds that $\sup_{b\in T(\theta)} d(b, \Psi(\theta)) = \sup_{b\in T(\theta)} d(b, T(\theta) + v) = \sup_{b\in T(\theta)} d(b - v, T(\theta))$; since $d(b - v, T(\theta)) \leq d(b, T(\theta)) + d(-v, T(\theta)) = d(-v, T(\theta))$ for any $b \in T(\theta)$, one obtains $\sup_{b\in T(\theta)} d(b, \Psi(\theta)) = d(-v, T(\theta))$. Combining both quantities leads to

$$\begin{aligned} d_H\left(\Psi(\theta), T(\theta)\right) &= \max\left\{\sup_{a\in\Psi(\theta)} d(a, T(\theta)),\ \sup_{b\in T(\theta)} d(b, \Psi(\theta))\right\} \\ &= \max\left\{d(v, T(\theta)),\ d(-v, T(\theta))\right\}. \end{aligned}$$

Now since $d_H(\Psi(\theta), T(\theta)) = \sup_{\psi\in\mathbb{R}^{d_\theta}} |d(\psi, \Psi(\theta) - d(\psi, T(\theta))|$, putting $\psi = v \in \Psi(\theta)$ or $0 \in T(\theta)$ leads to $d(v, T(\theta))$ or $d(0, \Psi(\theta)) = d(-v, T(\theta))$; that is, $\psi \in \{0, v\}$ achieves $d_H(\Psi(\theta), T(\theta))$, which shows (B.29).

Next I use the result (the "minimum norm duality theorem" (e.g. Luenberger (1969) section 5.13)) $d(x, T) = \sup\{\langle x, u\rangle - s(u, T) : \|u\| \leq 1\}$, where $x \in \mathbb{R}^{d_\theta}$, $T \subset \mathbb{R}^{d_\theta}$ is a closed convex set and $s(u, T) := \sup_{t\in T}\langle u, t\rangle$ (i.e. support function of T). In particular, for closed convex cone $T(\theta)$, it follows that $s(u, T(\theta)) = 0$ if $u \in T(\theta)^o := \{\psi \in \mathbb{R}^{d_\theta} : \langle\psi, t\rangle \leq 0,\ \text{for any } t \in T(\theta)\}$ (polar cone of $T(\theta)$), and $= +\infty$ otherwise. Here note that $T(\theta) = \{\psi \in \mathbb{R}^{d_\theta} : \langle\phi_j(\theta), \psi\rangle \leq 0,\ j \in \mathcal{A}(\theta)\}$ implies $T(\theta)^o = \{\sum_{j\in\mathcal{A}(\theta)} \alpha_j \cdot \phi_j(\theta) : \alpha_j \geq 0\} =: N(\theta)$. Thus,

$$d(x, T(\theta)) = \sup\left\{\langle x, u\rangle : u \in N(\theta), \|u\| \leq 1\right\}.$$

Combining this with the representation of $d_H(\Phi(\theta), T(\theta))$ shows

$$\begin{aligned} d_H\left(\Psi(\theta), T(\theta)\right) &= \max\left\{d(v, T(\theta)),\ d(-v, T(\theta))\right\} \\ &= \max\left\{\sup_{u\in N(\theta):\|u\|\leq 1} \langle v, u\rangle,\ \sup_{u\in N(\theta):\|u\|\leq 1} \langle -v, u\rangle\right\} \\ &= \sup_{u\in N(\theta):\|u\|\leq 1} |\langle v, u\rangle| \\ &= \sup_{u\in N(\theta):\|u\|=1} |\langle v, u\rangle|. \end{aligned}$$

Finally putting $u^t = \alpha^t G_{\mathcal{A}(\theta)}(\theta)$ with $\|\alpha^t G_{\mathcal{A}(\theta)}(\theta)\| = 1$ and noting that $G_{\mathcal{A}(\theta)}(\theta)v = -D_{\mathcal{A}(\theta)}(\theta)$ give

$$\langle v, u\rangle = v^t G_{\mathcal{A}(\theta)}(\theta)^t\alpha = -D_{\mathcal{A}(\theta)}(\theta)^t\alpha,$$

which shows the result in (B.28). □

B.6.2. *Proofs for Proposition B.10.*

Lemma B.17. *Given any subset $\mathcal{I} \subset \mathcal{J}_2$, let $F_{\mathcal{I}} = \{\theta \in \partial\Theta_I : \mathcal{A}(\theta) = \mathcal{I}\}$. Then*

$$\partial\Theta_I = \bigsqcup\nolimits_{\mathcal{I}\subset\mathcal{J}_2} F_{\mathcal{I}},$$

where the number of disjoint unions is finite.

Proof. This comes from the relation that $|\mathcal{J}_2| \le d_q < \infty$ (indeed $|\mathcal{J}_2| = d_q$ as $\mathcal{J}_1 = \emptyset$ is assumed here). □

Lemma B.18. *For every $\alpha \in \mathbb{R}^{|\mathcal{A}(\theta)|}$ satisfying*

$$\alpha \ge 0, \quad \left\|\alpha^t G_{\mathcal{A}}(\theta)\right\| = 1,$$

it holds that $\|\alpha\| \le 1/c$, where c is the lower bound in Assumption 5.4.

Proof. Pick $\alpha \ge 0$ with $\|\alpha^t G_{\mathcal{A}}(\theta)\| = 1$, and let $s = \sum_{j \in [|\mathcal{A}(\theta)|]} \alpha_j$; by Assumption 5.4 it holds that $s > 0$. Then $\alpha^t G_{\mathcal{A}}(\theta) = s \cdot w$ where $w = \sum_{j \in [|\mathcal{A}(\theta)|]} (\alpha_j / s) \phi_j$, which is in $\mathrm{conv}\{\phi_j(\theta) : j \in \mathcal{A}(\theta)\}$. Thus $\|w\| \ge c$ by Assumption 5.4 and $1 = \|\alpha^t G_{\mathcal{A}}(\theta)\| = s\|w\| \ge s \cdot c$. Since $\|\alpha\|^2 \le (\sum_{j \in [|\mathcal{A}(\theta)|]} \alpha_j)^2$ noting that $\alpha \ge 0$, one obtains $\|\alpha\| \le \sum_{j \in [|\mathcal{A}(\theta)|]} \alpha_j = s \le 1/c$. □

Lemma B.19. *Given an index set $\mathcal{I}$, a $|\mathcal{I}| \times d_\theta$-matrix $G \in \mathbb{R}^{|\mathcal{I}| \times d_\theta}$ and $D \in \mathbb{R}^{|\mathcal{I}|}$, let*

$$F(G, D) := \sup\left\{\left|\alpha^t D\right| : \left\|\alpha^t G\right\| = 1\right\}. \tag{B.30}$$

In particular, this represents $V(\theta) = F(G_{\mathcal{A}(\theta)}(\theta), D_{\mathcal{A}(\theta)}(\theta))$ which is defined in Lemma B.16. For any $G \in \mathbb{R}^{|\mathcal{I}| \times d_\theta}$ such that $d(0, \mathrm{conv}\{\text{rows of } G\}) \ge c > 0$ for some $c > 0$ and for any $D_1, D_2 \in \mathbb{R}^{|\mathcal{I}|}$, we have

$$|F(G, D_1) - F(G, D_2)| \le \frac{1}{c} \|D_1 - D_2\|.$$

Proof. Note that

$$\begin{aligned} |F(G, D_1) - F(G, D_2)| &\le \sup\left\{\left|\alpha^t (D_1 - D_2)\right| : \left\|\alpha^t G\right\| = 1\right\} \\ &\le \|\alpha\| \cdot \|D_1 - D_2\| \\ &\le \frac{1}{c} \|D_1 - D_2\|, \end{aligned}$$

where the last inequality follows by Lemma B.18. □

Lemma B.20. *For a continuous function $D : \partial\Theta_I \mapsto \mathbb{R}^{d_q}$, let*

$$\Gamma(D) := \max_{\mathcal{I}} \sup_{\theta \in \overline{F_{\mathcal{I}}}} F(G_{\mathcal{I}}(\theta), D_{\mathcal{I}}(\theta)), \tag{B.31}$$

where $F_{\mathcal{I}}$ is as in Lemma B.17 and F is as in Lemma B.19. Then $\sup_{\theta \in \partial\Theta_I} V(\theta) = \Gamma(D(\theta)) = \Gamma(\nabla\Phi(\theta, P)(\xi(\theta)))$ P-a.s., and

$$|\Gamma(D) - \Gamma(D')| \le \frac{1}{c} \cdot \sup_{\theta \in \partial\Theta_I} \|D(\theta) - D(\theta')\|.$$

Proof. Recall that $V(\theta) = F(G_{\mathcal{A}(\theta)}(\theta), D_{\mathcal{A}(\theta)}(\theta))$ from the construction of F in (B.30). Since Lemma B.17 shows that for any $\theta \in \partial\Theta_I$ there exists $\mathcal{I} \subset \mathcal{J}_2$ such that $\theta \in F_{\mathcal{I}} \subset \overline{F_{\mathcal{I}}}$, the first result follows from (23) by noting that $\nabla\Phi(\theta, P)(\xi(\theta))$ is continuous P-a.s., by Assumption 5.4(i) and Assumption 5.1.

For the second result, fix $\mathcal{I}$ and $\theta \in \overline{F_{\mathcal{I}}}$. Then

$$|F(G_{\mathcal{I}}(\theta), D_{\mathcal{I}}(\theta)) - F(G_{\mathcal{I}}(\theta), D'_{\mathcal{I}}(\theta)| \leq \frac{1}{c} \left\| D(\theta) - D'(\theta) \right\|,$$

by Lemma B.19. Since

$$\begin{aligned}
&\left| \max_{\mathcal{I}} \sup_{\theta \in \overline{F}_{\mathcal{I}}} F(G_{\mathcal{I}}(\theta), D_{\mathcal{I}}(\theta)) - \max_{\mathcal{I}} \sup_{\theta \in \overline{F}_{\mathcal{I}}} F(G_{\mathcal{I}}(\theta), D'_{\mathcal{I}}(\theta)) \right| \\
&\qquad \leq \max_{\mathcal{I}} \left| \sup_{\theta \in \overline{F}_{\mathcal{I}}} F(G_{\mathcal{I}}(\theta), D_{\mathcal{I}}(\theta)) - \sup_{\theta \in \overline{F}_{\mathcal{I}}} F(G_{\mathcal{I}}(\theta), D'_{\mathcal{I}}(\theta)) \right| \\
&\qquad \leq \max_{\mathcal{I}} \sup_{\theta \in \overline{F}_{\mathcal{I}}} |F(G_{\mathcal{I}}(\theta), D_{\mathcal{I}}(\theta)) - F(G_{\mathcal{I}}(\theta), D'_{\mathcal{I}}(\theta)| ,
\end{aligned}$$

the second result follows. □

Lemma B.21. *Let $\theta \in \partial\Theta_I$ and $\mathcal{I} \subset \mathcal{A}(\theta)$ non-empty. For $j \in \mathcal{I}$ let $s_j(\theta) = -\sqrt{n} \cdot \Phi_j(\theta, P) \geq 0$ and $s = (s_j(\theta))_{j \in \mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|}$. For vector $D_{\mathcal{I}} = (D_j)_{j \in \mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|}$, define*

$$\begin{aligned}
T^s(\theta) &:= \left\{ \psi \in \mathbb{R}^{d_\theta} : \langle \phi_j(\theta), \psi \rangle \leq s_j(\theta),\ j \in \mathcal{I} \right\}, \\
\Psi^s(\theta) &:= \left\{ \psi \in \mathbb{R}^{d_\theta} : \langle \phi_j(\theta), \psi \rangle \leq -D_j(\theta) + s_j(\theta),\ j \in \mathcal{I} \right\}.
\end{aligned}$$

Then $G_{\mathcal{I}} \in \mathbb{R}^{|\mathcal{I}| \times d_\theta}$ is of full row rank, implying that there exist $v, w \in \mathbb{R}^{d_\theta}$ such that $G_{\mathcal{I}}(\theta)v = -D_{\mathcal{I}}(\theta)$ and $G_{\mathcal{I}}(\theta) = s$. Moreover, it follows that $\Psi^s(\theta) = T^s(\theta) \oplus v$ and $T^s(\theta) = T^0(\theta) \oplus w$. Hence,

$$\sup_{\psi \in \overline{B_M(0)}} |d(\psi, \Psi^s(\theta)) - d(\psi, T^s(\theta))| = d_H(\Psi^s(\theta), T^s(\theta)) = d_H\left(T^0(\theta) \oplus v, T^0(\theta)\right).$$

Proof. Assumption 5.4 (iii) implies that $G_{\mathcal{A}(\theta)}(\theta)$ are of full row rank; since $G_{\mathcal{I}}(\theta)$ is a submatrix of $G_{\mathcal{A}(\theta)}(\theta)$, $G_{\mathcal{I}}(\theta)$ has also full row rank. The first results are then immediate. For the second results, note that

$$\begin{aligned}
\Psi^s(\theta) &= \left\{ \psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta)\psi \leq -D_{\mathcal{I}}(\theta) + s \right\} \\
&= \left\{ \psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta)(\psi - v) \leq s \right\} \\
&= T^s(\theta) \oplus v,
\end{aligned}$$

and likewise,

$$T^s(\theta) = \left\{ \psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta)\psi \leq s \right\} = \left\{ \psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta)(\psi - w) \leq 0 \right\} = T^0(\theta) \oplus w.$$

For the last equivalence, since the Hausdorff distance is invariant for shift transformation,

$$d_H(\Psi^s(\theta), T^s(\theta)) = d_H(T^0(\theta) \oplus w \oplus v, T^0(\theta) \oplus w) = d_H(T^0(\theta) \oplus v, T^0(\theta)).$$

□

Lemma B.22. *Let $D_n(\theta) := \sqrt{n} \cdot (\Phi(\theta, \mathbb{P}_n) - \Phi(\theta, P))$. Then*

$$\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I) = \Gamma(D_n) + o_P(1).$$

Proof. First let

$$V_n(\theta) := d_H(\Psi_n(\theta), T_n(\theta)), \quad \theta \in \partial\Theta_I, \tag{B.32}$$

which can be seen as the sample counterpart of $V(\theta)$ in (B.27). As in Lemma B.15, one can get

$$\sup_{(\theta,\psi)\in\partial\Theta_I\times\overline{B_M(0)}} |Z_n(\theta,\psi)| = \sup_{\theta\in\partial\Theta_I} V_n(\theta).$$

Since it follows with probability approaching 1 that

$$\sqrt{n}\cdot\left(d_H(\widehat{\Theta}_I,\Theta_I)\right) = \sup_{x\in\mathcal{T}_n(M)} |Z_n(x)| = \sup_{(\theta,\psi)\in\partial\Theta_I\times\overline{B_M(0)}} |Z_n(\theta,\psi)|,$$

where the first equality follows from Lemma B.12 and the second one follows from Lemma B.14, it holds that $\sqrt{n}\cdot(d_H(\widehat{\Theta}_I,\Theta_I)) = \sup_{\theta\in\partial\Theta_I} V_n(\theta)$.

Next note that $D_n(\theta) \rightsquigarrow D(\theta) = (D_j(\theta))_{j\in\mathcal{J}_2}$ in $\ell^\infty(\Theta)$ by the Donsker property in Assumption 5.1 and $\Phi(\cdot,P)\in C^2(\Theta)$ in Assumption 5.4. In particular, this implies $\sup_{\theta\in\partial\Theta_I}\|D_n - D\| = o_P(1)$. By Lemma B.20, it follows that $\Gamma(D) = \Gamma(D_n) + o_P(1)$. Here recall that $\Gamma(D) = \sup_{\theta\in\partial\Theta_I} V(\theta)$ as seen in Lemma B.20. Combining the arguments, in order to prove the result in Lemma B.22 it suffices to show

$$\limsup_{n\to\infty} \sup_{\theta\in\partial\Theta_I} V_n(\theta) \le \sup_{\theta\in\partial\Theta_I} V(\theta) \le \liminf_{n\to\infty} \sup_{\theta\in\partial\Theta_I} V_n(\theta).$$

For the second inequality, note that

$$|Z_n(\theta,\psi)| \le \sup_{\theta\in\partial\Theta_I} V_n(\theta)$$

for any fixed $(\theta,\psi)\in\partial\Theta_I\times\overline{B_M(0)}$. Note also that

$$\begin{aligned} Z_n(\theta,\psi) &= d(\psi,\Psi_n(\theta)) - d(\psi,T_n(\theta)) \\ &= d(\psi,\Psi(\theta)) - d(\psi,T(\theta)) + o_P(1) = Z(\theta,\psi) + o_P(1), \end{aligned}$$

where the first equality comes from the proof in Lemma B.14, the second one comes from Lemma B.13 combined with the fact that the metric $d(\psi,\cdot)$ is continuous, and the third one follows by the definition of $Z(\theta,\psi)$ in (B.26). Hence one gets $|Z(\theta,\psi)| \le \liminf_{n\to\infty}\sup_{\theta\in\partial\Theta_I} V_n(\theta)$, and the second inequality follows by taking the supremum over (θ,ψ) and using Lemma B.15.

For the first inequality, one can prove it by contradiction. Suppose that the first inequality would not hold. Then there exist some $\epsilon > 0$ and sequences $\{\theta_n\}_{n\in\mathbb{N}}\subset\partial\Theta_I$ and $\{\psi_n\}_{n\in\mathbb{N}}\subset\overline{B_M(0)}$ such that $|Z_n(\theta_n,\psi_n)| > \sup_{\theta\in\partial\Theta_I} V(\theta) + \epsilon$. Recall $s_j(\theta) = -\sqrt{n}\cdot\Phi_j(\theta,P)$ defined in Lemma B.21; these take values in $[0,\infty]$ which is compact with the standard ordered topology. Since $\partial\Theta_I$, $\overline{B_M(0)}$ and $[0,\infty]$ are compact, one can take a subsequence such that $\theta_{n_k}\to\theta^*\in\partial\Theta_I$, $\psi_{n_k}\to\psi^*\in\overline{B_M(0)}$ and $s_j(\theta_{n_k})\to s_j^*\in[0,\infty]$.

Now let $\mathcal{I} := \{j \in \mathcal{J}_2 : s_j^* < \infty\}$; clearly $\mathcal{I} \subset \mathcal{A}(\theta^*)$. Define

$$\begin{aligned} T^{s^*} &:= \{\psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta^*)\psi \leq s_{\mathcal{I}}^*\}, \\ &= \{\psi \in \mathbb{R}^{d_\theta} : \langle \phi_j(\theta^*), \psi \rangle \leq s_j^*,\ j \in \mathcal{I}\}, \\ \Psi^{s^*} &:= \{\psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta^*)\psi \leq -D_{\mathcal{I}}(\theta^*) + s_{\mathcal{I}}^*\}, \\ &= \{\psi \in \mathbb{R}^{d_\theta} : \langle \phi_j(\theta^*), \psi \rangle \leq -D_j(\theta^*) + s_j^*,\ j \in \mathcal{I}\}, \end{aligned}$$

which are the slack-local sets as in Lemma B.21. Similar to Lemma B.13, the Taylor expansion of $\Phi_j(\cdot, \mathbb{P}_n)$ around $\{\theta_{n_k} + \psi/\sqrt{n_k}\}$ shows

$$\sqrt{n_k} \cdot \Phi_j(\theta_{n_k} + \psi/\sqrt{n_k}, \mathbb{P}_n) = -s_j(\theta_{n_k}) + \langle \phi_j(\theta^*), \psi \rangle + D_j(\theta^*) + o_P(1),$$

which holds uniformly over $\psi \in \overline{B_M(0)}$. Applying the similar logic as in Lemma B.13 implies that $\Psi_n(\theta_{n_k}) \cap \overline{B_M(0)} \rightsquigarrow \Psi^{s^*} \cap \overline{B_M(0)}$ and $T_n(\theta_{n_k}) \cap \overline{B_M(0)} \to T^{s^*} \cap \overline{B_M(0)}$. Accordingly, it follows that

$$\begin{aligned} |Z_{n_k}(\theta_{n_k}, \psi_{n_k})| &= \left| d(\psi^*, \Psi^{s^*}) - d(\psi^*, T^{s^*}) \right| + o_P(1) \\ &\leq \sup_{\psi^* \in \overline{B_M(0)}} \left| d(\psi^*, \Psi^{s^*}) - d(\psi^*, T^{s^*}) \right| + o_P(1) \\ &\leq d_H(T^{0^*} + v, T^{0^*}) + o_P(1) \\ &= F(G_{\mathcal{I}}(\theta^*), D_{\mathcal{I}}(\theta^*)) + o_P(1), \end{aligned}$$

where $T^{0^*} = \{\psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta^*)\psi \leq 0\}$ and $\Psi^{0^*} = \{\psi \in \mathbb{R}^{d_\theta} : G_{\mathcal{I}}(\theta^*)\psi \leq -D_{\mathcal{I}}(\theta^*)\} = T^{0^*} \oplus v$ with $G_{\mathcal{I}}(\theta^*)v = D_{\mathcal{I}}(\theta^*)$. Since $F(G_{\mathcal{I}}(\theta^*), D_{\mathcal{I}}(\theta^*)) = V(\theta^*)$ as seen in Lemma B.19, this further implies $V(\theta^*) > \sup_{\theta \in \partial\Theta_I} V(\theta) + \epsilon$ with probability approaching 1, which is a contradiction. □

Proof. Proposition B.10.
Lemma B.22 shows that $\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I) = \Gamma(D_n) + o_P(1)$. Since $\Gamma : \ell^\infty(\partial\Theta_I) \mapsto \mathbb{R}_+$ is continuous as seen in Lemma B.20 and that $D_n \rightsquigarrow D$ in $\ell^\infty(\partial\Theta_I)$ by Assumption 5.1 (ii) and Assumption 5.4 (i), one then gets $\Gamma(D_n) \rightsquigarrow \Gamma(D)$ by the continuous mapping theorem. It then follows that

$$\sqrt{n} \cdot d_H(\widehat{\Theta}_I, \Theta_I) \rightsquigarrow \Gamma(D) = \sup_{\theta \in \partial\Theta_I} V(\theta) = \sup_{(\theta,\psi) \in \partial\Theta_I \times \mathbb{R}^{d_\theta}} |Z(\theta, \psi)|,$$

where the first equality follows from Lemma B.20 and the second one from Lemma B.15. This shows the desired result. □

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